



ANS MH10.8.2-2016
(Continuous Maintenance of ANS MH10.8.2-2016)

Normative reference:

ISO/IEC 15418

Rev: 2020-08-27

American National Standard
***Data Identifier and
Application Identifier Standard***

Approved: June 10, 2016
Updated: August 27, 2020

Abstract

This standard provides a comprehensive dictionary of MH 10/SC 8 Data Identifiers and GS1 Application Identifiers, provides for the assignment of new Data Identifiers, as required, and provides a document detailing the correlation, or mapping, of Data Identifiers to Application Identifiers, where a correlation exists.



Material Handling Industry
8720 Red Oak Blvd., Suite 201
Charlotte, NC 28217-3992
Published:

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Published by:

Material Handling Industry for:

**MH10, Unit-Loads & Transport-Packages,
MH10 is an ANSI Accredited Standards Committee**

Secretariat: Material Handling Industry

8720 Red Oak Blvd., Suite 201, Charlotte, NC 28217-3992

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Printed in the United States of America

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Foreword

It is the mission of this committee to develop a comprehensive dictionary of Data and Application Identifiers, assign new Data Identifiers, as required, and to provide a document detailing the correlation, or mapping, of Data Identifiers to Application Identifiers, where a correlation exists.

As with any American National standard, as new requirements are identified, interested parties request the assignment of new Data Identifiers and Application Identifiers to meet the needs of a particular industry or activity. ANSI has designated this standard as being “Under Continuous Maintenance”. Proposed changes to the standard that are accepted by the MH10.8.2 Data Identifier Committee shall be integrated into the previously published version at the recommendation of the committee. Upon approval of the new version by MH10 Subcommittee 8 and the full MH10 committee, the standard will be published as a new version.

The committee plans to incorporate accepted revisions into the standard as frequently as necessary, but in no case will a published revised standard be issued more frequently than yearly, in line with indicated needs and industry developments. Each accepted revision since the last published version shall be identified in a “Document Maintenance Summary” appearing immediately before the Table of Contents of the standard.

This standard has been updated from the last published issue of ANS MH10.8.2 representing the fourth five-year revision of the standard, published in 2015, published in 2006, published in 2002; the first revision occurring in 1995. Requests received subsequent to the date of the standard will be added to the draft standard for trial use and will be considered for incorporation at the fifth five-year revision of the standard.

Users desiring assignment of new Data Identifiers may submit their request to:

EMAIL: DIrequests@mhi.org

MAIL: DIMC
c/o MHI
8720 Red Oak Blvd – Suite #201
Charlotte, NC 28007-3992
USA

PHONE: +1 704.676.1190 Ask for Director of Standards

Users desiring assignment of new Application Identifiers may submit their request to <http://www.gs1.org/>.

Note:

The following annexes are provided:

Annex A	Quick Reference to Data Identifier (DI) Categories
Annex B	Annotated Listing of Assigned Data Identifier (DI) Categories
Annex C	Data Identifier (DI) Application Notes
Annex D	ANS X12.3 Data Element Number 355 Unit of Measure Code
Annex E	ANS X12.3 Data Element Number 374 Date/Time Codes
Annex F	ANS X12.3 Data Element Numbers 208 & 209 Hazardous Material Codes
Annex G	ISO 4217 Currencies and Funds Codes
Annex H	ISO 3166-1 Country Codes
Annex I	Data Identifier and Application Identifier Request Forms & Metadata
Annex J	User Guidance
Annex K	System Identifiers
Annex L	Identifiers for Returnable Packaging Items
Annex M	Material categories and material codes

At the time of approval, the MH10 committee consisted of the following members:

American Trucking Associations	Material Handling Industry
American Wood Packaging Association	Material Handling Management Society
APA - The Engineered Wood Association	Millwood, Inc.
Association of American Railroads	National Wooden Pallet & Container Association
Assoc. of Professional Material Handling Consultants	Packaging Machinery Manufacturers Institute
ASTM	Paper Shipping Sack Mfg. Association
Automotive Industry Action Group	Plastic Drum Institute
Containerization & Intermodal Institute, Inc.	Rack Manufacturers Institute
Fibre Box Association	Reusable Industrial Packaging Association
Flexible Intermediate Bulk Containers Association	Steel Shipping Container Institute
Glass Packaging Institute	The Coca-Cola Company
GS1 US	The Soap & Detergent Association
IDEAlliance	U.S. Air Force
Institute of Packaging Professionals	U.S. Dept. of Agriculture
Integrated Business Communications Alliance	U.S. Dept. of Defense AIT Office
Intermec Technologies Corporation	U.S. Forest Products Laboratory
International Association of Movers	United Parcel Service
International Cargo Handling Coordination Association	Virginia Tech – Center for Unit Load Design
International Foodservice Distributors Association	
International Safe Transit Association	

Data Identifier Maintenance Committee

ANSI MH10.8.2 is a reference standard to ISO/IEC 15418 (GS1 Application Identifiers and MH 10/SC 8 Data Identifiers). As such a Data Identifier Maintenance Committee was established representing diverse interests from various countries. Data Identifier Maintenance Committee Members are:

Bill Hoffman, Automotive Industry Action Group (AIAG) Chair	Heinrich Oehlmann, Eurodata Council, DIN
Joo-Sang Park, KAIST	Clive Hohberger, Consultant
Sten Lindgren, ODX Consulting AB	Richard Fisher, DoD (retired)
Erich Guenter, IBM (Germany) & EDIFICE	

DOCUMENT MAINTENANCE SUMMARY

Any changes to this document will be reflected in the Document Maintenance Summary, below.

Date	Action	Summary
2016-07-19	Add	30B – Packaging Item Number. Number to identify the type of packaging item (material) used when packing products and packages The number will enable packaging item (material) be identified and separated from products, packages, Returnable Transport Items (RTIs) and Returnable Packaging Items (RPIs) during packing. The number is constructed as a sequence of minimum 1 data element: Packaging item (material) number that is unique within the holder's domain.
2016-07-19	Add	31B – Global Unique Packaging Number Global unique number to identify the type of packaging item (material) used when packing products and packages. The global unique number will enable packaging items (materials) be identified and separated from products, packages, Returnable Transport Items (RTIs) and Returnable Packaging Items (RPIs) during packing. The number is constructed as a sequence of 3 concatenated data elements: The IAC, followed by the CIN, followed by the Packaging item (material) number that is unique within the CIN holder's domain.
2016-08-12	Corrected	10N - Due to an error in the assignment of DI “10N” (there is no central authority for data-definition nor maintenance), no new uses of DI “10N” should be implemented. The function of DI “10N” is established in Category 18, MISCELLANEOUS with DI “5R”. It is strongly recommended that existing applications that use DI “10N” migrate to DI 5R”.
2016-08-12	Added	5R - Data in the format and using semantics defined by the holder of a Company Identification Number (CIN) that has been issued by an Issuing Agency Code (IAC) in accordance with ISO/IEC 15459, defined as a sequence of concatenated data elements: IAC, followed by CIN, followed by the separator character “:” (colon) followed by the data in the format and using semantics as defined by the CIN holder. NOTE: Only the data syntax rules (if any) as provided by the declared IAC+CIN within each DI “10N” data stream shall be applied to the data following DI 10N+IAC+CIN.
2016-08-29	Added	6R - ISO/IEC 20248 digital signature data construct. If the underlying data carrier encoding is 7 bits, then only the ISO/IEC 20248 raw format may be used.
2016-09-22	Added	11N - Data fields related to reverse logistics, which covers returns, repair, refurbishing and recycling.

Date	Action	Summary
Original Date: 2015-11-03 Added back: 2016-12-06	Added back (This DI was missed in the 2016 document update)	4I - Globally unique transport vehicle identifier (e.g., Trucks, Trailer) with exclusive assignment - Vehicle Identification Number (VIN) as defined in the U.S. under 49 CFR, §§ 565 and internationally by ISO 3779 followed by the “+” character and the (mandatory) field for the Vehicle Registration License Plate Number.
Original Date: 2016-01-28 Added back: 2016-12-06	Added back (This DI was missed in the 2016 document update)	5I - Unique production vehicle identifier that will be used during the vehicle production processes, consisting of the Body Tag Number (BTN; or any other descriptor used to identify the raw car body, or stated another way, the assemblage of parts that are used to start the vehicle's production), followed by the “+” character, then followed by the Production Order Number (PON), followed by the “+” character, and then followed by the Manufacturer-assigned Serial Number (SN). NOTE: The SN component shall be replaced by the VIN as soon as the VIN is available in the assembly process.
2016-12-06	Corrected	11N - Due to an error in the assignment of DI “11N” (the language which states: ‘DI “11N” shall never be encoded in a 2D or RFID tag together with any other DI elements.’ is not a valid statement), no new uses of DI “11N” should be implemented. The function of DI “11N” is established in DI “12N”. It is strongly recommended that existing applications that use DI “11N” migrate to DI “12N”.
2016-12-06	Added	12N - The Data construct is defined and controlled by the RLA, comprised of 2 segments: the field identifier (FI) code, immediately followed by the data as defined for that element according to the data dictionary of the RLA. It is essentially a catalog of fields with standardized content. The Field Identifiers are posted at http://rla.org/11ncodes The use and structure of these codes are defined at: http://rla.org/11nformat Examples can be found at that site.
Original Date: 2015-09-08 Added back: 2016-12-06	Added back (This DI was missed in the 2016 document update)	13E - Moisture Sensitivity Level, class of time period in which a moisture sensitive device can be exposed to ambient room conditions, according to IPC/JEDEC J-STD-020E.
2017-08-02	Added	25D – Best before date (YYYYMMDD).
2017-08-02	Added	26D - First freeze date: (YYYYMMDD). The first freeze date is defined as the date on which products are frozen directly after slaughtering, harvesting, catching or after initial processing.
2017-08-02	Added	27D - Harvest date: (YYYYMMDD). The date when an animal was slaughtered or killed, a fish has been harvested, or a crop was harvested.
2017-08-02	Added	28D - Harvest date range: (YYYYMMDDYYYYMMDD). The start date and end date range over which harvesting occurred.
2017-08-02	Added	8J - Maritime Mobile Service Identity (MMSI), a nine digit number regulated by the International Telecommunications Union to uniquely identify a ship or a coast radio station.

Date	Action	Summary
2017-08-02	Added	18L - FAO fishing area code: Fishing area code as defined by the Fisheries and Aquaculture Department of the Food and Agricultural Organization (FAO) (http://www.fao.org . Search for Fishing Area Code sub-site). All characters of the GS1 General Specification-defined sub-set of ISO/IEC 646 are allowed.
2017-08-02	Added	35L - A government-assigned approval number of vessel / aquaculture site / farm / processor, starting with an ISO 3166-1 alpha-2 country code, followed by the approval number.
2017-08-02	Added	36L - A government-assigned approval number of producer or farm or first deboning / cutting hall, starting with an ISO 3166-1 alpha-2 country code, followed by the approval number.
2017-08-02	Added	7R - Aquatic Sciences and Fisheries Information System (ASFIS) 'Inter-agency 3-alpha species code', maintained by the Food and Agriculture Organization of the United Nations (www.fao.org).
2017-08-02	Added	8R - Food and Agricultural Organization (FAO) International Standard Classification of Fishing Gears (ISSCFG) code. (www.fao.org).
2017-08-02	Added	9R - Production method for fish and seafood as specified by the Fisheries and Aquaculture Department of the Food and Agricultural Organization (FAO) of the United Nations, according to EU Regulation 1379/2013. (www.fao.org)
2017-08-02	Added	23V- Government-assigned Value Added Tax identification number identifying supplier, starting with an ISO 3166-1 alpha-2 country code (except for Greece, which uses the ISO 639-1 language code EL), followed by the government-assigned VAT number.
2017-08-02	Added	24V - Government-assigned Value Added Tax identification number identifying customer, starting with an ISO 3166-1 alpha-2 country code (except for Greece, which uses the ISO 639-1 language code EL), followed by the government-assigned VAT number.
2017-8-10	Corrected	- Numerous punctuation and grammar errors/omissions. - Added DIs "27Q" through "31Q" to Section III. - DIs "28Q", "29Q" and "30Q"; corrected the examples to reflect the proper DIs (e.g., the example for DI "29Q" said "31Q10"). - Updated DI Request Form to latest version.
2017-10-5	Corrected	Additional corrections to DIs "28Q", "29Q" and "30Q" that were not corrected in the 10AUG2017 correction, above.
2018-1-31	Added	26H - Globally Unique Personal ID, with a "Party Qualifier" code value from EDIFACT Code List 3035, assigned by a holder of a Company Identification Code (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 5 concatenated data elements: IAC followed by CIN, followed by an ID unique within the CIN holder's domain, followed by the Plus character (+) and a code value from EDIFACT Code List 3035 "Party Qualifier".

Date	Action	Summary
2020-5-27	Added	53P – Identifier for Specific Marine Equipment approved under the European Union Directive on Marine Equipment (2014/90/EU and Implementing Regulation (EU) 2018/608
2020-5-27	Revised	Revised DIMC roster to remove Martin Treder and Mark Lewis
2020-8-27	Added	54P – Identifier for UDI-DI (Unique Device Identification - Device Identifier) for Medical Devices (MD) and In-vitro-Diagnostics (IvD) as the unique key to public UDI data bases (GUDID, EUDAMED, etc.)

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Data Identifier and Application Identifier Standard

1. Scope

This standard provides a comprehensive dictionary of MH 10/SC 8 Data Identifiers and GS1 Application Identifiers, provides for the assignment of new Data Identifiers, as required, and provides a document detailing the correlation, or mapping, of Data Identifiers to Application Identifiers, where a correlation exists.

This standard does not supersede or replace any applicable safety or regulatory marking or labeling requirements. The standard is to be applied in addition to any other mandated labeling requirements.

Unless otherwise stated within the document, the allowable character set for data fields identified by an ANS MH10.8.2 Data Identifier are the upper case alphabetic characters A to Z and the numeric characters 0 to 9.

2. Normative References

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 646	Information technology -- ISO 7-bit coded character set for information interchange
ISO 3166-1	Codes for the representation of names of countries and their subdivisions - Part 1: Country codes
ISO 4217	Codes for the representation of currency and funds
ISO/IEC 15418	Information technology – Automatic identification and data capture techniques – GS1 Application Identifiers and ASC MH10 Data Identifiers and maintenance
ISO/IEC 15424	Information technology – Automatic identification and data capture techniques – Data carrier identifiers (including Symbolology Identifiers)
ISO/IEC 15459-1	Information technology - Automatic Identification and Data Capture Techniques — Unique identification -- Part 1: Individual transport units
ISO/IEC 15459-2	Information technology – Unique identifiers--- Part 2: Registration procedures
ISO/IEC 19762	Information technology - Automatic identification and data capture (AIDC) techniques -- Harmonized vocabulary
UN/EDIFACT Code List 8053	United Nations Directories for Electronic Data Interchange for Administration, Commerce and Transport – Equipment Type Code Qualifier
UN/EDIFACT Code List 3035	United Nations Directories for Electronic Data Interchange for Administration, Commerce and Transport – Party Function Code Qualifier
UPU Standard M82-3	Universal Postal Union – Attribute Definitions
ANS X12.3	Electronic Data Interchange Data Element Dictionary
ANSI HIBC 2.3	The Health Industry Bar Code (HIBC) Supplier Labeling Standard
IEEE 802.11	Wireless LANs

GS1 General Specifications	GS1 General Specifications
GR-485-CORE	COMMON LANGUAGE® Equipment Codes (CLEI™ Codes) - Generic Requirements for Processes and Guidelines

3. Terms and Definitions

For the purposes of this document, the terms and definitions given in ISO/IEC 19762 and the following apply.

3.1 "+" (plus sign)

The "+" is used with specific Data Identifiers defined within this document (e.g., 14K and 3W) to separate different types of data that are encoded within a single field (e.g., a single bar code symbol). The "+" is also referenced as a flag character used by the HIBCC. The "+" may also be used to concatenate multiple data fields using Data Identifiers.

3.2 allocated

set aside for a specific purpose, such as a set of Data Identifiers allocated for a specific Category

3.3 assigned

designated for a specific purpose

NOTE: Example; "Container Type" has been assigned the Data Identifier "B".

3.4 carrier

party that provides transportation services

3.5 category

class or division in a scheme of classification of Data Identifiers

*NOTE: Example; the Category for date formats is **Category 4: Date.***

3.6 security seal

pre-numbered device designed to secure a container to preclude the doors being opened without detection

3.7 DUNS Number

nine-digit site-specific trading partner identification code assigned by Dun & Bradstreet

3.8 fixed asset

durable or non-consumable item owned by a company or agency

3.9 flag character

character that is used to signify that the data, which follows, conforms to a specific industry standard

*NOTE: these standards do not conform to the overall ANSI MH10.8.2 DAI Standard. See **Category 0.***

3.10 mutually defined

meaning that has been agreed upon by all appropriate parties to the transaction

3.11 package ID

code that provides the ability to differentiate one package from any other package (e.g., carton or label serial number)

NOTE: Also see "Serial Number" and "License Plate".

3.12 PRO number (PRO #)

unique number assigned by a motor freight carrier and placed on a freight bill for internal billing purposes

NOTE: The PRO (PROgressive) number is usually the freight bill (invoice) number. May also be affixed to a container (or containers) in a shipment for tracking purposes.

3.13 returnable (container, packaging item, pallet, etc.)

materials, whose ownership does not transfer, that are shipped with full expectation that such devices will be returned to the supplier (owner)

NOTE: See Annex L.

3.14 VMRS

Vehicle Maintenance Reporting Standard

SECTION I

DATA IDENTIFIERS

(DIs)

DEFINED CATEGORIES

Notes:

1. The usage of the term "number" below is not intended to be restricted to numeric characters only, but generically refers to a code structure which may contain numeric and/or alphabetic data.
2. The following Data Identifiers are assigned to the usages described.
3. The usage of any alphabetic, numeric, or special character in a leading position (as a "Data Identifier or Application Identifier") not defined herein is reserved for future assignment by the body controlling these guidelines.
4. Unless otherwise specified leading zeroes (0's) are non-significant and not to be employed (e.g., 0A, 00A, 000A, 01A, 011A).
5. References to other Standards are to the most current version of that standard.
6. Where field lengths are provided, they are provided along with the length of the identifier, for example, Container serial number "7B" is shown to have a length of "11an" or 11 alphanumeric characters. When the DI is included, the combined fields are 13 characters in length (an2+an11).
7. The META DATA details are interpreted as;
 - a. Examples:
 - i. an2+an1...25 = two alphanumeric characters followed by from 1 to 25 alphanumeric characters.
 - ii. an4+n3+an...5 = four alphanumeric characters followed by three numeric digits followed by from zero (no data) to five alphanumeric characters.

CATEGORY 0:	FLAG CHARACTERS NOT ASSIGNED OR CONTROLLED BY ANSI/MH10.8 NOTE1 & 2	
	The usage of any alphabetic, numeric, or special character in a leading position (as a "Data Identifier or Application Identifier") not defined herein is reserved for future assignment by the body controlling these guidelines.	
Allocation:	All Non-Alphanumeric Characters	
Assigned:		
META DATA	DI	EXPLANATION
	+	The "PLUS" character. Health Industry Business Communications Council (HIBCC).
	-	The "DASH" character. Reserved.
	&	The "AMPERSAND" character. ICCBBA ³ .
	=	The "EQUAL" character. ICCBBA ⁴ .
	FNC1	Function 1 (FUNC1) character. Appears in the first position following the symbology start character of a Code 128, Code 49, or Code 16K Symbol to signify a GS1-controlled symbol.

¹ See Annex K

² This is not an exhaustive list. It is not advisable to assign special characters in a "closed" system unless an exhaustive search has been accomplished that ensures that the special characters in question will never be scanned on items supplied from outside the closed system.

³ Formerly "American Association of Blood Banks (AABB)"

⁴ Formerly "International Society for Blood Transfusion (ISBT)"

	D]>^{Rs}	"Left Square Bracket", "Right Parenthesis", "Greater Than Sign", "Record Separator" characters. Data structure compliant to ISO/IEC 15434, <i>Information technology — Automatic Identification and Data Capture Techniques — Syntax for High Capacity ADC Media</i> .
	-	The "HYPHEN" or "MINUS" character. Pharmaceutical Central Number (PZN), controlled by IFA-ABDATA, Germany (Registration of this system identifier expires on 2016-07-01). Replaced by "9N".
	!	The "EXCLAMATION MARK". Eurocode-IBLS.
CATEGORY 1:	RESERVED	
Allocation:	A - 999A	
Assigned:		
	A - 999A	Reserved
CATEGORY 2:	CONTAINER INFORMATION	
Allocation:	B - 999B	
Assigned:		
META DATA	DI	EXPLANATION
	B	Container Type (internally assigned or mutually defined).
	1B	Returnable Container Identification Code assigned by the container owner or the appropriate regulatory agency (e.g., a metal tub, basket, reel, unit load device (ULD), trailer, tank, or intermodal container) (excludes gas cylinders See "2B").
	2B	Gas Cylinder Container Identification Code assigned by the manufacturer in conformance with U.S. Department of Transportation (D.O.T.) standards.
	3B	Motor Freight Transport Equipment Identification Code assigned by the manufacturer in conformance with International Organization for Standardization (ISO) standards.
	4B	Standard Carrier Alpha Code (SCAC) (4 alphanumeric characters) and an optional carrier assigned trailer number (one to ten alphanumeric characters). When used, the carrier assigned trailer number is separated from the SCAC by a dash "-".

an2+an1...35	5B	Receptacle Asset Number – Consisting of two joined parts: — Identification of an organization in accordance with ISO/IEC 15459 and a unique entity identification assigned in accordance with rules established by the issuing agency. — A unique serial number assigned by the entity, ending with a 3-character container type code taken from EDIFACT Code List 8053 or UPU standard M82-3. (If the container type code listed is less than three characters in length, the field will be dash "-" filled left to the length of three characters).
	6B	Reserved.
an2+an11	7B	Container Serial Number. According to ISO 6346. OC EI CSN CD, where the OC is the three letter owner code assigned in cooperation with BIC, the EI is the one letter equipment category identifier, the CSN is a 6-digit unique container identification assigned by the equipment owner, and CD is a modulus 11 check digit calculated in accordance with Annex A, ISO 6346.
an2+an3	8B	Identification of a Returnable Container owner assigned in cooperation with BIC.
an2+an4	9B	Container Size/Type Code. According to ISO 6346, §4.2.
an3+an4	10B	Container Ownership Code. Actual four-character abbreviation marked on the container by the owner. For DOD owned containers see Defense Transportation Regulation App EE-6.
	11B	Van Number (complete number minus check digit).
	12B	Check digit of Van Number identified in 11B.
	13B	Container Number Code (last 5 digits of number not counting check digit).
an3+a1	14B	Tag Status. Y=Authorized / N=Unauthorized
an3+an1...4	15B	Dangerous Cargo Class. IMDG Class in the format "n.na" where n = numeric, decimal point expressly encoded, and a = conditional alphabetic qualifier. http://docs.imo.org/
an3+an4	16B	UN Code for Dangerous Goods. For dangerous cargo provided by shipper in accordance with UN Code. www.unece.org/trans/danger/publi/unrec/English/part3.pdf
an3+an1...35	17B	Name Of Transportation Subject. Vessel name or vehicle code/train trip number in English.

an3+an3+n7	18B	Vessel Registration Number. The three letters "IMO" followed by the seven-digit number assigned to all ships by IHS Fairplay when constructed. http://www.imonumbers.lrfairplay.com/
an3+an18	19B	Voyage number/Trip number. Letter and number.
an3+an2	20B	Vessel Country. ISO 3166-1 Alpha 2 Code.
an3+6	21B	Reserved for Electronic Seal Numbers. Comprised of the 18185-1 seal tag ID - 32 bits and the ISO 14816 16-bit manufacturers ID (ISO 646).
an3+an11+n2	22B	Entry Number/Type. Comprised of the three-digit filer code, followed by the seven-digit entry number, and completed with the one digit check digit. Entry Filer Code represents the three-character alphanumeric filer code assigned to the filer or importer by CBP. Entry Number represents the seven-digit number assigned by the filer. The number may be assigned in any manner convenient, provided that the same number is not assigned to more than one CBP Form 7501. Leading zeros must be shown. Check Digit is computed on the previous 10 characters. The formula for calculating the check digit can be found in Appendix 1, CBP 7501 Instructions. Entry type is a two-digit code compliant to Block 2, CBP 7501 Instructions.
an3+n3	23B	Surety Number. The three-digit numeric code that identifies the surety company on the Customs Bond. This code can be found in block 7 of the CBP Form 301, or is available through CBP's automated system to ABI filers, via the importer bond query transaction. For U.S. Government importations and entry types not requiring surety, code 999 should appear in this block. When cash or Government securities are used in lieu of surety, use code 998.
an3+n5	24B	Foreign Port of Lading. "Schedule K" (Classification of Foreign Ports by Geographic Trade Area and Country) for the foreign port at which the merchandise was actually laden on the vessel that carried the merchandise to the U.S. http://www.navigationdatacenter.us/wcsc/scheduleK/schedulek.htm
an3+an1...35	25B	Identification of a Party to a Transaction as defined in ISO 17364, assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by the RTI serial number that is unique within the CIN holder's domain.

	26B	Unique Returnable Transport Item Identifier comprised of a sequence of 5 data elements: "IAC", followed by "CIN", followed by "RTI Number" (RTIN), followed by the "+" character, followed by the supplier assigned (or managed) "RTI Serial Number" (RTISN) that is globally unique within the CIN holder's domain; in the format IAC CIN RTIN + RTISN (spaces added for visual clarity only; they are not part of the data). See Annex C.11.
an3+an20...50	27B	Globally Unique Asset Identifier of a Large Load Carrier (LLC) Returnable Transport Item (RTI) with a side base of ≥ 1000 mm, as defined in ISO 17365:2013, tertiary packaging, layer 3 comprised of a sequence of 5 data elements: "IAC", followed by "CIN", followed by RTI Type Code "RTITC", followed by the "+" character, followed by the owner assigned (or managed) RTI Serial Number "RTISN" that is globally unique within the CIN holder's domain in the format IAC CIN RTITC + RTISN (spaces added for visual clarity only; they are not part of the data).
an3+an20...50	28B	Globally Unique Asset Identifier of a Small Load Carrier (SLC) Returnable Transport Item with a side base of < 1000 mm, as defined in ISO 17364:2013 (RTI), tertiary packaging, layer 2 comprised of a sequence of 5 data elements: "IAC", followed by "CIN", followed by RTI Type Code "RTITC", followed by the "+" character, followed by the owner assigned (or managed) RTI Serial Number "RTISN" that is globally unique within the CIN holder's domain in the format IAC CIN RTITC + RTISN (spaces added for visual clarity only; they are not part of the data).
an3+an1...50	29B	Globally Unique Returnable Packaging Item (RPI) identifier of the category packaging aid (lid, blister, inlay, ...) comprised of a sequence of 5 data elements: "IAC", followed by "CIN", followed by "RPI Number" RPIN, followed by the "+" character, followed by the owner assigned (or managed) "RPI Serial Number" RPISN that is globally unique within the CIN holder's domain in the format IAC CIN RPIN + RPISN (spaces added for visual clarity only; they are not part of the data).
an3+an2...35	30B	Packaging Item Number. Number to identify the type of packaging item (material) used when packing products and packages The number will enable packaging item (material) be identified and separated from products, packages, Returnable Transport Items (RTIs) and Returnable Packaging Items (RPIs) during packing. The number is constructed as a sequence of minimum 1 data element: Packaging item (material) number that is unique within the holder's domain.

an3+an6...35	31B	Global Unique Packaging Number Global unique number to identify the type of packaging item (material) used when packing products and packages. The global unique number will enable packaging items (materials) be identified and separated from products, packages, Returnable Transport Items (RTIs) and Returnable Packaging Items (RPIs) during packing. The number is constructed as a sequence of 3 concatenated data elements: The IAC, followed by the CIN, followed by the Packaging item (material) number that is unique within the CIN holder's domain.
	32B – 54B	Reserved.
an3+an1...50	55B	Global Unique Returnable Packaging Item (RPI) as defined in ISO 17364, assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by the RPI serial number that is unique within the CIN holder's domain.
	56B – 999B	Reserved for future assignment.
CATEGORY 3:	FIELD CONTINUATION	
Allocation:	C - 999C	
Assigned:		
META DATA	DI	EXPLANATION
	C	Continuation of an Item Code (Category 16) assigned by Customer that is too long for a required field size.
	1C	Continuation of Traceability Code (Category 20) assigned by Supplier.
	2C	Continuation of Serial Number (Category 19) assigned by Supplier.
	3C	Continuation of Free Text (Category 26) mutually defined between Supplier/Carrier/Customer.
	4C	Continuation of Transaction Reference (Category 11) mutually defined between Supplier/Carrier/Customer.
	5C	Continuation of Item Code (Category 16) Assigned by Supplier.
	6C - 999C	Reserved for future assignment.
CATEGORY 4:	DATE	
Allocation:	D - 999D	
Assigned:		

META DATA	DI	EXPLANATION
an2+n6	D	Format YYMMDD. Note 5
an2+n6	1D	Format DDMMYY. Note 5
an2+n6	2D	Format MMDDYY. Note 5
an2+n4	3D	Format YDDD (Julian). Note 5
an2+n5	4D	Format YYDDD (Julian). Note 5
an2+n6+an3	5D	ISO format YYMMDD, immediately followed by an ANSI X12.3 Data Element Number 374 Qualifier providing a code specifying type of date (e.g., ship date, manufacture date).
an2+n8+an3	6D	ISO format YYYYMMDD, immediately followed by an ANSI X12.3 Data Element Number 374 Qualifier providing a code specifying type of date (e.g., ship date, manufacture date).
an2+n4	7D	Format MMY. Note 5

⁵ Mutually Defined Significance

an2+n14...15	8D	Event, Date, And Time. ISO format YYYYMMDDHHMM (24 hour clock - UTC) immediately followed by a UN/EDIFACT Code Qualifier 2005 providing a code specifying type of date), e.g., 11 [Date when goods are expected to be dispatched/shipped message is issued.] 17 [Estimated delivery date/time when goods are expected to be delivered] 35 [Date on which goods are delivered to their destination.] 118 [Booking Confirmed] 129 [Date when the vessel/merchandise departed the last foreign port in the exporting country.] 132 [Date/time when the carrier estimates that a means of transport should arrive at the port of discharge or place of destination.] 133 [Date/time when carrier estimates that a means of transport should depart at the place of departure] 137 [Date/time when the supplier ships parts based on the customer's request. (Date when DESADV message is issued. Recommendation is the DESADV is issued within 30 minutes of goods being picked up at Ship-From party)] 146 [Estimated Entry date (Customs) date on which the official date of a Customs Entry is anticipated.] 151 [Import Date (Arrived at port with intent to unlade)] 186 Departs a Facility ("Gate-out") 204 [Date on which Customs releases merchandise to the carrier or importer] 253 [Departs from a Port ("Vessel Departure")] 252 [Arrives at a Port ("Vessel Arrival")] 283 [Arrives at a Facility ("Gate-in")] 342 [Conveyance Loaded] 351 [Terminal Gate Inspection] 411 [Ordered Stuffed] 412 [Ordered Stripped] 420 [Conveyance unloaded] 534 [Repaired] 677 [Confirmed Stuffed] 678 [Confirmed Stripped] 696 [Filing Date]
	9D	Date (structure and significance mutually defined).
an3+n4	10D	Format YYWW. Note 5
an3+n6	11D	Format YYYYWW. Note 5
an3+n8	12D	Format YYYYMMDD. Note 5
an3+n8	13D	Oldest and Newest Manufacturing Date in the format YYWWYYWW.
an3+n8	14D	Expiration Date (YYYYMMDD).

an3+n8	15D	Expiration Date (DDMMYYYY).
an3+n8	16D	Production Date (YYYYMMDD) – Date of manufacture.
an3+n8	17D	Production Date (DDMMYYYY).
an3+n12	18D	Tag Activation Time. YYYYMMDDHHMM (24 hour clock - UTC).
an3+n12	19D	Tag Deactivation Time. YYYYMMDDHHMM (24 hour clock - UTC).
an3+n8	20D	Inspection Date (DDMMYYYY).
	21D	Required Delivery Date (DDD Julian) or DOD MILSTAMP Code.
an3+n12	22D	Record Time. YYYYMMDDHHMM (24 hour clock - UTC).
	23D	Date, represented in modified UTC compliant form: yyyy[mm[dd[hh[mm[ss[fff]]]]]]][poooo] where square brackets indicate optionality and yyyy is the year, mmdd the month and day, hhmmss the time of day in hours minutes and seconds, fff the fractions of sections and poooo the offset from UTC expressed in hours and minutes, the offset being positive if p is a point (.), negative if P is a minus sign (-). EXAMPLE: 2005 (UTC) calendar year 2005 200505 (UTC) calendar month May 2005 20050518 (UTC) 18 May 2005 200505181247 12:47 UTC on 18 May 2005 200505181247.0100 12:47 local time, being 11:47 UTC, on 18 May 2005 20050518124723099 99 milliseconds after UTC 12:47:23 on 18 May 200
	24D	Qualified Date, comprising the concatenation of: — an ISO/IEC 15459 issuing agency code; — a date qualifier conforming to the specifications of that issuing agency; — a date whose format and interpretation comply with the specifications of the issuing agency for that date qualifier.
an3+n8	25D	Best before date: (YYYYMMDD). Example: 25D20170202 = February 2, 2017
an3+n8	26D	First freeze date (YYYYMMDD). The first freeze date is defined as the date on which products are frozen directly after slaughtering, harvesting, catching or after initial processing. Example: 26D20170721 = July 21, 2017
an3+n8	27D	Harvest date (YYYYMMDD). The date when an animal was slaughtered or killed, a fish has been harvested, or a crop was harvested. Example: 27D20170615 = June 15, 2017

an3+n16	28D	Harvest date range (YYYYMMDDYYYYMMDD). The start date and end date range over which harvesting occurred. For example; animals were slaughtered or killed, fish were harvested, or a crop was harvested. The data stream is defined as the first YYYYMMDD as the start date and the last YYYYMMDD as the end date. Example: 28D2017012320170214 = Start; January 23, 2017. End; February 14, 2017
	29D– 999D	Reserved for future assignment.
CATEGORY 5:	ENVIRONMENTAL FACTORS	
Allocation:	E - 999E	
Assigned:		
META DATA	DI	EXPLANATION
an1+a2	E	Restricted Substances Classification – “Environmental Classification Code” including Lead-Free (Pb-Free) finish categories defined in JESD97 (IPC JEDEC J-STD-609), and future industry or governmental agency assigned codes related to environmental regulatory compliance and hazardous material content.
	1E	Air Pressure – (altitude) expressed in Pascal’s as the standard international measure.
an2+n1...4	2E	Maximum Allowed Temperature. Maximum permitted temperature; Degrees Celsius, "-" (minus) encoded, if required.
an2+n1...4	3E	Minimum Allowed Temperature. Minimum permitted temperature; Degrees Celsius, "-" (minus) encoded, if required.
an2+n1...2	4E	Maximum Allowed Relative Humidity. Maximum permitted relative humidity, implied as percent.
an2+n1...2	5E	Minimum Allowed Relative Humidity. Maximum permitted relative humidity, expressed as percent.
an2+n1...4	6E	Refrigerator Container Temperature. For temperature-controlled cargo, target specified by shipper, Degrees Celsius, "-" (minus) encoded, if required.
	7E – 9E	Reserved.
	10E	Cumulative Time Temperature Index – expressed as the number of measurements or counts.
	11E	Time Temperature Index – Next Higher Assembly – expressed as the number of measurements or counts.

an3+an9...15	12E	Declaration of Packaging Material Category*, Code* and Weight for a given packaging material used in a given packaging according to the EU packaging and packaging waste directive. (Material category and code defined in Annex M). 12ECCMMMMMMNNNNNUU where - "12E" (an3) is the Data Identifier, - "CC" (n2) is the Material Category per Annex M, - "MMMMMM" (an1...6) is the Material Code per Annex M, - "NNNNN" (n5) Material Weight, including decimal point (e.g., 12.12), - "UU" (an2) is the Unit of measure for weight (e.g., KG, GR, LB or OZ per ANSI X12.3 as in Annex D).
	13E	The data following DI 13E will be one of the MSL indicators (1, 2, 2a, 3, 4, 5, 5a, 6) as shown in the LEVEL column in Table 5-1 of JEDEC standard IPC/JEDEC J-STD-020E. The Table is shown below for reference only; the currently released version of the referenced standard shall be used to obtain the correct MSL for the actual component. Example: 13E2a
Assigned:	14E – 999E	Reserved for future assignment.
CATEGORY 6:	LOOPING	
Allocation:	F - 999F	
Assigned:		
META DATA	DI	EXPLANATION
	F	Looping Header as defined as Section VI of this document.
	1F	My "parent" is . . . Unique identifier followed by a Data Identifier and associated data (for use with returnable packaging – See Annex L). This Data Identifier must immediately follow the field (constructed of a Data Identifier, data and a group separator) with which it is associated.
	2F	Reserved (prior use).
	3F	I have _____ children . . . (for use with returnable packaging, e.g.; 3F10, for ten children – See Annex L). This Data Identifier must immediately follow the field (constructed of a Data Identifier, data and a group separator) with which it is associated.

	4F	Logical Assignment of a Page of Information within a group of pages that are spread across several data carriers, structured as a sequence of up to three (3) concatenated data elements, separated by a slash (/) : Page number (required), followed by page count (optional, required for the last page), followed by an alphanumeric group ID (optional; if used then required for all pages and structured in accordance with ISO/IEC 15459-3 as a sequence of 3 data elements: Issuing Agency Code, followed by the Company Identification Number, followed by an alphanumeric code unique within the issuer's domain). Trailing slashes are optional.
	5F	I have _____ children and they are . . . (for use with returnable packaging – See Annex L) This Data Identifier must immediately follow the field (constructed of a Data Identifier, data and a group separator) with which it is associated.
	6F – 999F	Reserved for future assignment.
CATEGORY 7:	RESERVED	
Allocation:	G - 999G	
Assigned:		
	G - 999G	Reserved.
CATEGORY 8:	HUMAN RESOURCES	
Allocation:	H - 999H	
Assigned:		
META DATA	DI	EXPLANATION
an1+an1...60	H	Name of Party. Name of a party followed by a plus (+) character followed by one or more code values from EDIFACT Code List 3035 "Party Qualifier", e.g., BY [Buyer] CF [Container operator] CN [Consignee] CS [Consolidator] DEI [Vessel operator/captain of vessel] FA [Operational staff code] IM [Importer] MF [Manufacturer] OS [Shipper] SE [Seller] ST [Ship To] UC [Ultimate consignee]
	1H	Employee Identification Code assigned by employer.

an2+n9	2H	U.S. Social Security Number.
	3H	ID Number for Non-Employee (internally assigned or mutually defined) (e.g., contract workers, vendors, service, and delivery personnel).
	4H	National Social Security Number.
	5H	Last Name.
an2+an1...35	6H	Party Name (Line 2).
an2+n10...15	7H	Contact Phone. Country Code, Area Code, Exchange, number [XX YYY ZZZ ZZZZ].
an2+an3...35	8H	Contact Email.
an2+an10...12	9H	Consignee Number. The unique identifying number can be the IRS, EIN, SSN, or the CBP assigned number, as required on the Security Filing. Only the following formats shall be used: IRS EIN: NN-NNNNNNNN IRS EIN w/ suffix: NN-NNNNNNNNXX SSN: NNN-NN-NNNN CBP assigned nbr: YYDDPP-NNNNN
	10H	Personal Identification Code (first initial, Last initial, last four of SSN).
	11H	First Name and Middle Initial.
an2+an2	12H	Military Grade (E1-E9, W1-W5, and O1-O10).
	13H – 14H	Reserved.
an3+an2+an1...an20	15H	A National Identification Number, National Identity Number, or National Insurance Number used as a means of identifying individuals within a country for the purposes of work, taxation, government benefits, health care, and other governmentally-related functions. This structure of the identifier is DI (15H) followed by the ISO 3166-1 Alpha2 Country Code followed by the predominant government assigned identification code for individuals.
	16H – 24H	Reserved.
	25H	Globally Unique Personal ID. assigned by a holder of a Company Identification Code (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as sequence of 3 concatenated data elements: IAC followed by CIN, followed by the ID unique within the holder's domain.

an3+an3...35+"+"a1...3	26H	Globally Unique Personal ID, with a "Party Qualifier" code value from EDIFACT Code List 3035, assigned by a holder of a Company Identification Code (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 5 concatenated data elements: IAC followed by CIN, followed by an ID unique within the CIN holder's domain, followed by the Plus character (+) and a code value from EDIFACT Code List 3035 "Party Qualifier", e.g.: BG Employer GP Packer LK Patient LL Patient companion LM Medical treatment executant MF Manufacturer of goods Example: 26HLHHIBC987XY65+LK
	27H – 999H	Reserved for future assignment.
CATEGORY 9:	RESERVED	
Allocation:	I - 999I	
Assigned:		
META DATA	DI	EXPLANATION
	I	Exclusive Assignment - Vehicle Identification Number (VIN) as defined in the U.S. under 49 CFR, §§ 565 and internationally by ISO 3779. (These are completely compatible data structures).
	1I	Reserved.
	2I	Abbreviated VIN Code.
	3I	Reserved – Prior assignment.
	4I	Globally unique transport vehicle identifier (e.g., Trucks) consisting of the Vehicle Identification Number (VIN) as defined in the U.S. under 49 CFR §§ 565, and internationally by ISO 3779, followed by the "+" character, then followed by the government-issued Vehicle Registration License Plate Number in the form of "4I" "VIN" "+" "government-issued Vehicle Registration License Plate Number" (quotes and spaces shown for clarity only; they are not part of the data).

	5I	Unique production vehicle identifier that will be used during the vehicle production processes, consisting of the Body Tag Number (BTN; or any other descriptor used to identify the raw car body, or stated another way, the assemblage of parts that are used to start the vehicle's production), followed by the "+" character, then followed by the Production Order Number (PON), followed by the "+" character, and then followed by the Manufacturer-assigned Serial Number (SN). NOTE: The SN component shall be replaced by the VIN as soon as the VIN is available in the assembly process. The construction will be as follows; "5I" "BTN" "+" "PON" "+" "SN" changing to (when VIN available) "5I" "BTN" "+" "PON" "+" "VIN" NOTE: Quotes and spaces are shown for clarity only; they are not part of the data. NOTE: This DI is never to be concatenated with other DIs in a linear symbol or other media where the concatenation character is a plus (+) character. Examples: SN version: 5IABCD1234+CO1234+W0L201600500001 VIN version: 5IABCD1234+CO1234+W0L0XAP68F4050901
	6I - 999I	Reserved - Not recommended for use due to similarity of "I" to "l".
CATEGORY 10:	LICENSE PLATE	
Allocation:	J - 999J	
Assigned:		
META DATA	DI	EXPLANATION
an1+an1...35	J	Unique license plate number ⁶ .
an2+an1...35	1J	Unique license plate number ⁶ assigned to a transport unit which is the lowest level of packaging, the unbreakable unit.
an2+an1...35	2J	Unique license plate number ⁶ assigned to a transport unit which contains multiple packages.
an2+an1...35	3J	Unique license plate number ⁶ assigned to a transport unit which is the lowest level of packaging, the unbreakable unit and which has EDI data associated with the unit.

⁶ For a license plate number to be unique world-wide requires: 1) A unique number assigned by the trading partner, 2) A unique code assigned to the trading partner by an organization, and 3) A unique code providing global identification of the assigning organization. ISO/IEC 15459-1:1999 describes the format and usage of these Data Identifiers.

an2+an1...35	4J	Unique license plate number ⁶ assigned to a transport unit which contains multiple packages and which is associated with EDI data.
an2+an1...20	5J	Unique license plate number ⁶ assigned to a mixed transport unit containing unlike items on a single customer transaction and may or may not have associated EDI data.
an2+an1...20	6J	Unique license plate number ⁶ assigned to a master transport unit containing like items on a single customer transaction and may or may not have associated EDI data.
	7J	Vehicle Registration License Plate Number (not unique without identification of country and issuing governmental region/authority) ⁷ .
an2+n9	8J	Maritime Mobile Service Identity (MMSI). A nine digit number regulated by the International Telecommunications Union to uniquely identify a ship or a coast radio station. Example: 8J211123456
	9J– 999J	Reserved for future assignment.
CATEGORY 11:	TRANSACTION REFERENCE USED IN TRADING RELATIONSHIPS	
Allocation:	K - 999K	
Assigned:		
META DATA	DI	EXPLANATION
	K	Order Number assigned by Customer to identify a Purchasing Transaction (e.g., purchase order number).
	1K	Order Number assigned by Supplier to identify a Purchasing Transaction.
	2K	Bill of Lading/Waybill/Shipment Identification Code assigned by Supplier/Shipper.
	3K	Bill of Lading/Waybill/Shipment Identification Code assigned by Carrier.
	4K	Line Number of the order assigned by Customer to identify a Purchasing Transaction. (See Annex C.9).
	5K	Reference Number assigned by the Customer to identify a Shipment Authorization (Release) against an established Purchase Order.
	6K	PRO# Assigned by Carrier.

⁷ The format of "7J" is such that while a Vehicle Registration License Plate Number may, in practice, be unique within a governmental sub-division, it may not be unique worldwide without having met the requirements of items 2 and 3 of Note 3, above.

	7K	Carrier Mode in Free Text format mutually defined between Customer and Supplier (e.g., Air, Truck, Boat, Rail).
	8K	Contract Number.
	9K	Generic Transaction Reference Code (internally assigned or mutually defined).
	10K	Invoice Number.
	11K	Packing List Number.
an3+an4+an1...25	12K	SCAC (Standard Carrier Alpha Code) (an4 - dash "-" filled left) and carrier assigned PROgressive number.
an3+an4+an1...12	13K	Bill of Lading Number /Transport Receipt Number SCAC + Container cargo's B/L number or waybill number.
	14K	Combined Order Number and Line Number in the format nn...nn+nn...n where a plus (+) symbol is used as a delimiter between the Order Number and Line Number.
	15K	KANBAN Number.
	16K	DELINS Number: code assigned to identify a document which contains delivery information.
	17K	Check Number.
	18K	Structured Reference. (See Annex C.10).
	19K	Foreign Military Sales Case Number.
	20K	License Identifier, being a globally unique identifier for a license or contract under which items are generated, submitted for processing and/or paid for, that is constructed by concatenating: — an ISO/IEC 15459 issuing agency code; — a license or contract number which accords with specifications of the issuing agency concerned; and that: — comprises only upper case alphabetic and/or numeric characters; — is unique (that is, is distinct from any other ISO/IEC 15459 compliant identifier) within the domain of the issuing agency⁶; — cannot be derived from any other ISO/IEC 15459 compliant identifier, issued under the same issuing agency, by the simple addition of characters to, or their removal from, its end⁸.

⁸ 20K license identifiers, 26S equipment identifiers and, subject to certain conditions, 18V party identifiers can be used as the root component of 26T batch identifiers and of ISO/IEC 15459 transport unit identifiers. To ensure uniqueness of the latter, it is essential that such identifiers differ not only from all other identifiers of the same class, but also from all other identifiers of other classes. That is, the specifications of the issuing agency concerned are required to ensure that a 20K license identifier is distinct both from other 20K license identifiers and from 26S equipment identifiers, 18V party identifiers, license plates, etc. Since component-based transport unit identifiers are constructed by simple concatenation, it is also required that one root component cannot be derived from another by adding characters to it.

	21K	Customer Data, being data that: — from a customer perspective, is related to or associated with an item or transaction, or to a batch or related items or transactions, and — comprises up to 35 printable characters and/or spaces, other than plus (+), drawn from the character set defined in ISO/IEC 646.
	22K	“22K” Transaction Authentication Information, being a value, constructed by concatenating: — an ISO/IEC 15459 issuing agency code; — a value which accords with specifications of the issuing agency concerned, that allows verification of the authenticity of the transaction concerned and, in particular, that the transaction was initiated by the party, claimed within the transaction to have been its initiator, by: — the recipient of a transaction, and/or — one or more of the parties involved in its handling or processing, and/or — a trusted third party.
	23K – 24K	Reserved
	25K	Global Unique Identification of Groupings of Transport Units Assigned by the Carrier, defined as: Identification of a Party to a Transaction as defined assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by the Bill of Lading or Waybill or Shipment Identification Code that is unique within the CIN holder's domain
	26K	Global Unique Identification of Groupings of Transport Units Assigned by the Shipper, defined as: Identification of a Party to a Transaction assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by the Bill of Lading or Waybill or Shipment Identification Code that is unique within the CIN holder's domain
an3+an1...35	27K	Supplier Assigned Quotation Number – Number assigned to a quotation by the supplier in response to a request for quote from the customer.
	28K – 999K	Reserved for future assignment.
CATEGORY 12:	LOCATION REFERENCE	
Allocation:	L - 999L	

Assigned:		
META DATA	DI	EXPLANATION
	L	Storage Location.
	1L	Location.
	2L	"Ship To:" Location Code defined by an industry standard or mutually defined.
	3L	"Ship From:" Location Code defined by an industry standard or mutually defined.
an2+a2	4L	Country of Origin, two-character ISO 3166 country code. <i>With agreement of trading partners and when the Country of Origin is mixed, Country Code "AA" shall be used.</i>
	5L	"Ship For:" Location Code defined by an industry standard or mutually defined.
	6L	Route Code assigned by the supplier to designate a specific transportation path.
an2+an6	7L	6-character Department of Defense Activity Code (DoDAAC).
	8L	Port of Embarkation – Mutually Defined.
	9L	Port of Debarkation – Mutually Defined.
	10L	Reserved.
an3+n5...27	11L	Location (Latitude/Longitude/Altitude) encoded in the format xnn.nnnnn/xnnn.nnnnnn/xnnnn ⁹ .
	12L	"Ship To:" Location (Latitude/Longitude/Altitude) encoded in the format xnn.nnnnn/xnnn.nnnnnn/xnnnn ⁹ .
	13L	"Ship From:" Location (Latitude/Longitude/Altitude) encoded in the format xnn.nnnnn/xnnn.nnnnnn/xnnnn ⁹ .
	14L	Reserved.
	15L	"Ship For:" Location (Latitude/Longitude/Altitude) encoded in the format xnn.nnnnn/xnnn.nnnnnn/xnnnn ⁹ .
an3+an1...60	16L	Tag Activation Location. English location name (character set: 0-9, A-Z <Space>).

⁹ One degree is equivalent to approximately 110 000 meters, depending upon one's location on the earth where, because of its spheroid shape, a degree is of greater distance at the equator than at the poles. Degrees can be reduced to minutes (1/60th of a degree) and then seconds (1/60th of a minute) to where a second of latitude or longitude is approximately 30.56 meters. Likewise, we can decimally represent a portion of a meter where to six significant digits one can get to a resolution of 0.11 meters. Secondly, many representations of latitude and longitude include either a North / South designation for latitude and an East / West designation for longitude. Alternately, some designations of South latitude and West longitude are differentiated from their North and East counterparts by a "-" (minus) prefix. Further, there are 360 degrees of longitude from 180° E to 180° W (alternately 180° to -180°) and 180 degrees of latitude from 90° N to 90° S (alternately, 90° to -90°). Also, the tallest building in 2011 is 828 meters, though plans do exist for structures greater than 1 000 meters in height. Finally, there are locations that exist below sea level, so a minus (-) is also required for altitude.

Consequently, the format recommended by this standard for Latitude / Longitude / Altitude is xnn.nnnnn/xnnn.nnnnnn/xnnnn is in a range of 5 (intersection of the Prime Meridian with the Equator - n/n/n) to 27 characters (xnn.nnnnn/xnnn.nnnnnn/xnnnn). The "x" value is to permit the inclusion of a "-" (minus) prefix for South latitudes and West longitudes. The "-" (minus) prefix, "." (decimal point), and "/" solidus are explicitly encoded.

an3+an1...60	17L	Tag Deactivation Location. English location name (character set: 0-9, A-Z <Space>).
an3+an2...12	18L	FAO fishing area code as defined by the Fisheries and Aquaculture Department of the FAO (http://www.fao.org . Search for Fishing Area Code sub-site). All characters of the GS1 General Specification-defined subset of ISO/IEC 646 are allowed. Examples: 18L37.1.3 Western Mediterranean Sea, Sardinia 18L47.B.1 Atlantic, Southeast, SEAFO Division, Namibia EEZ 18L67 Pacific, Northeast
	19L– 19L	Reserved .
<i>The following DIs can be used to provide for Location identification, which is different than or in addition to Location Reference provided by "L".</i>		
	20L	First Level (internally assigned).
	21L	Second Level (internally assigned).
	22L	Third Level (internally assigned).
	23L	Fourth Level (internally assigned).
	24L	Fifth Level (internally assigned).
an3+an1...35	25L	Identification of a Party to a Transaction, e.g., 25L IAC CIN LOC assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by the physical internal location (LOC) that is unique within the CIN holder's domain.

	26L	“26L” Location Code, being a code identifying a location or geographic area, or an associated group of such locations or areas, that has relevance to a related transaction and that complies with one or the structures defined in (a) to (f) below: a) two upper case alphabetic character corresponding to the ISO 3166-1 two alpha country code of the country in which, or consisting of which, the location(s) or area(s) are situated; b) three upper case alphabetic characters corresponding to the IATA code of the airport or city in, close to, or consisting of which the location(s) or area(s) are situated; c) four or more characters of which the first three correspond to an ISO 3166-1 country code followed by a dash (-), with the balance being a postcode in the country concerned; d) four or more characters of which the first three correspond to an ISO 3166-1 country code followed by a dot (.), with the balance being an ISO 3166-2 country subdivision code in the country concerned; e) five upper case alphabetic characters corresponding to the UN/LOCODE of the area in, close to, or consisting of which, the location(s) or area(s) are situated; f) the concatenation, being not less than seven or more than 35 characters in length, of: — an ISO/IEC 15459 issuing agency code; — a location code, consisting of characters drawn from the set {A-Z; 0-9} which accords with specifications of the issuing agency concerned.
an3+an5+n1...3	27L	Event Location UN/LOCODE. UN/LOCODE followed by a plus (+) character followed by one or more code values from EDIFACT Code List 3227 “Location function code qualifier”, e.g., 7 Place of Final Delivery 5 Port of Departure 9 Port of Lading 11 Port of Unlading 13 Place of transshipment 24 Port of Entry 35 Exportation country 88 Place of Carrier Receipt 125 Foreign Port prior to Depart to U.S 147 Stowage cell/position 159 Place of delivery (to consignee) 248 Loading Location http://www.unece.org/cefact/locode/
an3+an1...35	28L	Number and Street Address. Used in conjunction with H, 6H, 28L, 29L, 30L, 31L, 32L.
an3+an1...35	29L	City Name. Used in conjunction with H, 6H, 28L, 29L, 30L, 31L, 32L.

an3+an1...9	30L	Country Sub-entity Details. Used in conjunction with H, 6H, 28L, 29L, 30L, 31L, 32L.
an3+an4...11	31L	Postal Code. Used in conjunction with H, 6H, 28L, 29L, 30L, 31L, 32L (If a "-" dash is used, it shall be expressly encoded).
an3+a2	32L	Country Code. ISO 3166-1 Alpha 2 Code Used in conjunction with H, 6H, 28L, 29L, 30L, 31L, 32L.
	33L	Uniform Resource Locator (URL). Includes all characters that form a URL, including header data such as e.g., http://. Character set as listed in RFC 1738.
	34L	Pointer to Process URL (P2P URL) for initiating a URL to carry all other data elements encoded in an AIDC media according to the following rule: Scan the code and initiate the URL starting with the P2P URL string, omitting DI 34L and ISO/IEC 15434 envelope syntax (prefix and postfix) and append all other data elements that have been scanned in same sequence as encoded in the media, including DIs and data element separators. Convert special characters in the appended data into RFC 1738 format (e.g., Group Separator "G _S " translated into RFC 1738 sequence %1D). Note that this does not apply to the P2P URL itself. Example: Encoded data string (using ISO/IEC 15434) J> ^{R_S} 06 ^{G_S} 25SUN123456789PA12345 ^{G_S} 4LUS ^{G_S} 16D20131108 ^{G_S} 34LHTTP://WWW.SECUREUID.COM/ITEMDATA/?SCAN= ^{R_S} 05 ^{G_S} 13131108 ^{R_S} EOT results in the following URL with the transmitted data: <u>HTTP://WWW.SECUREUID.COM/ITEMDATA/?SCAN=25SUN123456789PA12345%1D4LUS%1D16D20131108</u> <i>Note: data from the "05" format envelope was not incorporated in the URL since the 34L was encoded in the "06" format envelope</i>
an3+a2+an3...27	35L	A government-assigned approval number of vessel / aquaculture site / farm / processor, starting with an ISO 3166-1 alpha-2 country code, followed by the approval number. All characters of the GS1 General Specification-defined subset of ISO/IEC 646 are allowed. Example: 35LIECK0107EC = Country; Ireland. Vessel Name; FV Endurance DA31.
an3+a2+an3...27	36L	A government-assigned approval number of producer or farm or first deboning / cutting hall, starting with an ISO 3166-1 alpha-2 country code, followed by the approval number. All characters of the GS1 General Specification-defined subset of ISO/IEC 646 are allowed. Example: 36LIECK0107EC = Country; Ireland. Vessel Name; FV Endurance DA31.
	37L– 50L	Reserved.

The following two Data Identifiers are to be used for shipments within the jurisdiction of a single postal authority.		
an3+an1...9	51L	"Ship From:" - Location code defined by a postal authority (e.g., 5-digit and 9-digit ZIP codes identifying U.S. locations or 6-character postal codes identifying Canadian locations).
an3+an1...9	52L	"Ship To:" - Location code defined by a postal authority (e.g., 5-digit and 9-digit ZIP codes identifying U.S. locations or 6-character postal codes identifying Canadian locations).
	53L	Reserved.
The following two Data Identifiers are to be used for shipments between locations governed by different postal authorities		
an3+an1...9	54L	"Ship From:" - Location code defined by a postal authority in the format: postal codes (e.g., 5-digit ZIP codes identifying U.S. locations or 6- or 7-character postal codes identifying United Kingdom locations) followed by two character ISO 3166 country code (e.g., US or GB).
an3+an1...9	55L	"Ship To:" - Location code defined by a postal authority in the format: postal codes (e.g., 5-digit ZIP codes identifying U.S. locations or 6- or 7-character postal codes identifying United Kingdom locations) followed by two character ISO 3166 country code (e.g., US or GB).
	56L - 999L	Reserved for future assignment.
CATEGORY 13:	RESERVED	
Allocation:	M - 999M	
Assigned:		
META DATA	DI	EXPLANATION
	M	Reserved.
	1M – 9M	Reserved.
	10M	Army Form 2410 data. Format is data value preceded by the block number of the form 2410. Field lengths and acceptable characters can be found at; http://www.apd.army.mil/pdf/p738_751.pdf .
	11M	Army Form 2408 data. Format is data value preceded by the block number of the form 2408. Field lengths and acceptable characters can be found at; http://www.apd.army.mil/pdf/p738_751.pdf .
	12M	Army Form 2407 data. Format is data value preceded by the block number of the form 2407. Field lengths and acceptable characters can be found at; http://www.apd.army.mil/pdf/p738_751.pdf .
	13M	Air Force Form 95 data. Format is data value preceded by the block number of the form 95. Field lengths and acceptable characters can be found at; http://www.gsa.gov/portal/forms/download/116418 .

	14M	Navy Form 4790 data. Format is data value preceded by the block number of the form 2410. Field lengths and acceptable characters can be found at; http://www.navair.navy.mil/logistics/4790/library/Chapter%2015.pdf .
	15M – 999M	Reserved for future assignment.
CATEGORY 14:	INDUSTRY ASSIGNED CODES	
Allocation:	N - 999N	
Assigned:		
META DATA	DI	EXPLANATION
an1+an13...15	N	National/NATO Stock Number (NSN).
	1N	Product Characteristic Data defined by the Chemical Industry Data Exchange (CIDX).
	2N	Reserved.
	3N	Coding Structure in Accordance with Format Defined by Electronic Industries Association Japan (EIAJ).
	4N	Coding Structure and Formats in Accordance with GS1 Application Identifiers (AI plus data) (GS1).
	5N	Coding Structure and Formats in Accordance with AIAG Recommendations. The full Data Identifier is in the form 5Nxx where the “xx” is found in the full code list that can be found at http://www.mhi.org/standards - see under “MH10 Data Identifiers (Continuous Maintenance Version)”.
	6N	U.S. DOD Requisition and Issue Procedure Codes. The format is the appropriate MILSTRIP code followed by the data value associated with that code. (The full list of codes is available at; http://www2.dla.mil/j-6/dlmsolibrary/Manuals/DLM/MILSTRIP/MILSTRIP.pdf .
	7N	U.S. Defense Transportation Regulation Codes. The format is the DTR code followed by the appropriate data value associated with that code. (The full list of codes is available at; http://www.transcom.mil/dtr/part-ii/dtr_part_ii_toc.pdf .
	8N	Production Animal Identification Codes. The format is the production animal code followed by the appropriate data value associated with that code. The Technical Report and the full list of Extended Data Elements (codes) is maintained at; http://www.aimglobal.org/store/view_product.asp?id=4926441 Extended Data Elements (Codes). http://www.aimglobal.org/store/view_product.asp?id=4926483 Technical Report.

an2+an5...22	9N	Pharmacy Product Number maintained by IFA (www.ifaffm.de) and structured as follows: Two-digit product registration agency code followed by the registered product number (assigned by product registration agencies) and two PPN check digits.
	10N	NOTE: Due to an error in the assignment of DI “10N” (there is no central authority for data-definition nor maintenance), no new uses of DI “10N” should be implemented. The function of DI “10N” is established in Category 18, MISCELLANEOUS with DI “5R”. It is strongly recommended that existing applications that use DI “10N” migrate to DI 5R”. Data in the format and using semantics defined by the holder of a Company Identification Number (CIN) that has been issued by an Issuing Agency Code (IAC) in accordance with ISO/IEC 15459, defined as a sequence of concatenated data elements: IAC, followed by CIN, followed by the separator character “:” (colon) followed by the data in the format and using semantics as defined by the CIN holder. NOTE: Only the data syntax rules (if any) as provided by the declared IAC+CIN within each DI “10N” data stream shall be applied to the data following DI 10N+IAC+CIN.
	11N	NOTE: Due to an error in the assignment of DI “11N” (the language which states: ‘DI “11N” shall never be encoded in a 2D or RFID tag together with any other DI elements.’ is not a valid statement), no new uses of DI “11N” should be implemented. The function of DI “11N” is established in DI “12N”. It is strongly recommended that existing applications that use DI “11N” migrate to DI “12N”. The Data construct is defined and controlled by the RLA, and is comprised of 2 segments: the field identifier code, immediately followed by the data as defined for that element according to the data dictionary of the RLA. It is essentially a catalog of fields with standardized content. The Field Identifiers are posted at http://rla.org/11ncodes . The use and structure of these codes are defined at: http://rla.org/11nformat Additional examples can be found at that site as well. DI “11N” shall never be encoded in a 2D or RFID tag together with any other DI elements.
	12N	The Data construct is defined and controlled by the RLA, comprised of 2 segments: the field identifier (FI) code, immediately followed by the data as defined for that element according to the data dictionary of the RLA. It is essentially a catalog of fields with standardized content. The Field Identifiers are posted at http://rla.org/12ncodes . The use and structure of these codes are defined at: http://rla.org/12nformat . Examples can be found at that site.
	13N – 999N	
CATEGORY 15:	RESERVED	

Allocation:	O - 999O	
Assigned:		
	O - 999O	Not recommended for use due to similarity of "O" (zero) to "O" (letter).
CATEGORY 16:	ITEM INFORMATION	
Allocation:	P - 999P	
Assigned:		
META DATA	DI	EXPLANATION
	P	Item Identification Code assigned by Customer.
	1P	Item Identification Code assigned by Supplier.
	2P	Code Assigned to Specify the Revision Level for an Item (e.g., engineering change level, edition, or revision).
an2+n13...14	3P	Combined Manufacturer Identification Code/Item Code Under the 12/13-digit GS1 Formats, plus supplemental codes, if any.
	4P	Item Code Portion of GS1 Formats.
	5P	Freight Classification Item Number Assigned by Carrier for Purposes of Rating Hazardous Materials (e.g., Motor Freight, Air, Boat, Rail Classification).
	6P	Combined Supplier Identification and Item Code (internally assigned or mutually defined).
	7P	Common Language Equipment Identification (CLEI) assigned by the manufacturer to some telecommunications equipment.
an2+n14	8P	14-digit GS1 format for GTIN-14 code structure.
	9P	Combined Manufacturer Identification Code (9-digit DUNS number assigned by Dun & Bradstreet) and the Item Code/Part Number (assigned by the manufacturer).
	10P	Hazardous Material Code as defined by ANSI X12.3 in the format Data Element 208 (1-character code qualifier) followed by Data Element 209 (Hazardous Material Code)
an3+an10	11P	10-character CLEI Code for telecommunications equipment.
	12P	Document Type (e.g., Pick List, Design Drawing, etc.) (internally assigned or mutually defined).
	13P	VMRS System Code.
	14P	VMRS System and Assembly Code.
	15P	VMRS System, Assembly, & Part Code.
	16P	VMRS System, Assembly, or Part Code. (User Modified).
	17P	Combined GS1 Supplier Identification and Item Code Assigned By The Supplier.
	18P	Combined VMRS supplier ID and Supplier Assigned Part Number.

	19P	Component of an Item. (One product contained in multiple packages).
<i>The following five DIs can be used to provide for Item identification (Item ID), which is different than or in addition to Item ID provided by "P".</i>		
	20P	First Level (Customer Assigned).
	21P	Second Level (Customer Assigned).
	22P	Third Level (Customer Assigned).
	23P	Fourth Level (Customer Assigned).
	24P	Fifth Level (Customer Assigned).
	25P	Identification of a Party to a Transaction Assigned by a Holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by the supplier assigned part number that is unique within the CIN holder's domain.
	26P	Part Number of Next Higher Assembly.
an3+n7...12	27P	Commodity HTS-6 Code; Using the format: 4012.11 or 4012.11.4000 (Decimal point is expressly encoded) The Harmonized System (HS) Classification is a 6-digit standardized numerical method of classifying traded products. HS numbers are used by customs authorities around the world to identify products for the application of duties and taxes. Additional digits are added to the HS number by some governments to further distinguish products in certain categories. In the United States, numbers used to classify exported products are called "Schedule B" numbers. The U.S. Census Bureau administers the Schedule B system. Schedule B numbers, not HS numbers, must be provided on the Shippers' Export Declaration (SED). http://www.niccomp.com/rohs/files/NIC_HTS1006.pdf Import codes are administered by the U.S. International Trade Commission (USITC). http://hts.usitc.gov/
an3+an1...100	28P	Cargo Name. Plain language description (English).
an3+n5	29P	Product Classification Code as defined with the GMDN (Global Medical Device Nomenclature - http://www.gmdnagency.org).
<i>The following five DIs can be used to provide for Item identification (Item ID), which is different than or in addition to Item ID provided by "1P".</i>		
	30P	First Level (Supplier Assigned).
	31P	Second Level (Supplier Assigned).
	32P	Third Level (Supplier Assigned).
	33P	Fourth Level (Supplier Assigned).

	34P	Fifth Level (Supplier Assigned).
	35P – 39P	Reserved.
	40P	A Code Assigned BY A Customer TO THE Identification Number OF THE Manufacturer's Material Safety Data Sheet (MSDS) document that describes the uses, hazards, and chemical composition of a hazardous material.
	41P – 48P	Reserved
an3+an3...9	49P	Export Controlled Item. Subject to export control and or restrictions as identified in the Wassenaar Arrangement. DI followed by the Alpha-2 ISO 3166 Country Code of the country that imposed the restriction followed by Wassenaar Code (http://www.wassenaar.org/controllists/index.html).
an3+an3...35	50P	Manufacturer-Assigned Item Identifier - comprising an item number assigned by the item manufacturer, followed by a plus (+) sign, followed - if required to uniquely identify the item within the manufacturer's product range - by a manufacturer-assigned item version. Example 50PABC+6 would represent item number ABC, item version 6 <i>Note: The item number shall always be followed by a plus sign, even if no item version is present. This is required to permit the unambiguous concatenation of manufacturer-assigned item identifier with another data construct using the concatenation character plus (+). For example, the combination of a 50P manufacturer-assigned item identifier with no item version and a serial number (Data identifier S) on an entity might be encoded as 50PDEF++S1234.</i>
	51P	Globally Unique Item Identifier comprising the Identification of a party to a transaction assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, followed by a plus (+) sign, followed by the Manufacturer-assigned item identifier as defined with 50P Example: 51PJ4LBE0431863103+ABC+ would represent the item with item number ABC and no version number manufactured by the company with Belgian VAT number 0431863103.
an3+an1...50	52P	Color Code. Color of an item/object identified by a code or term mutually agreed upon between trading partners.

an3+a1+n4+an5...20	53P	Identifier for Specific Marine Equipment approved under the European Union Directive on Marine Equipment (2014/90/EU) and Implementing Regulation (EU) 2018/608 Format: – DI (an3); – Type of conformity assessment (CA) module(s) set out in Annex II to Directive 2014/90/EU used for the conforming assessment (a1); – Notified body (NB) identification number assigned by the Commission in accordance with point 3.1 of Annex IV to Directive 2014/90/EU (n4); – Certificate (an5...20)
an3+an1...35	54P	UDI-DI (Unique Device Identification - Device Identifier) for Medical Devices (MD) and In-vitro-Diagnostics (IvD) as the unique key to public UDI data bases (GUDID, EUDAMED, etc.), according to national regulatory requirements, as outlined by the International Medical Device Regulators Forum (IMDRF). All printable characters of the UTF-8 character set are allowed.
	55P - 999P	Reserved for future assignment.
CATEGORY 17:	MEASUREMENT	
Allocation:	Q - 999Q NOTE: If decimal points are to be used, they should be included within the data.	
Assigned:		
META DATA	DI	EXPLANATION
	Q	Quantity, Number of Pieces, or Amount (numeric only) (unit of measure and significance mutually defined).
	1Q	Theoretical Length/Weight (numeric only).
	2Q	Actual Weight (numeric only).
an2+an2	3Q	Unit of Measure, as defined by the two character ANSI X12.3 Data Element Number 355 Unit of Measurement Code.
	4Q	Gross Amount.
	5Q	Net Amount.
	6Q	Where Multiple Containers Comprise a Single Product (the contents of each container must be combined with the content of the other containers to constitute a single product) the Data Identifier "6Q" shall be used to link the various containers. The format # of # ("this is the nth piece of x pieces to define the product") Presented in the format "n/x", where the "/" (slash) is used as a delimiter between two values.

an2+n1...8+an2	7Q	Quantity, Amount, or Number of Pieces in the format: Quantity followed by the two character ANSI X12.3 Data Element Number 355 Unit of Measurement Code.
an2+n4...6	8Q	Container Rated Weight. Manufacturer-assigned weight carrying capability of the container. Assigned at time of manufacture. Unit of measure is kg.
	9Q	Piece Weight: weight of a single item.
	10Q	Reserved .
an3+n4...6	11Q	Tare Weight: weight of an empty container Container body weight. Manufacturer-assigned weight of the empty container. Assigned at time of manufacture. Unit of measure is kg (Tare weight).
an3+n1...10+an3	12Q	Monetary Value established by the Supplier in the format of: the value followed by an ISO 4217 data element code for representing unit of value of currencies and funds (e.g., 12Q2.50USD) (2.50 Monetary Value in USA Dollars) significance mutually defined. Entry Value; Value followed by an ISO 4217 data element code for representing unit of value of currencies and funds (e.g., 12Q2.50USD) (2.50 Monetary Value in USA Dollars)
	13Q	# of # ("this is the <i>n</i> th piece of <i>x</i> pieces in this shipment") Presented in the format " <i>n/x</i> ", where the "/" (slash) is used as a delimiter between two values. See Annex C.6.3 for further information.
	14Q	Beginning Secondary Quantity.
	15Q	Ending Secondary Quantity.
	16Q	Number Of Pieces in Van.
	17Q	Number Of Shipments in Van.
	18Q	Cube expressed in cubic meters or cubic feet followed by the ANSI X12.3 data element number 355 unit of measure code (CR or CF). No implied decimal point.
	19Q	Width expressed in linear meters or linear feet followed by the ANSI X12.3 data element number 355 unit of measure code (LC or LF). No implied decimal point.
	20Q	Height expressed in linear meters or linear feet followed by the ANSI X12.3 data element number 355 unit of measure code (LC or LF). No implied decimal point.
	21Q	Length expressed in linear meters or linear feet followed by the ANSI X12.3 data element number 355 unit of measure code (LC or LF). No implied decimal point.
	22Q	Net Weight Of Shipment expressed in pounds or kilograms (kilos) followed by the ANSI X12.3 data element number 355 unit of measure (LB or KG). No implied decimal point.

	23Q	Van Length expressed in linear meters or linear feet followed by the ANSI X12.3 data element number 355 unit of measure (LC or LF). No implied decimal point.
	24Q	Inside Cube of a Van expressed in cubic meters or cubic feet followed by the ANSI X12.3 data element number 355 of unit measure code (CR or CF). No implied decimal point.
	25Q	Net Explosive Weight (a computed value of explosive equivalent expressed in pound of TNT). The measure of NEW is used internationally for explosive safety quantity distance arc computations. No implied decimal point.
	26Q	Packaging Level, specifying the hierarchical level of packaging in accordance with HIBC (Health Industry Bar Code) specifications.
an3+an1...20	27Q	Single Product Price Value, Net, "." (dot) used as decimal point (e.g., 27Q1000.5 for the price value of 1000.50) Structure: an3+an1...20 <DI><price value> Character set: 0 to 9, dot (ISO 646 ASCII value decimal 46, hexadecimal 2E). Example of encoding using a net price value of 1000: 27Q1000 Example of encoding using a net price value of 1000.50: 27Q1000.5 <i>NOTE: If currency is required it can be taken from another data element used in same code, e.g., 12Q.</i>
an3+an1...10	28Q	Single Price Charge Value For Postage And Packaging, "." (dot) represents the position of a comma (e.g., 28Q100.50 for the value of 100,50) Structure: an3+an1...10 <DI><price value> Character set: 0 to 9, dot Example of encoding using postage & packing value of 100: 28Q100 Example of encoding using postage & packing value of 100,50: 28Q100.50 <i>NOTE: If currency is required it can be taken from another data element used in same code, e.g., 12Q.</i>
an3+n1...6	29Q	Discount Percentage, "." (dot) represents the position of a comma (e.g., 29Q8.5 for a discount value of 8,5%) Structure: an3+n1...6 (12.456) <DI><discount percentage (%)> Character set: 0 to 9, dot Example of encoding using discount percentage of 10%: 29Q10 Example of encoding using discount percentage of 8,5%: 29Q8.5

an3+an1...5	30Q	VAT Percentage, "." (dot) represents the position of a comma (e.g., 30Q8.5 for the VAT value of 8,5%) Structure: an3+an1...5 (12.45) <DI><VAT percentage (%)> Character set: 0 to 9, dot Example of encoding using VAT percentage of 19%: 30Q19 Example of encoding using VAT percentage of 8,5%: 30Q8.5
an3+an3	31Q	Currency, ISO 4217 currency code. Structure: an3+an3 <DI><Currency, e.g., EUR> Character set: A-Z, 0 to 9 Example of encoding using ISO alphabetic code of US Dollar: 31QUSD Example of encoding using ISO alphabetic code of EURO: 31QEUR Example of encoding using ISO numeric code of EURO: 31Q978
	32Q – 999Q	Reserved for future assignment.
CATEGORY 18:	MISCELLANEOUS	
Allocation:	R - 999R	
Assigned:		
META DATA	DI	EXPLANATION
	R	Reserved.
	1R	Return Authorization Code (RMA) assigned by the Supplier.
	2R	Return Code Assigned by the Customer.
	3R	Reserved.
an4	4R	U.S. Department of Defense Identification Code (DoDIC).
	5R	Data in the format and using semantics defined by the holder of a Company Identification Number (CIN) that has been issued by an Issuing Agency Code (IAC) in accordance with ISO/IEC 15459, defined as a sequence of concatenated data elements: IAC, followed by CIN, followed by the separator character ":" (colon) followed by the data in the format and using semantics as defined by the CIN holder. NOTE: Only the data syntax rules (if any) as provided by the declared IAC+CIN within each DI "5R" data stream shall be applied to the data following DI 5R+IAC+CIN.4.

	6R	ISO/IEC 20248 digital signature data construct. If the underlying data carrier encoding is 7 bits, then only the ISO/IEC 20248 raw format may be used. Example with an URL format: <6R><https://20248.sigvr.it/?Oo586eJAMEYCIQCf31EqIJMLGclBpHLIRgBdO> Example with a raw format: <6R><Oo586eJAMEYCIQCf31EqIJMLGclBpHLIRgBdO> An ISO/IEC 20248 data structure contains a digital signature which is used to verify the specified data elements of the message of data elements. The value of 6R, as the first parameter, and the data elements to be verified (stripped from all non-printable characters), as the second parameter, is passed to the ISO/IEC 20248 DecoderVerifier - which will return the verification result: ACCEPT, REJECT or ERROR(error code), and the JSON object of decoded ISO/IEC 20248 additional fields. The ISO/IEC 20248 data structure may contain additional fields and instructions to decode and verify one or more messages of data elements. These instructions will be processed by the ISO/IEC 20248 DecoderVerifier.
an3+an1...3	7R	Aquatic Sciences and Fisheries Information System (ASFIS) 'Inter-agency 3-alpha species code', maintained by the Food and Agriculture Organization of the United Nations (www.fao.org, then search for "ASFIS"). Examples; 7RMUC = Mud carp 7RPCD = Australian freshwater herring 7RWSH = Great white shark
an3+an1...10	8R	Food and Agricultural Organization (FAO) International Standard Classification of Fishing Gears (ISSCFG) code. (www.fao.org) All characters of the GS1 General Specification-defined subset of ISO/IEC 646 are allowed. Examples: 8R02.1.0 = Beach seines 8R03.1.5 = Shrimp trawls 8R05.1.0 = Portable lift net
an3+an2	9R	Production method for fish and seafood as specified by the Fisheries and Aquaculture Department of the Food and Agricultural Organization (FAO) of the United Nations, according to EU Regulation 1379/2013. (www.fao.org). All characters of the GS1 General Specification-defined subset of ISO/IEC 646 are allowed. Examples; 9R01 = Caught at sea 9R02 = Caught in fresh water 9R03 = Farmed
	10R- 999R	Reserved for future assignment.

CATEGORY 19: TRACEABILITY NUMBER FOR AN ENTITY		
Allocation:	S - 999S	
Assigned:		
META DATA	DI	EXPLANATION
	S	Serial Number or Code Assigned by the Supplier to an Entity for its Lifetime, (e.g., computer serial number, traceability number, contract tool identification).
	1S	Additional Code Assigned by the Supplier to an Entity for its Lifetime (e.g., traceability number, computer serial number).
an2+an2...30	2S	Advance Shipment Notification (ASN) Shipment ID (SID) corresponds to ANSI ASC X12 Data Element 396.
	3S	Unique Package Identification Assigned by Supplier (lowest level of packaging which has a package ID code; shall contain like items).
	4S	Package Identification Assigned by Supplier to master packaging containing like items on a single customer order. (See Annex C.7).
	5S	Package Identification Assigned by Supplier to master packaging containing unlike items on a single customer order. (See Annex C.7).
	6S	Package Identification Assigned by Supplier to master packaging containing like items over multiple customer orders. (See Annex C.7).
	7S	Package Identification Assigned by Supplier to master packaging containing unlike items over multiple customer orders. (See Annex C.7).
an2+n18	8S	Supplier ID/Unique Container ID presented in the data format specified by the GS1 SSCC-18.
	9S	Package Identification, Generic (mutually defined).
	10S	Machine, Cell, or Tool ID Code.
	11S	Fixed Asset ID Code.
	12S	Document Number (internally assigned or mutually defined).
	13S	Container Security Seal ¹⁰ .
	14S	4th Class Non-identical parcel post manifesting.
	15S	Serial Number Assigned by the Vendor Entity, that can only be used in conjunction with "13V".
	16S	Version Number, e.g., Software Version.
	17S	Combined 6-digit GS1 Supplier Identification and Unique Package Identification Assigned by the Supplier.

¹⁰ For Freight Containers, this refers to a mechanical seal. See ISO 17712.

an3+an5 + an1...20 ¹¹	18S	CAGE Code & Serial Number Unique Within CAGE.
	19S	Combined Dun & Bradstreet company identification of the supplier followed by a unique package identification assigned by the supplier, in the format nn...nn+nn...n where a plus symbol (+) is used as a delimiter between the DUNS Number and unique package identification.
	20S	Traceability Code for an Entity Assigned by the Customer.
	21S	Tire Identification Number as defined by the U.S. Department of Transportation (D.O.T) under U.S. Code 49 CFR 574.5.
	22S	Unique Individual Identity for Cellular Mobile Telephones.
an3+an12	23S	Media Access Control (MAC) Address conforming with IEEE 802.11.
an3+n6...26	24S	According to ISO/IEC 15963 (value is a conversion of its bit value to 8-bit ASCII values). This Data Identifier could possibly assume any ASCII-256 value. For freight container tags the Registration Authority (RA) for manufacturers is the RA for ISO 14816. (ISO 646).
	25S	Identification of a party to a transaction assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by the supplier assign serial number that is unique within the CIN holder's domain. (See Annex C.11).

¹¹ For the purposes of DI 18S, the characters dash "-" and slash "/" are part of the allowable character set.

an3+an1...35	26S	Equipment Identifier, being a globally unique identifier for a device, an item of equipment or instance of a computer application used in the production, transport, processing or other handling of items, that is constructed by concatenating: — an ISO/IEC 15459 issuing agency code; — an equipment number which accords with specifications of the issuing agency concerned; and that: — comprises only upper case alphabetic and/or numeric characters; — is unique (that is, is distinct from any other ISO/IEC 15459 compliant identifier) within the domain of the issuing agency⁶; — cannot be from any other ISO/IEC 15459 compliant identifier, issued under the same issuing agency, by the simple addition of characters to, or their removal from, its end⁶. Reader ID. Equipment identifier, being a globally unique identifier for a device, an item of equipment or instance of a computer application used in the production, transport, processing or other handling of items.
	27S	Item Number Within Batch, being a string of numeric digits: — that uniquely distinguishes an item, within an identifiable batch of related items, from all other items in the same batch; — whose length is the same for all items within the batch concerned.
	28S	Batch-and-Item Number, being the concatenation of a data identifier 27T batch number and the data identifier 27S item number of an item belonging to the batch concerned.
	29S	Reserved.
	30S	Additional Traceability Code For An Entity Assigned by the Supplier in addition to or different from the traceability code(s) provided by "S" or "1S".
	31S	Beginning Serial Number for serial numbers in sequence.
	32S	Ending Serial Number for serial numbers in sequence.
	33S	Serial Number of Next Higher Assembly.
	34S	Serial Number or Part Number of End Item.
	35S	Bumper Number. (Used in Unit DOD Move).
	36S	Pallet Identifier. (Used for loaded 463L air pallets).

	37S	Unique Item Identifier comprised of a sequence of 5 data elements: "IAC", followed by "CIN", followed by "Part Number (PN)", followed by the "+" character, followed by the supplier assigned (or managed) "Part Serial Number (PSN)" that is globally unique within the CIN holder's domain; in the format IAC CIN PN + PSN (spaces provided for visual clarity only; they are not part of the data). See Annex C.11.
	38S - 41S	Reserved.
an3+n2+an1...3+an1...9+an1.. .30	42S	Unique Item Identifier (UII) in 25S format preceded by numeric value indicating serial number element length for use by systems that require the "serial number" component of a concatenated Serial Number element (IAC+CIN+SN). Format: DI+LI+IAC+CIN+SN (LI=length of SN).
an3+n1...7+n12...18+n1	43S	Integrated Circuit Card Identifier (ICCID) in accordance with ITU-T Recommendation E.118 and ETSI Recommendation GSM 11.11; a maximum of 20 digits consisting of Issuer identification number (IIN; maximum of 7 digits), Individual account identification (variable; length determined by IIN, but the same length within individual IINs), Check digit (single digit calculated using Luhn algorithm. http://en.wikipedia.org/wiki/Luhn_algorithm). 43Siiiiiiiiiiiiiiiiiiiiiiiinnnnnnnnnnnc (i = IIN, n = account identification, c = check digit)
	44S - 49S	Reserved
<i>The following five DIs can be used to provide for identification of entities within a single unit that is different than or in addition to identification provided by "S".</i>		
an3+an1...20	50S	First Level (Supplier Assigned).
an3+an1...20	51S	Second Level (Supplier Assigned).
an3+an1...20	52S	Third Level (Supplier Assigned).
an3+an1...20	53S	Fourth Level (Supplier Assigned).
an3+an1...20	54S	Fifth Level (Supplier Assigned).
	55S - 95S	Reserved.
an3+16...26	96S	EPC number (Typically Serialized Global Trade Identification Number - SGTIN).
an3+an4...25	97S	Encrypted serial number assigned by the Supplier to an entity, which can be authenticated by an independent trusted third party. The encrypted serial number does not describe any parameters of the entity without decryption by an independent third party.
	98S - 999S	Reserved for future assignment.
CATEGORY 20:	TRACEABILITY NUMBER FOR GROUPS OF ENTITIES	
Allocation:	T - 999T	
Assigned:		

META DATA	DI	EXPLANATION
	T	Traceability Number assigned by the Customer to identify/trace a unique group of entities (e.g., lot, batch, heat).
	1T	Traceability Number assigned by the Supplier to identify/trace a unique group of entities (e.g., lot, batch, heat).
	2T	Reserved.
	3T	Exclusive Assignment. (U.S. EPA vehicle identification for emissions testing).
	4T - 19T	Reserved.
<i>The following five DIs can be used to provide for identification of a group of entities, which is different than or in addition to identification provided by "T".</i>		
	20T	First Level (Customer Assigned).
	21T	Second Level (Customer Assigned).
	22T	Third Level (Customer Assigned).
	23T	Fourth Level (Customer Assigned).
	24T	Fifth Level (Customer Assigned).
	25T	Identification of a party to a transaction assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by the supplier assigned traceability number that is unique within the CIN holder's domain.
	26T	Batch Identifier comprising the concatenation of either: — a data identifier 26S mail processing equipment identifier, or — a data identifier 20K license identifier, or — a data identifier 18V party identifier that: — is distinct from any other ISO/IEC 15459 compliant identifier within the domain of the issuing agency concerned⁶; — cannot be derived from another party identifier or any other ISO/IEC 15459 compliant identifier, issued under the same issuing agency, by the simple addition of characters to, or their removal from, its end⁶; with a data identifier 27T batch number, the two being separated by a dash (-) character¹².

¹² Note that the dash character cannot occur in either of the two components and can thus be used to support decomposition of the batch identifier into these components. A transport unit identifier constructed from the same two components and a "27S" item number contains no such separator and cannot be decomposed.

	27T	Batch Number, issued under the control of an identified party or unit of processing equipment, or under the provisions of an identified license, that: — uniquely distinguishes one batch of related items from all other batches to which a batch number is assigned by the party or equipment, or under the license, concerned; — comprises a string of maximum length 10 characters, of which the first (numeric) character indicates the number of following characters, each of which is taken from the set {0-9; A-Z}.
	28T – 29T	Reserved.
<i>The following five DIs can be used to provide for identification of a group of entities, which is different than or in addition to identification provided by "1T".</i>		
	30T	First Level (Supplier Assigned).
	31T	Second Level (Supplier Assigned).
	32T	Third Level (Supplier Assigned).
	33T	Fourth Level (Supplier Assigned).
	34T	Fifth Level (Supplier Assigned).
	35T - 999T	Reserved for future assignment.
CATEGORY 21:	UPU/MH 10/SC8 AGREED UPON CODES	
Allocation:	U - 999U	
Assigned:		
META DATA	DI	EXPLANATION
	U-4U	Reserved.
	5U	Specification of a postal service and associated process data in accordance with UPU standard S25 data construct "Service Data".
	6U	Licensing Post Data, in accordance with the specification in UPU standard S25.
	7U – 14U	Reserved for Assignment for UPU needs in collaboration with ASC MH10/SC 8/WG 2.
	15U	Specification of supplementary postal service and associated process data in accordance with UPU standard S25 data construct "Supplementary Service Data".
	16U	Postal Administration Identifications, being the identification, expressed in accordance with the specification in UPU standard S25, of one or more postal administrations involved in the processing of a mail item or batch.

	17U	UPU Location Code, being a code identifying a location or geographic area, or an associated group of such locations or areas, that has relevance to a related transaction and that complies with one of the structures defined in a) to g) below: a) two upper case alphabetic characters corresponding to the ISO 3166-1 two alpha country code of the country in which, or consisting of which, the location(s) or area(s) are situated; b) three upper case alphabetic characters corresponding to the IATA code of the airport or city in, close to, or consisting of which the location(s) or area(s) are situated; c) four or more characters of which the first three correspond to an ISO 3166-1 country code followed by a dash (-), with the balance being a postcode in the country concerned; d) four or more characters of which the first three correspond to an ISO 3166-1 country code followed by a dot (.), with the balance being an ISO 3166-2 country subdivision code in the country concerned; e) five upper case alphabetic characters corresponding to the UN/LOCODE of the area in, close to, or consisting of which, the location(s) or area(s) are situated; f) six upper case alphanumeric characters corresponding to a UPU IMPC code allocated in accordance with UPU standard S34; g) the concatenation, being not less than seven nor more than 25 characters in length, of: — an issuer code allocated in accordance with UPU standards S31; — a location code, consisting of characters drawn from the set {A-Z; 0-9} which accords with specifications of the issuer concerned.
	18U	Qualified UPU Location Code, concatenation of: — a location category drawn from UPU code list 139; — a data identifier 17U UPU location code.
	19U	License Plate with Service Data and Location Code is a compound data construct, compliant with the specification in UPU standard S25, which includes specification of: — an ISO/IEC 15459-compliant item identifier; — a data identifier 5U compliant specification of the service to be provided in respect of the item; — a data identifier 17U compliant UPU location code or a data identifier 18U compliant qualified UPU location code. <i>Note: For further details, please refer to UPU standard S25. The distinction between a simple UPU location code (DI 17U) and a qualified UPU location code (DI 18U) can be determined from the first character. If this is numeric, 18U applies; if it is alphabetic, 17U applies.</i>
	20U – 54U	Reserved for Assignment for UPU needs in collaboration with ASC MH 10/SC 8/WG 2.

	55U	OCR Data Locator.
	56U – 999U	Reserved for future assignment.
CATEGORY 22:	PARTY TO THE TRANSACTION	
Allocation:	V - 999V	
Assigned:		
META DATA	DI	EXPLANATION
	V	Supplier Code Assigned by Customer.
	1V	Supplier Code Assigned by Supplier.
an2+n8	2V	U.P.C. Company Prefix.
an2+n9	3V	GS1 Company Prefix.
	4V	Carrier Identification Code assigned by an industry standard mutually defined by the Supplier, Carrier, and Customer.
	5V	Financial Institution Identification Code (mutually defined).
	6V	Manufacturer's Identification Code (mutually defined).
	7V	Code assigned to a party which has financial liability for an entity or group of entities (e.g., owner of inventory) (mutually defined).
	8V	Customer Code Assigned by the Customer.
	9V	Customer Code Assigned by the Supplier.
an3+an10...15	10V	Manufacturer ID. <i>NOTE: See Appendix 2, CBP 7501 Instructions.</i>
	11V	Organization with budget responsibility for an entity, process, or procedure (e.g., shop, division, department)(internally assigned).
an3+n9...13	12V	DUNS Number Identifying Manufacturer.
an3+n9...13	13V	DUNS Number Identifying Supplier.
an3+n9...13	14V	DUNS Number Identifying Customer.
	15V	Carrier-Assigned Shipper Number.
	16V	VMRS Supplier ID.
an3+an5	17V	U.S. DoD CAGE Code.
	18V	Identification of a party to a transaction in which the data format consists of two concatenated segments. The first segment is the Issuing Agency Code (IAC) in accordance with ISO/IEC 15459, the second segment is a unique entity identification Company Identification Number (CIN) assigned in accordance with rules established by the issuing agency (see http://www.aimglobal.org/?page=Reg_Authority15459&hhSearchTerms=%22IAC%22).

	19V	Specification of a party's role(s), in a transaction, consisting of one or more code values from EDIFACT Code List 3035 "Party Qualifier", separated by plus (+) characters (Never to be concatenated with other DIs in a linear symbol or other media where the concatenation character is a plus (+) character).
	20V	Identification of a party to a transaction assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by a plus (+) character followed by one or more code values from EDIFACT Code List 3035 "Party Qualifier", separated by plus (+) characters (Never to be concatenated with other DIs in a linear symbol or other media where the concatenation character is a plus (+) character).
an3+an1...35	21V	Identification of a party to a transaction, e.g., 21V IAC CIN OSU, assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by the organizational sub-unit identification assigned by the CIN that is unique within the CIN holder's domain.
an3+an4	22V	Carrier SCAC. Standard Carrier Alpha Code - The National Motor Freight Traffic Association, Inc., (NMFTA) assigns SCACs for all companies except those codes used for identification of freight containers not operating exclusively in North America, intermodal chassis and trailers, non-railroad owned rail cars, and railroads. http://www.nmfta.org/Pages/welcome.aspx Companies seeking identification codes for freight containers not operating in North America should contact the Bureau International des Containers, 38, rue des Blancs Manteaux, F-75004 Paris, France, email: bic@bic-code.org , web www.bic-code.org . Railroads and owners of intermodal chassis, trailers and non-railroad owned rail cars should contact Railinc Customer Service, Attn: Private Marks, 7001 Weston Parkway, Suite 200, Cary, NC 27513, (800) 544-7245, email: private.marks@railinc.com .
an3+a2+an3...18	23V	Government-assigned Value Added Tax identification number identifying supplier, starting with an ISO 3166-1 alpha-2 country code (except for Greece, which uses the ISO 639-1 language code EL), followed by the government-assigned VAT number. Example: 23VIE6388047V assigned to Google Ireland

an3+a2+an3...18	24V	Government-assigned Value Added Tax identification number identifying customer, starting with an ISO 3166-1 alpha-2 country code (except for Greece, which uses the ISO 639-1 language code EL), followed by the government-assigned VAT number. Example: 24VIE6388047V assigned to Google Ireland
	25V– 999V	Reserved for future assignment.
CATEGORY 23:	ACTIVITY REFERENCE	
Allocation:	W - 999W	
Assigned:		
META DATA	DI	EXPLANATION
	W	Work Order Number (e.g., "Production Paper") (internally assigned).
	1W	Operation Sequence Number. A number that defines the order of a particular operation in a series of operations, generally in a manufacturing or assembly process.
	2W	Operation Code/Work Code - the type of work to be performed (internally assigned or mutually defined).
	3W	Combined Work Order Number and Operation Sequence Number in the format nn...n+nn...n where a plus symbol (+) is used as a delimiter between the Work Order Number and the Operation Sequence Number.
	4W	Status Code (internally assigned or mutually defined).
	5W	Work Unit Code – identifies system, subsystem, assembly, component etc. on which maintenance is performed.
	6W	Nomenclature – (internally assigned or mutually defined).
	7W – 9W	Reserved.
	10W	Form Control Number – Preprinted control number on forms.
	11W	Quality Assurance Inspector – Last Name.
	12W	Telephone Number of the Person/Activity Completing the Form – expressed in the format (country code) city or area code plus local number i.e. (1) 319 555 1212.
	13W – 999W	Reserved for future assignment.
CATEGORY 24:	RESERVED	
Allocation:	X - 999X	
Assigned:		
	X - 999X	Reserved.

CATEGORY 25:			INTERNAL APPLICATIONS
Allocation:	Y - 999Y		
Assigned:			
	Y - 999Y	Never to appear on item/document which leaves a closed system environment.	
CATEGORY 26:			MUTUALLY DEFINED
Allocation:	Z - 999Z		
Assigned:			
META DATA	DI	EXPLANATION	
	Z	Mutually Defined Between Customer and Supplier.	
	1Z	Mutually Defined Between Carrier and Supplier.	
	2Z	Mutually Defined Between Customer and Carrier.	
	3Z	Free Text.	
	4Z	Mutually Defined Between Carrier and Trading Partner.	
	5Z - 9Z	Reserved.	
	10Z	Structured Free Text (Header Data).	
	11Z - 99Z	Structured Free Text (Line 1-89 Data).	
	100Z - 999Z	Reserved for future assignment.	

SECTION II

GS1 APPLICATION

IDENTIFIERS

(AIs)

The AIs listed in Section II of this standard represent the assignments made through January 2015. Those wishing further information should contact the GS1 for the current list of AI assignments and relevant standards. Those requesting new AI assignments should use the GS1 Application Identifier Standard Request Form attached to this document.

GS1 Application Identifiers as of 28 January 2015

AI	DATA TITLE	DATA CONTENT	FORMAT (*)	FNC1 REQUIRED (****)
00	SSCC	Serial Shipping Container Code	N2+N18	
01	GTIN	Global Trade Item Number	N2+N14	
02	CONTENT	GTIN of Contained Trade Items	N2+N14	
10	BATCH/LOT	Batch or Lot Number	N2+X1...20	✓
11 (**)	PROD DATE	Production Date (YYMMDD)	N2+N6	
12 (**)	DUE DATE	Due Date (YYMMDD)	N2+N6	
13 (**)	PACK DATE	Packaging Date (YYMMDD)	N2+N6	
15 (**)	BEST BEFORE or BEST BY	Best Before Date (YYMMDD)	N2+N6	
16 (**)	SELL BY	Sell By Date (YYMMDD)	N2+N6	
17 (**)	USE BY OR EXPIRY	Expiration Date (YYMMDD)	N2+N6	
20	VARIANT	Variant Number	N2+N2	
21	SERIAL	Serial Number	N2+X1...20	✓
240	ADDITIONAL ID	Additional Item Identification	N3+X1...30	✓
241	CUST. PART NO.	Customer Part Number	N3+X1...30	✓
242	MTO VARIANT	Made-to-Order Variation Number	N3+N1...6	✓
243	PCN	Packaging Component Number	N3+X1...20	✓
250	SECONDARY SERIAL	Secondary Serial Number	N3+X1...30	✓
251	REF. TO SOURCE	Reference to Source Entity	N3+X1...30	✓
253	GDTI	Global Document Type Identifier	N3+N13+X1...17	✓
254	GLN EXTENSION COMPONENT	GLN Extension Component	N3+X1...20	✓
255	GCN	Global Coupon Number	N3+N13+N1...12	✓
30	VAR. COUNT	Count of Items (Variable Measure Trade Item)	N2+N1...8	✓
310 (***)	NET WEIGHT (kg)	Net weight, kilograms (Variable Measure Trade Item)	N4+N6	

AI	DATA TITLE	DATA CONTENT	FORMAT (*)	FNC1 REQUIRED (****)
311 (***)	LENGTH (m)	Length or first dimension, metres (Variable Measure Trade Item)	N4+N6	
312 (***)	WIDTH (m)	Width, diameter, or second dimension, metres (Variable Measure Trade Item)	N4+N6	
313 (***)	HEIGHT (m)	Depth, thickness, height, or third dimension, metres (Variable Measure Trade Item)	N4+N6	
314 (***)	AREA (m ²)	Area, square metres (Variable Measure Trade Item)	N4+N6	
315 (***)	NET VOLUME (l)	Net volume, litres (Variable Measure Trade Item)	N4+N6	
316 (***)	NET VOLUME (m ³)	Net volume, cubic metres (Variable Measure Trade Item)	N4+N6	
320 (***)	NET WEIGHT (lb)	Net weight, pounds (Variable Measure Trade Item)	N4+N6	
321 (***)	LENGTH (i)	Length or first dimension, inches (Variable Measure Trade Item)	N4+N6	
322 (***)	LENGTH (f)	Length or first dimension, feet (Variable Measure Trade Item)	N4+N6	
323 (***)	LENGTH (y)	Length or first dimension, yards (Variable Measure Trade Item)	N4+N6	
324 (***)	WIDTH (i)	Width, diameter, or second dimension, inches (Variable Measure Trade Item)	N4+N6	
325 (***)	WIDTH (f)	Width, diameter, or second dimension, feet (Variable Measure Trade Item)	N4+N6	
326 (***)	WIDTH (y)	Width, diameter, or second dimension, yards (Variable Measure Trade Item)	N4+N6	
327 (***)	HEIGHT (i)	Depth, thickness, height, or third dimension, inches (Variable Measure Trade Item)	N4+N6	

AI	DATA TITLE	DATA CONTENT	FORMAT (*)	FNC1 REQUIRED (****)
328 (***)	HEIGHT (f)	Depth, thickness, height, or third dimension, feet (Variable Measure Trade Item)	N4+N6	
329 (***)	HEIGHT (y)	Depth, thickness, height, or third dimension, yards (Variable Measure Trade Item)	N4+N6	
330 (***)	GROSS WEIGHT (kg)	Logistic weight, kilograms	N4+N6	
331 (***)	LENGTH (m), log	Length or first dimension, metres	N4+N6	
332 (***)	WIDTH (m), log	Width, diameter, or second dimension, metres	N4+N6	
333 (***)	HEIGHT (m), log	Depth, thickness, height, or third dimension, metres	N4+N6	
334 (***)	AREA (m ²), log	Area, square metres	N4+N6	
335 (***)	VOLUME (l), log	Logistic volume, litres	N4+N6	
336 (***)	VOLUME (m ³), log	Logistic volume, cubic metres	N4+N6	
337 (***)	KG PER m ²	Kilograms per square metre	N4+N6	
340 (***)	GROSS WEIGHT (lb)	Logistic weight, pounds	N4+N6	
341 (***)	LENGTH (i), log	Length or first dimension, inches	N4+N6	
342 (***)	LENGTH (f), log	Length or first dimension, feet	N4+N6	
343 (***)	LENGTH (y), log	Length or first dimension, yards	N4+N6	
344 (***)	WIDTH (i), log	Width, diameter, or second dimension, inches	N4+N6	
345 (***)	WIDTH (f), log	Width, diameter, or second dimension, feet	N4+N6	
346 (***)	WIDTH (y), log	Width, diameter, or second dimension, yard	N4+N6	
347 (***)	HEIGHT (i), log	Depth, thickness, height, or third dimension, inches	N4+N6	
348 (***)	HEIGHT (f), log	Depth, thickness, height, or third dimension, feet	N4+N6	

AI	DATA TITLE	DATA CONTENT	FORMAT (*)	FNC1 REQUIRED (****)
349 (***)	HEIGHT (y), log	Depth, thickness, height, or third dimension, yards	N4+N6	
350 (***)	AREA (i ²)	Area, square inches (Variable Measure Trade Item)	N4+N6	
351 (***)	AREA (f ²)	Area, square feet (Variable Measure Trade Item)	N4+N6	
352 (***)	AREA (y ²)	Area, square yards (Variable Measure Trade Item)	N4+N6	
353 (***)	AREA (i ²), log	Area, square inches	N4+N6	
354 (***)	AREA (f ²), log	Area, square feet	N4+N6	
355 (***)	AREA (y ²), log	Area, square yards	N4+N6	
356 (***)	NET WEIGHT (t)	Net weight, troy ounces (Variable Measure Trade Item)	N4+N6	
357 (***)	NET VOLUME (oz)	Net weight (or volume), ounces (Variable Measure Trade Item)	N4+N6	
360 (***)	NET VOLUME (q)	Net volume, quarts (Variable Measure Trade Item)	N4+N6	
361 (***)	NET VOLUME (g)	Net volume, gallons U.S. (Variable Measure Trade Item)	N4+N6	
362 (***)	VOLUME (q), log	Logistic volume, quarts	N4+N6	
363 (***)	VOLUME (g), log	Logistic volume, gallons U.S.	N4+N6	
364 (***)	VOLUME (i ³)	Net volume, cubic inches (Variable Measure Trade Item)	N4+N6	
365 (***)	VOLUME (f ³)	Net volume, cubic feet (Variable Measure Trade Item)	N4+N6	
366 (***)	VOLUME (y ³)	Net volume, cubic yards (Variable Measure Trade Item)	N4+N6	
367 (***)	VOLUME (i ³), log	Logistic volume, cubic inches	N4+N6	
368 (***)	VOLUME (f ³), log	Logistic volume, cubic feet	N4+N6	
369 (***)	VOLUME (y ³), log	Logistic volume, cubic yards	N4+N6	
37	COUNT	Count of Trade Items	N2+N1...8	✓

AI	DATA TITLE	DATA CONTENT	FORMAT (*)	FNC1 REQUIRED (****)
390 (***)	AMOUNT	Applicable Amount Payable or Coupon Value, local currency	N4+N1...15	✓
391 (***)	AMOUNT	Applicable Amount Payable with ISO Currency Code	N4+N3+N1...15	✓
392 (***)	PRICE	Applicable Amount Payable, single monetary area (Variable Measure Trade Item)	N4+N1...15	✓
393 (***)	PRICE	Applicable Amount Payable with ISO Currency Code (Variable Measure Trade Item)	N4+N3+N1...15	✓
400	ORDER NUMBER	Customer's Purchase Order Number	N3+X1...30	✓
401	GINC	Global Identification Number for Consignment (GINC)	N3+X1...30	✓
402	GSIN	Global Shipment Identification Number (GSIN)	N3+N17	✓
403	ROUTE	Routing Code	N3+X1...30	✓
410	SHIP TO LOC	Ship to - Deliver to Global Location Number	N3+N13	✓
411	BILL TO	Bill to - Invoice to Global Location Number	N3+N13	✓
412	PURCHASE FROM	Purchased from Global Location Number	N3+N13	✓
413	SHIP FOR LOC	Ship for - Deliver for - Forward to Global Location Number	N3+N13	✓
414	LOC No	Identification of a Physical Location - Global Location Number	N3+N13	✓
415	PAY TO	Global Location Number of the Invoicing Party	N3+N13	✓
420	SHIP TO POST	Ship to - Deliver to Postal Code Within a Single Postal Authority	N3+X1...20	✓
421	SHIP TO POST	Ship to - Deliver to Postal Code with ISO Country Code	N3+N3+X1...9	✓
422	ORIGIN	Country of Origin of a Trade Item	N3+N3	✓

AI	DATA TITLE	DATA CONTENT	FORMAT (*)	FNC1 REQUIRED (****)
423	COUNTRY - INITIAL PROCESS.	Country of Initial Processing	N3+N3+N1...12	✓
424	COUNTRY - PROCESS.	Country of Processing	N3+N3	✓
425	COUNTRY - DISASSEMBLY	Country of Disassembly	N3+N3	✓
426	COUNTRY – FULL PROCESS	Country Covering full Process Chain	N3+N3	✓
427	ORIGIN SUBDIVISION	Country Subdivision of Origin	N3+X1...3	✓
7001	NSN	NATO Stock Number (NSN)	N4+N13	✓
7002	MEAT CUT	UN/ECE Meat Carcasses and Cuts Classification	N4+X1...30	✓
7003	EXPIRY TIME	Expiration Date and Time	N4+N10	✓
7004	ACTIVE POTENCY	Active Potency	N4+N1...4	✓
7005	CATCH AREA	Catch Area	N4+X1...12	✓
7006	FIRST FREEZE DATE	First Freeze Date	N4+N6	✓
7007	HARVEST DATE	Harvest Date	N4+N6...12	✓
7008	AQUATIC SPECIES	Species for Fishery Purposes	N4+X1...3	✓
7009	FISHING GEAR TYPE	Fishing Gear Type	N4+N1...10	✓
7010	PROD METHOD	Production Method	N4+X1...2	✓
703s	PROCESSOR # s	Number of Processor with ISO Country Code	N4+N3+X1...27	✓
710	NHRN PZN	National Healthcare Reimbursement Number (NHRN) – Germany PZN	N3+X1...20	✓
711	NHRN CIP	National Healthcare Reimbursement Number (NHRN) – France CIP	N3+X1...20	✓
712	NHRN CN	National Healthcare Reimbursement Number (NHRN) – Spain CN	N3+X1...20	✓
713	NHRN DRN	National Healthcare Reimbursement Number (NHRN) – Brasil DRN	N3+X1...20	✓

AI	DATA TITLE	DATA CONTENT	FORMAT (*)	FNC1 REQUIRED (****)
nnn (****)	NHRN xxx	National Healthcare Reimbursement Number (NHRN) – Country “A” NHRN	N3+X1...20	✓
8001	DIMENSIONS	Roll Products (Width, Length, Core Diameter, Direction, Splices)	N4+N14	✓
8002	CMT No	Cellular Mobile Telephone Identifier	N4+X1...20	✓
8003	GRAI	Global Returnable Asset Identifier	N4+N14+X1...16	✓
8004	GIAI	Global Individual Asset Identifier	N4+X1...30	✓
8005	PRICE PER UNIT	Price Per Unit of Measure	N4+N6	✓
8006	GCTIN	Identification of the Components of a Trade Item	N4+N14+N2+N2	✓
8007	IBAN	International Bank Account Number	N4+X1...34	✓
8008	PROD TIME	Date and Time of Production	N4+N8+N1...4	✓
8010	CPID	Component / Part Identifier	N4 + X1...30	✓
8011	CPID SERIAL	Component / Part Identifier Serial Number (CPID SERIAL)	N4 + N1...12	✓
8017	GSRN - PROVIDER	Global Service Relation Number to identify the relationship between an organisation offering services and the provider of services	N4+N18	✓
8018	GSRN - RECIPIENT	Global Service Relation Number to identify the relationship between an organisation offering services and the recipient of services	N4+N18	✓
8019	SRIN	Service Relation Instance Number	N4+N1...10	✓
8020	REF No	Payment Slip Reference Number	N4+X1...25	✓
8110	-	Coupon Code Identification for Use in North America	N4+X1...70	✓
8200	PRODUCT URL	Extended Packaging URL	N4+X1...70	✓

AI	DATA TITLE	DATA CONTENT	FORMAT (*)	FNC1 REQUIRED (****)
90	INTERNAL	Information Mutually Agreed Between Trading Partners	N2+X1...30	✓
91 to 99	INTERNAL	Company Internal Information	N2+X1...30	✓

NOTES:

(*): The first position indicates the length (number of digits) of the GS1 Application Identifier. The following value refers to the format of the data content. The following convention is applied:

- N numeric digit
- X any character in "Table 1", below
- N3 3 numeric digits, fixed length
- N1...3 from one up to 3 numeric digits
- X1...3 from one up to 3 characters in "Table 1", below

(**): If only year and month are available, DD must be filled with two zeroes.

(***): The fourth digit of this GS1 Application Identifier indicates the implied decimal point position.

Example:

- 3100 Net weight in kg without a decimal point
- 3102 Net weight in kg with two decimal points

(****): All GS1 Application Identifiers indicated with (FNC1) are defined as of variable length and shall be delimited unless this Element String is the last one to be encoded in the symbol. The delimiter shall be a Function 1 Symbol Character in GS1-128 Symbology, GS1 DataBar Expanded Versions and GS1 Composite Symbology and should be a Function 1 Symbol Character in GS1 DataMatrix and GS1 QR Code Symbology.

(*****) An example to illustrate future additional NHRNs. If additional NHRN AIs are required, a request for a new NHRN AI shall be made through the GS1 GSMP (See contact information below).

To request a new GS1 Application Identifier, or a change to an existing one, you should contact your local GS1 Member Organization (MO) for more information. To locate the contact information on your local GS1 MO, please go to: <http://www.gs1.org/contact>

The GS1 Subset of International Standard ISO/IEC 646

Table 1 lists all characters allowed for use in GS1 Application Identifier (AI) Element Strings with the exception of the Component and Parts Identifier. Table 1 corresponds to *ISO/IEC 646* Table 1. All other ISO 646 characters that are not listed here are not allowed in GS1 Application Identifier (AI) Element Strings. Table 2 lists all the characters allowed for use in the GS1 Application Identifier for Component and Parts Identifier.

NOTE: Tables 1 and 2 can be found in the GS1 General Specification version 15, as Figures 7.11-1 and 7.11-2, respectively.

Table 1: GS1 AI Encodable Character Set 82

GRAPHIC SYMBOL	NAME	CODED REPRESENTATION	GRAPHIC SYMBOL	NAME	CODED REPRESENTATION
!	Exclamation mark	2/1	M	Capital letter M	4/13
"	Quotation mark	2/2	N	Capital letter N	4/14
%	Percent sign	2/5	O	Capital letter O	4/15
&	Ampersand	2/6	P	Capital letter P	5/0
'	Apostrophe	2/7	Q	Capital letter Q	5/1
(	Left parenthesis	2/8	R	Capital letter R	5/2
)	Right parenthesis	2/9	S	Capital letter S	5/3
*	Asterisk	2/10	T	Capital letter T	5/4
+	Plus sign	2/11	U	Capital letter U	5/5
,	Comma	2/12	V	Capital letter V	5/6
-	Hyphen/Minus	2/13	W	Capital letter W	5/7
.	Full stop	2/14	X	Capital letter X	5/8
/	Solidus	2/15	Y	Capital letter Y	5/9
0	Digit zero	3/0	Z	Capital letter Z	5/10
1	Digit one	3/1	_	Low line	5/15
2	Digit two	3/2	a	Small letter a	6/1

GRAPHIC SYMBOL	NAME	CODED REPRESENTATION	GRAPHIC SYMBOL	NAME	CODED REPRESENTATION
3	Digit three	3/3	b	Small letter b	6/2
4	Digit four	3/4	c	Small letter c	6/3
5	Digit five	3/5	d	Small letter d	6/4
6	Digit six	3/6	e	Small letter e	6/5
7	Digit seven	3/7	f	Small letter f	6/6
8	Digit eight	3/8	g	Small letter g	6/7
9	Digit nine	3/9	h	Small letter h	6/8
:	Colon	3/10	i	Small letter i	6/9
;	Semicolon	3/11	j	Small letter j	6/10
<	Less-than sign	3/12	k	Small letter k	6/11
=	Equals sign	3/13	l	Small letter l	6/12
>	Greater-than sign	3/14	m	Small letter m	6/13
?	Question mark	3/15	n	Small letter n	6/14
A	Capital letter A	4/1	o	Small letter o	6/15
B	Capital letter B	4/2	p	Small letter p	7/0
C	Capital letter C	4/3	q	Small letter q	7/1
D	Capital letter D	4/4	r	Small letter r	7/2
E	Capital letter E	4/5	s	Small letter s	7/3
F	Capital letter F	4/6	t	Small letter t	7/4
G	Capital letter G	4/7	u	Small letter u	7/5

GRAPHIC SYMBOL	NAME	CODED REPRESENTATION	GRAPHIC SYMBOL	NAME	CODED REPRESENTATION
H	Capital letter H	4/8	v	Small letter v	7/6
I	Capital letter I	4/9	w	Small letter w	7/7
J	Capital letter J	4/10	x	Small letter x	7/8
K	Capital letter K	4/11	y	Small letter y	7/9
L	Capital letter L	4/12	z	Small letter z	7/10

Table 2: GS1 AI Encodable Character Set 39

GRAPHIC SYMBOL	NAME	CODED REPRESENTATION	GRAPHIC SYMBOL	NAME	CODED REPRESENTATION
#	Number Sign	2/3	H	Capital letter H	4/8
-	Hyphen/Minus	2/13	I	Capital letter I	4/9
/	Solidus	2/15	J	Capital letter J	4/10
0	Digit zero	3/0	K	Capital letter K	4/11
1	Digit one	3/1	L	Capital letter L	4/12
2	Digit two	3/2	M	Capital letter M	4/13
3	Digit three	3/3	N	Capital letter N	4/14
4	Digit four	3/4	O	Capital letter O	4/15
5	Digit five	3/5	P	Capital letter P	5/0
6	Digit six	3/6	Q	Capital letter Q	5/1
7	Digit seven	3/7	R	Capital letter R	5/2
8	Digit eight	3/8	S	Capital letter S	5/3
9	Digit nine	3/9	T	Capital letter T	5/4
A	Capital letter A	4/1	U	Capital letter U	5/5
B	Capital letter B	4/2	V	Capital letter V	5/6
C	Capital letter C	4/3	W	Capital letter W	5/7
D	Capital letter D	4/4	X	Capital letter X	5/8
E	Capital letter E	4/5	Y	Capital letter Y	5/9
F	Capital letter F	4/6	Z	Capital letter Z	5/10
G	Capital letter G	4/7	Intentionally left blank		

SECTION III

MAPPING

ANS MH10.8.2 DIs & GS1 AIs

DEFINED CATEGORIES

Editor's Note: The usage of the term "number" below is not intended to be restricted to numeric characters only, but to generically refer to a code structure which may contain numeric and/or alphabetic data. The following Application and Data Identifiers are assigned to the usages described. The usage of any alphabetic, numeric, or special character in a leading position (as a "Data Identifier") not defined herein is Reserved for future assignment by the body controlling these guidelines. Unless otherwise specified, leading zeroes (0s) are non-significant and not to be employed (e.g., 0A, 00A, 000A, 01A, 011A). References to other ANSI Standards are to the most current version of that standard.

"n/e" means no equivalent.

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
CATEGORY 1: RESERVED		
Reserved	A - 999A	n/e
CATEGORY 2: CONTAINER INFORMATION		
Container Type (internally assigned or mutually defined)	B	n/e
Returnable container identification code assigned by the container owner or the appropriate regulatory agency (e.g., a metal tub, basket, reel, unit load device (ULD), trailer, tank, or intermodal container) (excludes gas cylinders See "2B")	1B	8003 or 8004
Gas Cylinder Container Identification Code assigned by the manufacturer in conformance with U.S. Department of Transportation (D.O.T.) standards	2B	n/e
Motor Freight Transport Equipment Identification Code assigned by the manufacturer in conformance with International Organization for Standardization (ISO) standards	3B	n/e
Standard Carrier Alpha Code (SCAC) (an4 - dash "-" filled left) and carrier assigned trailer number	4B	n/e
Receptacle Asset Number – Consisting of two joined parts: • Identification of an organization in accordance with ISO/IEC 15459 and a unique entity identification assigned in accordance with rules established by the issuing agency • A unique serial number assigned by the entity, ending with a 3-character container type code taken from EDIFACT Code List 8053 or UPU standard M82-3. (If the container type code listed is less than three characters in length, the field will be dash "-" filled left to the length of three characters)	5B	8003
Reserved	6B	n/e
Container serial number	7B	n/e
Identification of a returnable container owner assigned in cooperation with BIC	8B	n/e
Container size/type code	9B	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Container Ownership Code. Actual four-character abbreviation marked on the container by the owner. For DOD owned containers see Defense Transportation Regulation App EE-6	10B	n/e
Van Number (complete number minus check digit)	11B	n/e
Check digit of Van Number identified in 11B	12B	n/e
Container Number Code (last 5 digits of number not counting check digit)	13B	n/e
Tag status	14B	n/e
Dangerous cargo class IMDG	15B	n/e
UN Code for Dangerous Goods	16B	n/e
Name of transportation subject	17B	n/e
Vessel registration number	18B	n/e
Voyage number/Trip number	19B	n/e
Vessel Country	20B	n/e
Seal Numbers	21B	n/e
Entry Number/Type	22B	n/e
Number Surety Number	23B	n/e
Foreign Port of Lading	24B	n/e
Identification of a party to a transaction as identified in 18V, followed by the supplier assigned serial number to a returnable transport item (RTI)	25B	8003
Unique RTI Identifier (format: IAC CIN RTIN + RTISN)	26B	n/e
Reserved	27B – 54B	n/e
Global Unique Returnable Packaging Item (RPI) as defined in ISO 17364	55B	n/e
Reserved for future assignment	56B – 999B	n/e
CATEGORY 3: FIELD CONTINUATION		
Continuation of an Item Code (Category 16) assigned by Customer that is too long for a required field size	C	n/e
Continuation of Traceability Code (Category 20) assigned by Supplier	1C	n/e
Continuation of Serial Number (Category 19) assigned by Supplier	2C	n/e
Continuation of Free Text (Category 26) mutually defined between Supplier/Carrier/Customer	3C	n/e
Continuation of Transaction Reference (Category 11) mutually defined between Supplier/Carrier/Customer	4C	n/e
Continuation of Item Code (Category 16) Assigned by Supplier	5C	n/e
Reserved – Prior Assignment (2009) – To be re-released upon publication of AIM IUIC-1, currently on hold.	6C	n/e
Reserved for future assignment	7C - 999C	n/e
CATEGORY 4: DATE		

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Format YYMMDD ^{Note 5}	D	n/e
Format DDMMYY ^{Note 5}	1D	n/e
Format MMDDYY ^{Note 5}	2D	n/e
Format YDDD (Julian) ^{Note 5}	3D	n/e
Format YYDDD (Julian) ^{Note 5}	4D	n/e
ISO format YYMMDD immediately followed by an ANSI X12.3 Data Element Number 374 Qualifier providing a code specifying type of date (e.g., ship date, manufacture date)	5D	n/e
Production Date (YYMMDD)	5D...405	11
Expiration Date (YYMMDD)	5D...036	17
Packaging Date (YYMMDD)	n/e	13
Best Before/Sell By Date (YYMMDD)	n/e	15
ISO format YYYYMMDD immediately followed by an ANSI X12.3 Data Element Number 374 Qualifier providing a code specifying type of date (e.g., ship date, manufacture date)	6D	
Format MMY ^{Note 5}	7D	n/e
Event, date, and time ISO format YYYYMMDDHHMM (24 hour clock - UTC) immediately followed by a UN/EDIFACT Code Qualifier 2005 providing a code specifying type of date)	8D	n/e
Date (structure and significance mutually defined)	9D	n/e
Format YYWW ^{Note 5}	10D	n/e
Format YYYYWW ^{Note 5}	11D	n/e
Format YYYYMMDD ^{Note 5}	12D	n/e
Oldest and Newest Manufacturing Date in the format YYWWYYWW	13D	n/e
Expiration Date (YYYYMMDD)	14D	n/e
Expiration Date (DDMMYYYY)	15D	n/e
Production Date (YYYYMMDD)	16D	n/e
Production Date (DDMMYYYY)	17D	n/e
Date and Time of Production (YYMMDDHHSS)	n/e	8008
Tag activation time	18D	n/e
Tag deactivation time	19D	n/e
Inspection Date (DDMMYYYY)	20D	n/e
Required Delivery Date (DDD Julian) or DOD MILSTAMP Code	21D	n/e
Record time	22D	n/e
Date represented in modified UTC compliant form	23D	n/e
Qualified date	24D	n/e
Best before date	25D	15
First freeze date	26D	7006

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Harvest date	27D	7007
Harvest date range	28D	7007
Expiration Date and Time (YYMMDDHHMM)	n/e	7003
Reserved for future assignment	29D – 999D	n/e
CATEGORY 5: ENVIRONMENTAL FACTORS		
Restricted Substance Classification – “Environmental Classification Code” including Lead-Free (Pb-Free) finish categories defined in JESD97 (IPC JEDEC J-STD-609), and future Industry or governmental agency assigned codes related to environmental regulatory compliance and hazardous material content	E	n/e
Air pressure – (altitude) expressed in Pascal's as the standard international measure	1E	n/e
Maximum Allowed Temperature	2E	n/e
Minimum Allowed Temperature	3E	n/e
Maximum Allowed Relative Humidity	4E	n/e
Minimum Allowed Relative Humidity	5E	n/e
Refrigerator container temperature	6E	n/e
Reserved	7E – 9E	n/e
Cumulative Time Temperature index – expressed as the number of measurements or counts	10E	n/e
Time Temperature Index – Next Higher Assembly – expressed as the number of measurements or counts	11E	n/e
Declaration of packaging material category*, type* and weight for a given packaging material used in a given packaging according to the EU packaging and packaging waste directive. (Material category and code defined in Annex M)	12E	n/e
Reserved for future assignment	13E – 999E	n/e
CATEGORY 6: LOOPING		
Looping Header as defined as Section VI of this document	F	n/e
My “parent” is . . . (for use with returnable packaging – See Annex L). This Data Identifier must immediately follow the field (constructed of a Data Identifier, data and a group separator) with which it is associated.	1F	n/e
Reserved (prior use)	2F	n/e
I have _____ children . . . (for use with returnable packaging, e.g., 3F10 for ten children – See Annex L). This Data Identifier must immediately follow the field (constructed of a Data Identifier, data and a group separator) with which it is associated	3F	n/e
Logical assignment of a page of information within a group of pages that are spread across several data carriers, structured as a sequence of up to three (3) concatenated data elements, separated by a slash (/)	4F	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
I have _____ children and they are . . . (for use with returnable packaging – See Annex L) This Data Identifier must immediately follow the field (constructed of a Data Identifier, data and a group separator) with which it is associated	5F	n/e
Reserved for future assignment	6F – 999F	
CATEGORY 7: RESERVED		
Reserved	G - 999G	
CATEGORY 8: HUMAN RESOURCES		
Name of a party followed by a plus (+) character followed by one or more code values from EDIFACT Code List 3035 “Party Qualifier”	H	n/e
Employee Identification Code assigned by employer	1H	n/e
U.S. Social Security Number	2H	n/e
ID Number for non-employee (internally assigned or mutually defined) (e.g., contract workers, vendors, service, and delivery personnel)	3H	n/e
National Social Security Number	4H	n/e
Last Name	5H	n/e
Party Name (Line 2)	6H	n/e
Contact Phone (Country Code, Area Code, Exchange, number [XX YYY ZZZ ZZZZ])	7H	n/e
Contact Email	8H	n/e
Consignee Number - The unique identifying number can be the IRS, EIN, SSN, or the CBP assigned number	9H	n/e
Personal Identification Code (first initial, Last initial, last four of SSN)	10H	n/e
First name and middle initial	11H	n/e
Military Grade (E1-E9, W1-W5, and O1-O10)	12H	n/e
Reserved	13H – 14H	n/e
A national identification number, national identity number, or national insurance number used as a means of identifying individuals within a country	15H	n/e
Reserved	16H – 24H	n/e
Globally Unique Personal ID	25H	n/e
Reserved for future assignment	26H – 999H	n/e
CATEGORY 9: RESERVED		
Exclusive Assignment - Vehicle Identification Number (VIN) as defined in the U.S. under 49 CFR, §§ 565 and internationally by ISO 3779. (These are completely compatible data structures)	I	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Reserved	1I	n/e
Abbreviated VIN Code	2I	n/e
Reserved – Prior assignment	3I	n/e
Reserved - Not recommended for use due to similarity of "1" to "I"	4I - 999I	n/e
CATEGORY 10: LICENSE PLATE		
Unique license plate number*	J	00
Unique license plate number* assigned to a transport unit which is the lowest level of packaging, the unbreakable unit	1J	00
Unique license plate number* assigned to a transport unit which contains multiple packages	2J	00
Unique license plate number* assigned to a transport unit which is the lowest level of packaging, the unbreakable unit and which has EDI data associated with the unit	3J	00
Unique license plate number* assigned to a transport unit which contains multiple packages and which is associated with EDI data	4J	00
Unique license plate number* assigned to a mixed transport unit containing unlike items on a single customer transaction and may or may not have associated EDI data.	5J	n/e
Unique license plate number* assigned to a master transport unit containing like items on a single customer transaction and may or may not have associated EDI data.	6J	n/e
Vehicle Registration License Plate Number (not unique without identification of country and issuing governmental region/authority)	7J	n/e
Maritime Mobile Service Identity (MMSI)	8J	n/e
Reserved for future assignment	9J– 999J	n/e
<i>*Note: For a license plate number to be unique world-wide requires: 1) A unique number assigned by the trading partner, 2) A unique code assigned to the trading partner by an organization, and 3) A unique code providing global identification of the assigning organization. ISO/IEC 15459-1:1999 describes the format and usage of these Data Identifiers.</i>		
CATEGORY 11: TRANSACTION REFERENCE USED IN TRADING RELATIONSHIPS		
Order number assigned by Customer to identify a Purchasing Transaction (e.g., purchase order number)	K	400
Order number assigned by Supplier to identify a Purchasing Transaction	1K	n/e
Bill of Lading/Waybill/Shipment Identification Code assigned by Supplier/Shipper	2K	402
Bill of Lading/Waybill/Shipment Identification Code assigned by Carrier	3K	n/e
Line number of the order assigned by Customer to identify a Purchasing Transaction (See Annex C.9)	4K	400

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Reference number assigned by the Customer to identify a Shipment Authorization (Release) against an established Purchase Order	5K	400
PRO# Assigned by Carrier	6K	n/e
Carrier Mode in Free Text format mutually defined between Customer and Supplier (e.g., Air, Truck, Boat, Rail)	7K	n/e
Contract Number	8K	n/e
Generic Transaction Reference Code (internally assigned or mutually defined)	9K	n/e
Invoice Number	10K	n/e
Packing List Number	11K	n/e
SCAC (Standard Carrier Alpha Code) (an4 - dash "-" filled left) and carrier assigned PROgressive number	12K	95 or 96
Bill of lading number /transport receipt number SCAC + Container cargo's B/L number or waybill number	13K	n/e
Combined Order Number and Line Number in the format nn...nn+nn...n where a plus (+) symbol is used as a delimiter between the Order Number and Line Number	14K	400
KANBAN Number	15K	n/e
DELINS Number: code assigned to identify a document which contains delivery information	16K	n/e
Check Number	17K	n/e
Structured Reference (See Annex C.10)	18K	n/e
Foreign Military Sales Case Number	19K	n/e
License identifier	20K	n/e
Customer data related to item or transaction	21K	n/e
Transaction authentication	22K	n/e
Reserved	23K – 24K	n/e
Carrier assigned unique identification of groupings of transport units	25K	n/e
Shipper assigned unique identification of groupings of transport units	26K	n/e
Supplier Assigned Quotation Number – Number assigned to a quotation by the supplier in response to a request for quote from the customer	27K	n/e
Reserved for future assignment	28K – 999K	n/e
CATEGORY 12: LOCATION REFERENCE		
Storage Location	L	n/e
Location	1L	n/e
"Ship To:" Location code defined by an industry standard or mutually defined	2L	410
"Ship From:" Location code defined by an industry standard or mutually defined	3L	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
GLN Extension component	n/e	254
"Bill To" (Invoice To) - GS1 Global Location Number	n/e	411
"Purchased From" - GS1 Global Location Number	n/e	412
Country of Origin, two-character ISO 3166 country code	4L	422
"Ship For:" Location code defined by an industry standard or mutually defined	5L	413
Route Code assigned by the supplier to designate a specific transportation path	6L	403
6-digit Department of Defense Activity Code (DoDAAC)	7L	n/e
Port of Embarkation – Mutually defined	8L	n/e
Port of Debarkation – Mutually defined	9L	n/e
Country of Initial Processing	n/e	423
Country of Processing	n/e	424
Country of Disassembly	n/e	425
Country covering full process chain	n/e	426
Reserved	10L	n/e
Location (Latitude/Longitude/Altitude)	11L	n/e
"Ship To:" Location (Latitude/Longitude/Altitude)	12L	n/e
"Ship From:" Location (Latitude/Longitude/Altitude)	13L	n/e
Reserved	14L	n/e
"Ship For:" Location (Latitude/Longitude/Altitude)	15L	n/e
Tag activation location	16L	n/e
Tag deactivation location	17L	n/e
Catch area	18L	7005
Reserved	19L– 19L	n/e
<i>The following DIs can be used to provide for Location identification, which is different than or in addition to Location Reference provided by "L".</i>		
First Level (internally assigned)	20L	n/e
Second Level (internally assigned)	21L	n/e
Third Level (internally assigned)	22L	n/e
Fourth Level (internally assigned)	23L	n/e
Fifth Level (internally assigned)	24L	n/e
Identification of a party to a transaction assigned in accordance with ISO/IEC 15459 (IAC, CIN), followed by an internal physical location of and assigned by the party identified in 18V, e.g., 25L IAC CIN LOC, where the IAC is the issuing agency code assigned by the ISO 15459-2 Registration Authority, the CIN is the company identification code assigned by the IAC, and the LOC is the physical internal location assigned by the CIN.	25L	414
Location code to a related transaction	26L	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Event location UN/LOCODE	27L	n/e
Number and Street Address	28L	n/e
City Name	29L	n/e
Country Sub-entity Details	30L	n/e
Postal Code	31L	n/e
Country Code - ISO 3166-1 Alpha 2 Code	32L	n/e
Uniform Resource Locator (URL). Includes all characters that form a URL, including header data such as e.g., http://. Character set as listed in RFC 1738.	33L	8200
Pointer to Process URL (P2P URL)	34L	n/e
Vessel ID / Farm ID	35L	7030
Producer ID / Farm ID	36L	7031
Reserved	37L– 50L	n/e
<i>The following two Data Identifiers are to be used for shipments within the jurisdiction of a single postal authority.</i>		
"Ship From:" - Location code defined by a postal authority (e.g., 5-digit and 9-digit ZIP codes identifying U.S. locations or 6-character postal codes identifying Canadian locations)	51L	n/e
"Ship To:" - Location code defined by a postal authority (e.g., 5-digit and 9-digit ZIP codes identifying U.S. locations or 6-character postal codes identifying Canadian locations)	52L	420
Reserved	53L	n/e
<i>The following two Data Identifiers are to be used for shipments between locations governed by different postal authorities</i>		
"Ship From:" - Location code defined by a postal authority in the format: postal codes (e.g., 5-digit ZIP codes identifying U.S. locations or 6- or 7-character postal codes identifying United Kingdom locations) followed by two character ISO 3166 country code (e.g., US or GB)	54L	n/e
"Ship To:" - Location code defined by a postal authority in the format: postal codes (e.g., 5-digit ZIP codes identifying U.S. locations or 6- or 7-character postal codes identifying United Kingdom locations) followed by two character ISO 3166 country code (e.g., US or GB)	55L	n/e
Ship To (Deliver To) Postal Code With 3-digit ISO Country Code Prefix	n/e	421
Reserved for future assignment	56L - 999L	n/e
CATEGORY 13: MAINTENANCE CODES		
Reserved	M	n/e
Reserved	1M – 9M	n/e
Army form 2410 data. Format is data value preceded by the block number of the form 2410. Field lengths and acceptable characters can be found at http://www.apd.army.mil/pdf/p738_751.pdf	10M	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Army form 2408 data. Format is data value preceded by the block number of the form 2408. field lengths and acceptable characters can be found at http://www.apd.army.mil/pdffiles/p738_751.pdf	11M	n/e
Army form 2407 data. Format is data value preceded by the block number of the form 2407. field lengths and acceptable characters can be found at http://www.apd.army.mil/pdffiles/p738_751.pdf	12M	n/e
Air Force Form 95 data. Format is data value preceded by the block number of the form 95. Field lengths and acceptable characters can be found at http://www.gsa.gov/portal/forms/download/116418	13M	n/e
Navy Form 4790 data. Format is data value preceded by the block number of the form (OPNAV 4790/60) 2410. Field lengths and acceptable character can be found at http://www.navair.navy.mil/logistics/4790/library/Chapter%2015.pdf	14M	n/e
Reserved for future assignment	15M – 999M	n/e
CATEGORY 14: INDUSTRY ASSIGNED CODES		
National/NATO Stock Number (NSN)	N	7001
Product Characteristic Data defined by the Chemical Industry Data Exchange (CIDX)	1N	n/e
Reserved	2N	n/e
Coding Structure in Accordance with Format Defined by Electronic Industries Association Japan (EIAJ)	3N	n/e
Coding Structure and Formats in Accordance with GS1 Application Identifiers (AI plus data) (GS1)	4N	n/e
Coding Structure and Formats in Accordance with AIAG Recommendations. The full code list can be found at list http://www.mhi/standards - see under “MH10 Data Identifiers (Continuous Maintenance Version)”	5N	n/e
U.S. DOD Requisition and Issue Procedure Codes. The format is the MILSTRIP code the appropriate followed by the data value associated with that code. (The full list of codes is available at http://www2.dla.mil/j-6/dlmsolibrary/Manuals/DLM/MILSTRIP/MILSTRIP.pdf	6N	n/e
U.S. Defense Transportation Regulation codes. The format is the DTR code followed by the appropriate data value associated with that code. (The full list of codes is available at http://www.transcom.mil/dtr/part-ii/dtr_part_ii_toc.pdf)	7N	n/e
Production Animal Identification Codes. The format is the production animal code followed by the appropriate data value associated with that code. The Technical Report and the full list of Extended Data Elements (codes) is maintained at; http://www.aimglobal.org/store/view_product.asp?id=4926441 Extended Data Elements (Codes) http://www.aimglobal.org/store/view_product.asp?id=4926483 Technical Report	8N	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Pharmacy Product Number maintained by IFA (www.ifaffm.de) and structured as follows: Two-digit product registration agency code followed by the registered product number (assigned by product registration agencies) and two PPN check digits.	9N	n/e
Reserved for future assignment	10N – 999N	n/e
UN/ECE Meat Carcasses and Cuts Classification	n/e	7002
Approval number of processor with ISO country code	n/e	703(s)
CATEGORY 15: RESERVED		
Not recommended for use due to similarity of "0" (zero) to "O"	O - 999O	n/e
CATEGORY 16: ITEM INFORMATION		
Item Identification Code assigned by Customer	P	241
Item Identification Code assigned by Supplier	1P	01
Code assigned to specify the revision level for an Item (e.g., engineering change level, edition, or revision)	2P	n/e
Combined manufacturer identification code/item code under the 12/13-digit GS1 formats, plus supplemental codes, if any	3P	GS1 GTIN
Roll products - Width, Length, Core Diameter, Direction, & Splices	n/e	8001
Item Code portion of GS1 formats	4P	n/e
Freight Classification Item Number assigned by Carrier for purposes of rating hazardous materials (e.g., Motor Freight, Air, Boat, Rail Classification)	5P	n/e
Combined supplier identification and item code (internally assigned or mutually defined)	6P	n/e
Common Language Equipment Identification (CLEI) assigned by the manufacturer to some telecommunications equipment	7P	n/e
14-digit GS1 format for SCC-14 code structure	8P	01
Combined manufacturer identification code (9-digit DUNS number assigned by Dun & Bradstreet) and the item code/part number (assigned by the manufacturer).	9P	n/e
Hazardous Material Code as defined by ANSI X12.3 in the format Data Element 208 (1-character code qualifier) followed by Data Element 209 (Hazardous Material Code)	10P	n/e
10-character CLEI Code for telecommunications equipment	11P	n/e
Document Type (e.g., Pick List, Design Drawing, etc.) (internally assigned or mutually defined)	12P	n/e
VMRS System Code	13P	n/e
VMRS System and Assembly Code	14P	n/e
VMRS System, Assembly, & Part Code	15P	n/e
VMRS System, Assembly, or Part Code (User Modified	16P	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Combined GS1 supplier identification and item code assigned by the supplier	17P	01
Combined VMRS supplier ID and supplier assigned part number	18P	n/e
Component of an Item (One product contained in multiple packages)	19P	8006
Product Variant	n/e	20
HIBCC - Quantity, Date, Batch, and Link	n/e	22
Made-to-Order Variation Number	n/e	242
<i>The following five DIs can be used to provide for Item identification (Item ID), which is different than or in addition to Item ID provided by "P".</i>		
First Level (Customer Assigned)	20P	n/e
Second Level (Customer Assigned)	21P	n/e
Third Level (Customer Assigned)	22P	n/e
Fourth Level (Customer Assigned)	23P	n/e
Fifth Level (Customer Assigned)	24P	n/e
Identification of a party to a transaction as identified in 18V, followed by the supplier assigned part number.	25P	n/e
Part Number of next higher assembly	26P	n/e
Commodity HTS-6 Code	27P	n/e
Cargo name	28P	n/e
Product classification code as defined with the GMDN (Global Medical Device Nomenclature - http://www.gmdnagency.org)	29P	n/e
<i>The following five DIs can be used to provide for Item identification (Item ID), which is different than or in addition to Item ID provided by "1P".</i>		
First Level (Supplier Assigned)	30P	240
Second Level (Supplier Assigned)	31P	n/e
Third Level (Supplier Assigned)	32P	n/e
Fourth Level (Supplier Assigned)	33P	n/e
Fifth Level (Supplier Assigned)	34P	n/e
Reserved	35P – 39P	n/e
A code assigned by a customer to the identification number of the manufacturer's Material Safety Data Sheet (MSDS) document that describes the uses, hazards, and chemical composition of a hazardous material.	40P	n/e
Reserved	41P – 48P	n/e
Export controlled item. Subject to export control and or restrictions as identified in the Wassenaar Arrangement. DI followed by the Alpha-2 ISO 3166 Country Code of the country that imposed the restriction followed by Wassenaar Code (http://www.wassenaar.org/controllists/index.html)	49P	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Manufacturer-assigned item identifier - Manufacturer-assigned item identifier comprising an item number assigned by the item manufacturer, followed by a plus (+) sign, followed - if required to uniquely identify the item within the manufacturer's product range - by a manufacturer-assigned item version.	50P	n/e
Globally unique item identifier comprising the Identification of a party to a transaction as identified in 18V, followed by a plus (+) sign, followed by the Manufacturer-assigned item identifier as defined with 50P	51P	n/e
Reserved for future assignment	52P - 999P	n/e
CATEGORY 17: MEASUREMENT		
Quantity, Number of Pieces, or Amount (numeric only) (unit of measure and significance mutually defined)	Q	30
Theoretical Length/Weight (numeric only)	1Q	n/e
Actual Weight (numeric only)	2Q	n/e
Unit of Measure, as defined by the two character ANSI X12.3 Data Element Number 355 Unit of Measurement Code	3Q	n/e
Gross Amount	4Q	n/e
Net Amount	5Q	n/e
Reserved	6Q	n/e
Quantity, Amount, or Number of Pieces in the format: Quantity followed by the two character ANSI X12.3 Data Element Number 355 Unit of Measurement Code	7Q	↓↓↓↓↓
Net Weight, Kilograms	7Q...58	310
Length or 1st Dimension, Meters	7Q...MR	311 or 331
Width, Diameter, or 2nd Dimension, Meters	7Q...MR	312 or 332
Depth, Height, or Thickness or 3rd Dimension, Meters	7Q...MR	313 or 333
Area, Square Meters	7Q...SM	314 or 334
Volume, Liters	7Q...LT	315 or 335
Volume, Cubic Meters (Net)	7Q...CO	316
Volume, Cubic Meters (Gross)	7Q...CR	336
Net Weight, Pounds	7Q...PN	320
Length or 1st Dimension, Inches	7Q...ED	321 or 341
Length or 1st Dimension, Feet	7Q...EZ	322 or 342
Length or 1st Dimension, Yards	7Q...YD	323
Length or 1st Dimension, Yards (Gross)	7Q...GY	343
Width, Diameter, or 2nd Dimension, Inches	7Q...ED	324 or 344
Width, Diameter, or 2nd Dimension, Feet	7Q...EZ	325 or 345
Width, Diameter, or 2nd Dimension, Yards	7Q...YD	326
Width, Diameter, or 2nd Dimension, Yards (Gross)	7Q...GY	346

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Depth, Thickness, Height or 3rd Dimension, Inches	7Q...ED	327 or 347
Depth, Thickness, Height or 3rd Dimension, Feet	7Q...EZ	328 or 348
Depth, Thickness, Height or 3rd Dimension, Yards	7Q...YD	329
Depth, Thickness, Height or 3rd Dimension, Yards	7Q...GY	349
Gross Weight, Kilograms	7Q...GT	330
Kilograms per Square Meter	7Q...KM	337
Gross Weight, Pounds	7Q...PG	340
Area, Square Inches	7Q...SI	350 or 353
Area, Square Feet	7Q...SF	351 or 354
Area, Square Yards	7Q...SY	352 or 355
Net Weight, Troy Ounces	7Q...TO	356
Net Weight, Ounces	7Q...OZ	357
Volume, Quarts	7Q...QT	360 or 362
Volume, Gallons	7Q...GA	361
Volume, Gallons (Gross)	7Q...GN	363
Volume, Cubic Inches	7Q...CI	364 or 367
Volume, Cubic Feet	7Q...CF	365 or 368
Volume, Cubic Yards	7Q...CY	366 or 369
Container rated weight. Unit of measure is kg	8Q	n/e
Piece Weight: weight of a single item	9Q	n/e
Reserved	10Q	n/e
Tare Weight: weight of an empty container Container body weight. Unit of measure is kg. (Tare weight)	11Q	n/e
Monetary Value/ Entry Value established by the Supplier in the format of: the value followed by an ISO 4217 data element code for representing unit of value of currencies and funds (e.g., 12Q2.50USD) (2.50 Monetary Value in USA Dollars) significance mutually defined	12Q	n/e
# of # ("this is the <i>n</i> th piece of <i>x</i> pieces in this shipment") Presented in the format " <i>n/x</i> ", where the "/" (slash) is used as a delimiter between two values. See Annex C.6.3 for further information	13Q	n/e
Beginning Secondary Quantity	14Q	n/e
Ending Secondary Quantity	15Q	n/e
Number of pieces in Van	16Q	n/e
Number of shipments in van	17Q	n/e
Cube expressed in cubic meters or cubic feet as indicated by the ANSI X12.3 data element number 355 unit of measure code (CR or CF). No implied decimal point.	18Q	n/e
Width expressed in linear meters or linear feet as indicated by the ANSI X12.3 data element number 355 unit of measure (LC or LF). No implied decimal point.	19Q	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Height expressed in linear meters or linear feet as indicated by the ANSI X12.3 data element number 355 unit of measure (LC or LF). No implied decimal point.	20Q	n/e
Length expressed in linear meters or linear feet as indicated by the ANSI X12.3 data element number 355 unit of measure (LC or LF). No implied decimal point.	21Q	n/e
Net weight of shipment expressed in pounds or kilograms (kilos) as indicated by the ANSI X12.3 data element number 355 unit of measure (LB or KG). No implied decimal point.	22Q	n/e
Van length expressed in linear meters or linear feet as indicated by the ANSI X12.3 data element number 355 unit of measure (LC or LF). No implied decimal point.	23Q	n/e
Inside cube of a van expressed in cubic meters or cubic feet as indicated by the ANSI X12.3 data element number 355 unit of measure code (CR or CF). No implied decimal point.	24Q	n/e
Net explosive weight (a computed value of explosive equivalent expressed in pounds of TNT). The measure of NEW, is used internationally for explosive safety quantity distance arc computations. No implied decimal point.	25Q	n/e
Packaging Level, specifying the hierarchical level of packaging in accordance with HIBC (Health Industry Bar Code) specifications.	26Q	n/e
Single Product Price Value, Net, “.” (dot) used as decimal point (e.g., 27Q1000.5 for the price value of 1000.50).	27Q	n/e
Single Price Charge Value For Postage And Packaging, “.” (dot) represents the position of a comma (e.g., 30Q100.50 for the value of 100,50).	28Q	n/e
Discount Percentage, “.” (dot) represents the position of a comma (e.g., 31Q8.5 for a discount value of 8,5%).	29Q	n/e
VAT Percentage, “.” (dot) represents the position of a comma (e.g., 27Q8.5 for the VAT value of 8,5%).	30Q	n/e
Currency, ISO 4217 currency code.	31Q	n/e
Reserved for future assignment	32Q – 999Q	n/e
CATEGORY 18: MISCELLANEOUS		
Reserved	R	n/e
Return Authorization Code (RMA) assigned by the Supplier	1R	n/e
Return Code assigned by the Customer	2R	n/e
Reserved	3R	n/e
U.S. Department of Defense Identification Code (DoDIC)	4R	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Data in the format and using semantics defined by the holder of a Company Identification Number (CIN) that has been issued by an Issuing Agency Code (IAC) in accordance with ISO/IEC 15459, defined as a sequence of concatenated data elements: IAC, followed by CIN, followed by the separator character ":" (colon) followed by the data in the format and using semantics as defined by the CIN holder	5R	n/e
ISO/IEC 20248 digital signature data construct. If the underlying data carrier encoding is 7 bits, then only the ISO/IEC 20248 raw format may be used.	6R	n/e
Aquatic species	7R	7008
Fishing gear type	8R	7009
Production method	9R	7010
Reserved for future assignment	10R- 999R	n/e
CATEGORY 19: TRACEABILITY NUMBER FOR AN ENTITY		
Serial number or code assigned by the Supplier to an entity for its lifetime, (e.g., computer serial number, traceability number, contract tool identification)	S	21
Additional code assigned by the Supplier to an entity for its lifetime (e.g., traceability number, computer serial number)	1S	n/e
Advance Shipment Notification (ASN) Shipment ID (SID) corresponds to ANSI ASC X12 Data Element 396	2S	n/e
Unique Package Identification assigned by Supplier (lowest level of packaging which has a package ID code; shall contain like items)	3S	n/e
Package Identification assigned by Supplier to master packaging containing like items on a single customer order (See Annex C.7)	4S	n/e
Package Identification assigned by Supplier to master packaging containing unlike items on a single customer order (See Annex C.7)	5S	n/e
Package Identification assigned by Supplier to master packaging containing like items over multiple customer orders (See Annex C.7)	6S	n/e
Package Identification assigned by Supplier to master packaging containing unlike items over multiple customer orders (See Annex C.7)	7S	n/e
Supplier ID/Unique Container ID presented in the data format specified by the GS1 SSCC-18	8S	00
Package Identification, Generic (mutually defined)	9S	n/e
Machine, cell, or tool ID code	10S	n/e
Fixed asset ID code	11S	n/e
Document Number (internally assigned or mutually defined)	12S	n/e
Container Security Seal	13S	n/e
4th Class Non-identical parcel post manifesting	14S	n/e
Serial Number Assigned by the Vendor Entity, that can only be used in conjunction with "13V"	15S	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Version Number, e.g., Software Version	16S	n/e
Combined 6-digit GS1 supplier identification and unique package identification assigned by the supplier	17S	n/e
Reserved (CAGE Code & Serial Number unique within CAGE)	18S	n/e
Combined Dun & Bradstreet company identification of the supplier followed by a unique package identification assigned by the supplier,	19S	n/e
Traceability code for an entity assigned by the customer	20S	n/e
Tire Identification Number as defined by the U.S. Department of Transportation (D.O.T) under U.S. Code 49 CFR 574.	21S	n/e
Electronic Serial Number for Cellular Mobile Telephones	22S	8002
Media Access Control (MAC) Address conforming with IEEE 802.11	23S	n/e
Reserved	24S	n/e
Identification of a party to a transaction as identified in 18V, followed by the supplier assigned serial number.	25S	n/e
Reader ID Equipment identifier, being a globally unique identifier for a device	26S	n/e
Reserved	27S - 29S	n/e
Global Identifier Serialized for Trade (GIST)	n/e	252
Additional traceability code for an entity assigned by the supplier in addition to or different from the traceability code(s) provided by "S" or "1S"	30S	250
Beginning Serial Number for serial numbers in sequence	31S	n/e
Ending Serial Number for serial numbers in sequence	32S	n/e
Serial number of Next higher assembly	33S	n/e
Serial number or Part number of End Item	34S	n/e
Bumper Number (Used in Unit DOD Move)	35S	n/e
Pallet Identifier (Used for loaded 463L air pallets)	36S	n/e
Unique Item Identifier (format IAC CIN PN + PSN)	37S	n/e
Reserved	38S – 41S	n/e
Unique item identifier (UII) in 25S format preceded by numeric value indicating serial number element length for use by systems that require the "serial number" component of a concatenated Serial Number element (IAC+CIN+SN)	42S	n/e
Integrated Circuit Card Identifier (ICCID) in accordance with ITU-T Recommendation E.118 and ETSI Recommendation GSM 11.11	43S	n/e
Reserved	44S – 49S	n/e
<i>The following five DIs can be used to provide for identification of entities within a single unit that is different than or in addition to identification provided by "S".</i>		
First Level (Supplier Assigned)	50S	n/e
Second Level (Supplier Assigned)	51S	n/e
Third Level (Supplier Assigned)	52S	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Fourth Level (Supplier Assigned)	53S	n/e
Fifth Level (Supplier Assigned)	54S	n/e
Reserved	55S - 95S	n/e
EPC number (Typically Serialized Global Trade Identification Number - SGTIN)	96S	n/e
Encrypted serial number	97S	n/e
Reserved for future assignment	98S – 999S	n/e
CATEGORY 20: TRACEABILITY NUMBER FOR GROUPS OF ENTITIES		
Traceability Number assigned by the Customer to identify/trace a unique group of entities (e.g., lot , batch , heat)	T	n/e
Traceability Number assigned by the Supplier to identify/trace a unique group of entities (e.g., lot , batch , heat)	1T	10
Reserved	2T	n/e
Exclusive Assignment (U.S. EPA vehicle identification for emissions testing)	3T	n/e
Reserved	4T - 19T	n/e
<i>The following five DIs can be used to provide for identification of a group of entities, which is different than or in addition to identification provided by "T".</i>		
First Level (Customer Assigned)	20T	n/e
Second Level (Customer Assigned)	21T	n/e
Third Level (Customer Assigned)	22T	n/e
Fourth Level (Customer Assigned)	23T	n/e
Fifth Level (Customer Assigned)	24T	n/e
Supplier assigned traceability number (format CIN IAC Trace#)	25T	n/e
Batch identifier comprising the concatenation of either: — a data identifier 26S mail processing equipment identifier, or — a data identifier 20K license identifier, or — a data identifier 18V party identifier that: — is distinct from any other ISO/IEC 15459 compliant identifier within the domain of the issuing agency concerned ⁶ ; — cannot be derived from another party identifier or any other ISO/IEC 15459 compliant identifier, issued under the same issuing agency, by the simple addition of characters to, or their removal from, its end ⁶ ; with a data identifier 27T batch number, the two being separated by a dash (-) character	26T	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Batch number, issued under the control of an identified party or unit of processing equipment, or under the provisions of an identified license, that: — uniquely distinguishes one batch of related items from all other batches to which a batch number is assigned by the party or equipment, or under the license, concerned; comprises a string of maximum length 10 characters, of which the first (numeric) character indicates the number of following characters, each of which is taken from the set {0-9; A-Z}	27T	n/e
Reserved	28T - 29T	n/e
<i>The following five DIs can be used to provide for identification of a group of entities, which is different than or in addition to identification provided by "1T".</i>		
First Level (Supplier Assigned)	30T	n/e
Second Level (Supplier Assigned)	31T	n/e
Third Level (Supplier Assigned)	32T	n/e
Fourth Level (Supplier Assigned)	33T	n/e
Fifth Level (Supplier Assigned)	34T	n/e
Reserved for future assignment	35T - 999T	n/e
CATEGORY 21: UPU/MH 10/SC8 AGREED UPON CODES		
Reserved	U-4U	n/e
Specification of a postal service and associated process data in accordance with UPU standard S25 data construct "Service Data"	5U	n/e
Licensing post data, in accordance with the specification in UPU standard S25.	6U	n/e
Reserved for Assignment for UPU needs in collaboration with ASC MH 10/SC 8	7U – 14U	n/e
Specification of supplementary postal service and associated process data in accordance with UPU standard S25 data construct	15U	n/e
Postal administration identifications, being the identification, expressed in accordance with the specification in UPU standard S25, of one or more postal administrations involved in the processing of a mail item or batch.	16U	n/e
UPU location code	17U	n/e
Qualified UPU location code, concatenation of: — a location category drawn from UPU code list 139; — a data identifier 17U UPU location code	18U	n/e
License plate with service data and location code is a compound data construct, compliant with the specification in UPU standard S25	19U	n/e
Reserved for Assignment for UPU needs in collaboration with ASC MH 10/SC 8	20U – 54U	n/e
OCR Data Locator	55U	n/e
Reserved for future assignment	56U – 999U	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
CATEGORY 22: PARTY TO THE TRANSACTION		
Supplier Code assigned by Customer	V	n/e
Supplier Code assigned by Supplier	1V	n/e
U.P.C. Company Prefix	2V	n/e
GS1 Company Prefix	3V	n/e
Carrier Identification Code assigned by an industry standard mutually defined by the Supplier, Carrier, and Customer	4V	n/e
Financial Institution Identification Code (mutually defined)	5V	n/e
Manufacturer's identification code (mutually defined)	6V	n/e
Code assigned to a party which has financial liability for an entity or group of entities (e.g., owner of inventory) (mutually defined)	7V	n/e
Customer code assigned by the customer	8V	n/e
Customer code assigned by the supplier	9V	n/e
Manufacturer ID (See Appendix 2, CBP 7501 Instructions)	10V	n/e
Organization with budget responsibility for an entity, process, or procedure (e.g., shop, division, department)(internally assigned)	11V	n/e
DUNS number identifying manufacturer	12V	n/e
DUNS number identifying supplier	13V	n/e
DUNS number identifying customer	14V	n/e
Carrier-assigned shipper number	15V	n/e
VMRS Supplier ID	16V	n/e
U.S. DoD CAGE Code	17V	n/e
Identification of a party to a transaction in which the data format consists of two concatenated segments. The first segment is the Issuing Agency Code (IAC) in accordance with ISO/IEC 15459, the second segment is a unique entity identification Company Identification Number (CIN) assigned in accordance with rules established by the issuing agency	18V	n/e
Specification of a party's role(s), in a transaction, consisting of one or more code values from EDIFACT Code List 3035 "Party Qualifier", separated by plus (+) characters (Never to be concatenated with other DIs in a linear symbol or other media where the concatenation character is a plus (+) character)	19V	n/e
Identification of a party to a transaction assigned by a holder of a Company Identification Number (CIN) and including the related Issuing Agency Code (IAC) in accordance with ISO/IEC 15459 and its registry, structured as a sequence of 3 concatenated data elements: IAC, followed by CIN, followed by a plus (+) character followed by one or more code values from EDIFACT Code List 3035 "Party Qualifier", separated by plus (+) characters (Never to be concatenated with other DIs in a linear symbol or other media where the concatenation character is a plus (+) character)	20V	n/e

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Identification of a party to a transaction as identified in 18V, followed by the organizational sub-unit of and assigned by the party identified in 18V, e.g., 21V IAC CIN OSU, where the IAC is the issuing agency code assigned by the ISO 15459-2 Registration Authority, the CIN is the company identification code assigned by the IAC, and the OSU is the organizational sub-unit identification assigned by the CIN.	21V	n/e
Carrier SCAC (Standard Carrier Alpha Code)	22V	n/e
SPLR VAT NO	23V	n/e
CUST VAT NO	24V	n/e
Reserved for future assignment	25V– 999V	251
Reference to Source Entity	n/e	251
CATEGORY 23: ACTIVITY REFERENCE		
Work Order Number (e.g., "Production Paper") (internally assigned)	W	n/e
Operation Sequence Number	1W	n/e
Operation Code/Work Code - the type of work to be performed (internally assigned or mutually defined)	2W	n/e
Combined Work Order Number and Operation Sequence Number in the format nn...n+nn...n where a plus symbol (+) is used as a delimiter between the Work Order Number and the Operation Sequence Number	3W	n/e
Status Code (internally assigned or mutually defined)	4W	n/e
Work Unit Code – identifies system, subsystem, assembly, component etc. on which maintenance is performed	5W	n/e
Nomenclature – (internally assigned or mutually defined)	6W	n/e
Reserved	7W – 9W	n/e
Form Control Number – Preprinted control number on forms	10W	n/e
Quality Assurance Inspector – Last Name	11W	n/e
Telephone number of person completing the form	12W	n/e
Reserved for future assignment	13W – 999W	n/e
CATEGORY 24: RESERVED		
Reserved	X - 999X	n/e
CATEGORY 25: INTERNAL APPLICATIONS		
Never to appear on item/document which leaves a closed system environment	Y - 999Y	91-99
CATEGORY 26: MUTUALLY DEFINED		

CATEGORY/DESCRIPTION	ANS MH10.8.2 DI	GS1 AI
Mutually Defined between Customer and Supplier	Z	90
Mutually Defined between Carrier and Supplier	1Z	90
Mutually Defined between Customer and Carrier	2Z	90
Free Text	3Z	n/e
Mutually Defined between Carrier and Trading Partner	4Z	90
Reserved	5Z - 9Z	n/e
Structured Free Text (Header Data)	10Z	n/e
Structured Free Text (Line 1-89 Data)	11Z - 99Z	n/e
Reserved for future assignment	100Z - 999Z	n/e

SECTION IV

MAPPING

GS1 AIs to ANS MH10.8.2 DIs

MAPPING GS1 AIs to ANS MH10.8.2 DIs

n/e = no-equivalent Data Identifier for Application Identifier or no equivalent Application Identifier for Data Identifier.

AI	Data Content	DI
00	Serial Shipping Container Code (SSCC)	J, 1J, 2J, 3J, 4J, 8S
01	Global Trade Item Number (GTIN) (f.k.a. SCC-14)	8P
02	GTIN of trade items contained in a logistic unit (Must be used with AI 37)	n/e
10	Batch or Lot Number	1T
11 (*)	Production Date (YYMMDD)	5D...405
12 (*)	Due Date (YYMMDD)	5D...013
13 (*)	Packaging Date (YYMMDD)	n/e
15 (*)	Best Before Date (YYMMDD) (f.k.a. Best Before / Quality)	25D
16 (**)	Sell By Date (YYMMDD)	
17 (*)	Expiration Date (YYMMDD) (f.k.a. Use By / Safety)	5D...036
20	Variant Number	n/e
21	Serial Number	S
240	Additional Item Identification	30P
241	Customer Part Number	P
242	Made-to-Order Variation Number	n/e
243	Packaging Component Number	n/e
250	Secondary Serial Number	30S
251	Reference to Source Entity	n/e
253	Global Document Type Identifier	n/e
254	GLN Extension component	n/e
255	Global Coupon Number	n/e
30	Count of Items (Variable Measure Trade Item)	Q
310 (**)	Net Weight, Kilograms	7Q...58
311 (**)	Length or 1st Dimension Trade, Meters	7Q...MR
312 (**)	Width, Diameter, or 2nd Dimension, Trade, Meters	7Q...MR
313 (**)	Depth, Thickness, Height or 3rd Dimension, Trade, Meters	7Q...MR
314 (**)	Area, Trade, Square Meters	7Q...SM
315 (**)	Net Volume, Liters	7Q...LT
316 (**)	Net Volume, Cubic Meters	7Q...CR
320 (**)	Net Weight, Pounds	7Q...PN
321 (**)	Length or 1st Dimension, Trade, Inches	7Q...ED
322 (**)	Length or 1st Dimension, Trade, Feet	7Q...EZ
323 (**)	Length or 1st Dimension, Trade, Yards	7Q...YD
324 (**)	Width, Diameter, or 2nd Dimension, Trade, Inches	7Q...ED

AI	Data Content	DI
325 (**)	Width, Diameter, or 2nd Dimension, Trade, Feet	7Q...EZ
326 (**)	Width, Diameter, or 2nd Dimension, Trade, Yards	7Q...YD
327 (**)	Depth, Thickness, Height or 3rd Dimension, Trade, Inches	7Q...ED
328 (**)	Depth, Thickness, Height or 3rd Dimension, Trade, Feet	7Q...EZ
329 (**)	Depth, Thickness, Height or 3rd Dimension, Trade, Yards	7Q...YD
330 (**)	Logistic Weight, Kilograms	7Q...GT
331 (**)	Length or 1st Dimension, Meters Logistics	7Q...MR
332 (**)	Width, Diameter, or 2nd Dimension, Meters Logistics	7Q...MR
333 (**)	Depth, Thickness, Height or 3rd Dimension, Meters, Logistics	7Q...MR
334 (**)	Area, Square Meters Logistics	7Q...SM
335 (**)	Logistic Volume, Liters	7Q...LT
336 (**)	Logistic Volume, Cubic Meters	7Q...CO
337 (**)	Kilograms per Square Meter	7Q...KM
340 (**)	Logistic Weight, Pounds	7Q...PG
341 (**)	Length or 1st Dimension, Inches Logistics	7Q...ED
342 (**)	Length or 1st Dimension, Feet Logistics	7Q...EZ
343 (**)	Length or 1st Dimension, Yards Logistics	7Q...GY
344 (**)	Width, Diameter, or 2nd Dimension, Inches Logistics	7Q...ED
345 (**)	Width, Diameter, or 2nd Dimension, Feet Logistics	7Q...EZ
346 (**)	Width, Diameter, or 2nd Dimension, Yards Logistics	7Q...GY
347 (**)	Depth, Thickness, Height or 3rd Dimension, Inches, Logistics	7Q...ED
348 (**)	Depth, Thickness, Height or 3rd Dimension, Feet, Logistics	7Q...EZ
349 (**)	Depth, Thickness, Height or 3rd Dimension, Yards, Logistics	7Q...GY
350 (**)	Area, Trade, Square Inches	7Q...SI
351 (**)	Area, Trade, Square Feet	7Q...SF
352 (**)	Area, Trade, Square Yards	7Q...SY
353 (**)	Area, Square Inches, Logistics	7Q...SI
354 (**)	Area, Square Feet, Logistics	7Q...SF
355 (**)	Area, Square Yards, Logistics	7Q...SY
356 (**)	Net Weight, Trade, Troy Ounces	7Q...TO
357 (**)	Net Volume, Trade, Ounces (U.S.)	7Q...OZ
360 (**)	Net Volume, Trade, Quarts	7Q...QT
361 (**)	Net Volume, Trade, Gallons (U.S.)	7Q...GA
362 (**)	Logistic Volume, Quarts	7Q...QT
363 (**)	Logistic Volume, Gallons (U.S.)	7Q...GN
364 (**)	Net Volume, Trade, Cubic Inches	7Q...CI
365 (**)	Net Volume, Trade, Cubic Feet	7Q...CF
366 (**)	Net Volume, Trade, Cubic Yards	7Q...CY
367 (**)	Logistic Volume, Cubic Inches	7Q...CI

AI	Data Content	DI
368 (**)	Logistic Volume, Cubic Feet	7Q...CF
369 (**)	Logistic Volume, Cubic Yards	7Q...CY
37	Count of Trade Items Contained in a Logistics Unit (For Use with AI 02 Only)	n/e
390 (**)	Applicable Amount Payable – local currency	n/e
391 (**)	Applicable Amount Payable – with ISO currency code	n/e
392 (**)	Applicable Amount Payable Variable Measure Trade Item – local currency	n/e
393 (**)	Applicable Amount Payable for a Variable Measure Trade Item – with ISO currency code	n/e
400 (+)	Customer's Purchase Order Number	K
401	Global Identification Number of Consignment	n/e
402	Global Shipment Identification Number	2K
403	Routing Code	n/e
410	"Ship To" (Deliver To) - GS1 Global Location Number	2L/12L
411	"Bill To" (Invoice To) - GS1 Global Location Number	n/e
412	"Purchased From" - GS1 Global Location Number	n/e
413	"Ship For - Deliver For - Forward To" GS1 Global Location Number	5L/15L
414	Identification of a Physical Location, GS1 Global Location Number	n/e
415	GS1 Global Location Number of the Invoicing Party	n/e
420	Ship To (Deliver To) Postal Code Within a Single Postal Authority	52L
421	Ship To (Deliver To) Postal Code With 3-digit ISO Country Code Prefix	55L
422	Country of Origin of a Trade Item	4L
423	Country of Initial Processing	n/e
424	Country of Processing	n/e
425	Country of Disassembly	n/e
426	Country covering full process chain	n/e
427	Country subdivision of origin	n/e
7001	NATO Stock Number (NSN)	N
7002	UN/ECE Meat Carcasses and Cuts Classification	n/e
7003	Expiration Date and Time (YYMMDDHHMM)	n/e
7004	Active Potency	n/e
7005	Catch Area	18L
7006	First Freeze Date	26D
7007	Harvest Date	27D, 28D
7008	Species for Fishery Purposes	7R
7009	Fishing Gear Type	8R
7010	Production Method	9R
703(s)	Approval number of processor with ISO country code	n/e
710	National Healthcare Reimbursement Number (NHRN) – Germany PZN	n/e

AI	Data Content	DI
711	National Healthcare Reimbursement Number (NHRN) – France CIP	n/e
712	National Healthcare Reimbursement Number (NHRN) – Spain CN	n/e
713	National Healthcare Reimbursement Number (NHRN) – Brasil DRN	n/e
nnn	National Healthcare Reimbursement Number (NHRN) – Country “A” NHRN	n/e
7030	Vessel ID / Farm ID	35L
7031	Producer ID / Farm ID	36L
8001	Roll products - Width, Length, Core Diameter, Direction, & Splices	n/e
8002	Cellular Mobile Telephone Identifier	22S
8003	Global Returnable Asset Identifier	25B
8004	Global Individual Asset Identifier	1B, 5B
8005	Price Per Unit of Measure	n/e
8006	Identification of the Component of a Trade Item	19P
8007	International Bank Account Number	n/e
8008	Date and Time of Production	n/e
8010	Component / Part Identifier Serial Number (CPID SERIAL)	n/e
8011	Global Service Relation Number to identify the relationship between an organisation offering services and the provider of services	n/e
8017	Global Service Relation Number to identify the relationship between an organization offering services and the provider of services	n/e
8018	Global Service Relation Number to identify the relationship between an organization offering services and the recipient of services	n/e
8019	Service Relation Instance Number (SRIN)	n/e
8020	Payment Slip Reference Number	n/e
8110	Coupon Code Identification for Use in North America	n/e
8200	Extended Packaging URL	33L
90	Information Mutually Agreed Between Trading Partners	Y, 3K, 6K, 12K, 1Z, 2Z, 4Z
91 to 99	Company Internal Information	

(*) : To indicate only year and month, DD can be filled with "00"

(**) : Plus one digit for decimal point indication

(+) : The definition of 400 has been modified to allow order, release, and line numbers, at the discretion of the issuer

Date Value Representation:

a	alphabetic characters (chars)	n1...3	from one up to 3 numeric chars
n	numeric chars	a1...3	from one up to 3 alphabetic chars
an	alphanumeric chars	an1...3	from one up to 3 alphanumeric chars
n3	3 numeric chars, fixed length	s	sequence in the process
an3	3 alpha-numeric chars, fixed length		

SECTION V

SHORT TITLES

The Short Titles listed herein are for guidance of developing standards. This list is not comprehensive or mandatory.

SHORT TITLES

When printing bar codes (or 2D symbols) it is recommended that each bar code have human readable text printed above the bar code (or adjacent to each 2D symbol) to identify what type of data is contained in the bar code (or 2D symbol). This is called a "short title" and should resemble one of the formats shown in Figure V-1.

Figure V-1: Examples of recommended formats for printing short titles

(S) Serial # 1234567  Bar code contains: S1234567	Serial # (S) 1234567  Bar code contains: S1234567
(13V) DUNS SPLR ID 987654321  Bar code contains: 13V987654321	DUNS SPLR ID (13v) 987654321  Bar code contains: 13V987654321

This Section lists the recommended short titles for some of the most common data identifiers. The Short Titles listed herein are for guidance of developing standards. This list is not comprehensive or mandatory.

SECTION V.A

ANS MH10.8.2 DI SHORT TITLES

(The following list is not a complete list of all identifiers)

DI	SHORT TITLE	Description
B	CONT TYPE	Container type
1B	CONT ID	Returnable container identification code
C	PART # Cont.	Continuation of an Item Code
D	DATE	Date
14D	EXP DATE	Expiration Date (YYYYMMDD)
16D	PROD DATE	Production Date (YYYYMMDD)
25D	BEST BEFORE DATE	Best before date (YYYYMMDD)
26D	FIRST FREEZE DATE	First freeze date (YYYYMMDD)
27D	HARVEST DATE	Harvest date (YYYYMMDD)
J	LIC PLATE	Unique license plate number
1J	LIC PLATE-UNIT	Unique license plate assigned to a transport unit which is the lowest level of packaging, the unbreakable unit.
2J	LIC PLATE-MULTI	Unique license plate assigned to a transport unit which contains multiple packages.
8J	MMSI	Maritime Mobile Service Identity
K	CUST PO #	Order number assigned by Customer
1K	SPLR ORDER #	Order number assigned by Supplier
2K	SPLR SHIP ID	Shipment Identification Code assigned by Supplier/Shipper
3K	BOL/WB	Bill of Landing/Waybill Code assigned by Carrier
4K	CUST LINE	Line number of the order assigned by Customer
5K	CUST REL	Reference number assigned by the Customer to identify a Shipment Authorization (Release) against an established Purchase Order
6K	CARRIER PRO	PRO # Assigned by Carrier
14K	PO = LINE	Combined Order Number and Line Number in the format nn...nn=nn...n where a plus symbol (+) is used as a delimiter between the Order Number and Line Number.
15K	PULL SIG	Pull signal (e.g., KANBAN) Number
16K	DELINS	DELINS Number. Code assigned to identify a document containing delivery information.
1L	LOC	Location
4L	ORIGIN or COO	Country of Origin, two-character ISO 3166 country code
18L	CATCH AREA	FAO fishing area code as defined by the Fisheries and Aquaculture Department of the FAO (http://www.fao.org . Search for Fishing Area Code sub-site)

DI	SHORT TITLE	Description
35L	VESSEL ID / FARM ID	A government-assigned approval number of vessel / aquaculture site / farm / processor, starting with an ISO 3166-1 alpha-2 country code, followed by the approval number.
36L	PRODUCER ID / FARM ID	A government-assigned approval number of producer or farm or first deboning / cutting hall, starting with an ISO 3166-1 alpha-2 country code, followed by the approval number.
51L	FROM POST CODE	"Ship From;" – Location code defined by a postal authority (e.g., 5-digit and 9 digit ZIP codes identifying U.S. locations or 6-character postal codes identifying Canadian locations).
52L	TO POST CODE	"Ship To;" – Location code defined by a postal authority (e.g., 5-digit and 9 digit ZIP codes identifying U.S. locations or 6-character postal codes identifying Canadian locations).
54L	FROM POST CODE + CTRY	" Ship To;" – Location code defined by a postal authority (e.g., 5-digit and 9 digit ZIP codes identifying U.S. locations or 6-character postal codes identifying United Kingdom locations) followed by two character ISO 3166 country code (e.g., US or GB)
55L	TO POST CODE+CTRY	" Ship From;" – Location code defined by a postal authority (e.g., 5-digit and 9 digit ZIP codes identifying U.S. locations or 6-character postal codes identifying United Kingdom locations) followed by two character ISO 3166 country code (e.g., US or GB)
P	CUST PART or CUST ITEM	Item Identification Code assigned by Customer
1P	SPLR PART or SPLR ITEM	Item Identification Code assigned by Supplier`
2P	EC #	Code assigned to specify the revision level for an Item (e.g., engineering change)
10P	HAZMAT CODE	Hazardous Material Code as defined by ANS X12.3 (Version 003000) in the format Data Element 208 (1-character code qualifier) followed by Data Element 209 (Hazardous Material Code)
11P	CLEI	10-character CLEI Code for telecommunications equipment
Q	QTY	Quantity, Number of Pieces or Amount (numeric only) (unit of measure and significance mutually defined)
1Q	THEO LGTH or THEO WT	Theoretical Length/Weight (numeric only)
2Q	ACT WT	Actual Weight (numeric only)
3Q	U/M	Unit of Measure, as defined by the two character ANS X12.3 (Version 003000) Data Element Number 355 Unit of Measurement Code
7Q	QTY + U/M	Quantity, Amount, or Number of Pieces in the format: Quantity followed by the two character ANS X12.3 (Version 003000) Data Element Number 355 Unit of Measurement Code
13Q	N OF X	# of # ("this is the nth piece of x pieces in this shipment"). Presented in the format "n/x:", where the "/"(slash) is used as a delimiter between two values. See Annex C.6.3 for further information.

DI	SHORT TITLE	Description
7R	AQUATIC SPECIES	Aquatic Sciences and Fisheries Information System (ASFIS) 'Inter-agency 3-alpha species code', maintained by the Food and Agriculture Organization of the United Nations (www.fao.org , then search for "ASFIS")
8R	FISHING GEAR TYPE	Food and Agricultural Organization (FAO) International Standard Classification of Fishing Gears (ISSCFG) code. (www.fao.org)
9R	PRODUCTION METHOD	Production method for fish and seafood as specified by the Fisheries and Aquaculture Department of the Food and Agricultural Organization (FAO) of the United Nations, according to EU Regulation 1379/2013. (www.fao.org)
S	SERIAL	Serial number or code assigned by the Supplier to an entity for its lifetime, (e.g.,) computer serial number, tractability number, contract tool identification)
2S	ASN ID	Advance Shipment Notification (ASN) Shipment ID (SOID) corresponds to ANS ASC X12 Data Element 396
3S	PKG ID	Unique Package Identification assigned by Supplier (lowest level of packaging which has a package ID code shall contain like items)
4S	PKG ID-MASTER-LIKE	Package Identification assigned by Supplier to master packaging containing like items on a single customer order
5S	PKG ID-MASTER MIXED	Package Identification assigned by Supplier to master packaging contain unlike items on a single customer order
6S	PKG ID-MASTER-LIKE MULTI	Package Identification assigned by Supplier to master packaging containing like items on over multiple customer orders
7S	PKG ID-MASTER MIXED MULTI	Package Identification assigned by supplier to master packaging containing unlike items on over multiple customer orders
T	CUST LOT or CUST BATCH or CUST HEAT	Tractability Number assigned by the Customer to identity/trace a unique group of entities (e.g., lot, batch, heat)
1T	SPLR LOT or SPLR BATCH or SPLR HEAT	Traceability Number assigned by the Supplier to identify/trace a unique group of entities (e.g., lot, batch, heat)
V	CUST ASG SPLR ID	Supplier Code assigned by Customer
1V	SPLR ASG SPLR ID	Supplier Code assigned by Supplier
12V	DUNS MFR ID	DUNS number identifying manufacturer
13V	DUNS SPLR ID	DUNS number identifying supplier
14V	DUNS CUST ID	DUNS number identifying customer
15V	SHIPPER	Carrier assigned shipper number

DI	SHORT TITLE	Description
23V	SPLR VAT NO	Government-assigned Value Added Tax identification number identifying supplier, starting with an ISO 3166-1 alpha-2 country code (except for Greece, which uses the ISO 639-1 language code EL), followed by the government-assigned VAT number.
24V	CUST VAT NO	Government-assigned Value Added Tax identification number identifying customer, starting with an ISO 3166-1 alpha-2 country code (except for Greece, which uses the ISO 639-1 language code EL), followed by the government-assigned VAT number.

SECTION VI

HIERARCHICAL LEVELS - Data Identifier “F”

When the Data Identifier “F” is used in Data Identifier looping structures the format shall follow the format defined in this Section. See Annex L for usage rules of Data Identifiers 1F, 3F, and 5F for Returnable Packaging Items.

As the application of automatic data capture (ADC) storage media became more sophisticated it became possible to store more item data about more items in a single medium. Data capacities increased from the single data element linear bar code to concatenated symbols to two-dimensional symbols to high capacity RF tags to contact memory buttons to optical memory cards and micro compact disks. It became possible to store information about multiple orders on a shipment, multiple containers or pallets per order, multiple part numbers per order, multiple containers per part number, and multiple serial numbers per part number.

As this sophistication increased so increased the need to provide a structure for such data in order to ensure that there was an unambiguous relationship of a serial number (or lot number / expiration date) all of the way up to the order and shipment level. It would have been possible to create a unique structure for ADC media. However, the world of electronic data interchange (EDI) has faced this issue for many years. After careful analysis ASC MH 10/SC 8 decided to follow the lessons learned from the EDI community, namely the creation of a structured looping of data.

The X12 EDI Ship Notice/Manifest (Transaction 856) is a hierarchical document, that is, the electronic document which can:

- Represent one or several shipments in a single Ship Notice/Manifest,
- Each shipment can consist of one or several orders in a single shipment,
- Each order can consist of one or several pallets (tares) in a single order,
- Each pallet can consist of one or several cartons (packs) in a single pallet,
- Each carton can consist of one or several inner packs (sub-packs) in a single carton,
- Each sub-pack can consist of one or several items in a single sub-pack, and
- Each item can consist of one or several components in a single item.

Data should be encoded at the hierarchical level to which it logically applies. For example, shipment data at the shipment level, order information at the order level, tare (pallet) information at the tare level, carton information at the carton level, etc. To avoid unnecessary data redundancy it may be preferable to encode data at a higher level. For example, if a shipment involves only one order, order information could be transmitted at the shipment level. Also, if the only package information needed is the label serial number (license plate) and there is one per item then the package data can be specified at the item level. As a general rule data can be specified at a higher level as long as it does not create confusion with similar data at the same level. Weights dimensions, quantities, and license plates are examples of data which are used in multiple levels and could create confusion if levels are combined.

The following example depicts the detail area of the Ship Notice/Manifest transaction in the traditional manner.

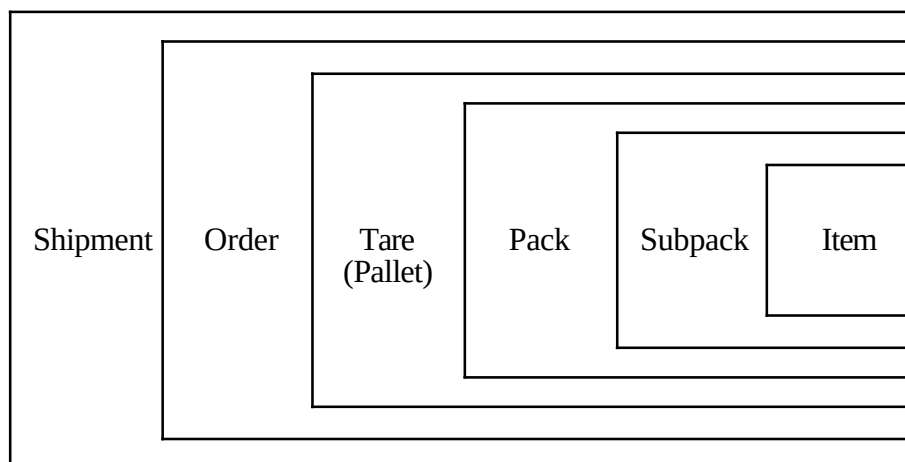


Figure VI – 1 – Typical X12 856 Nested Packaging Levels

The purpose of such structure looping is to facilitate the identification of dependencies among the content of related groups of data segments. Several methods existed, however, the SC 8 committee settled on the use of hierarchical structures similar to the ASC X12 EDI 856 Ship Notice/Manifest transaction.

Within the X12 856 transaction set the “HL segment” is comprised of four data elements (DE). These are

Table VI – 1 – ASC X12 856 “HL Segment”

DE Reference	DE Identifier	DE Name	DE Requirement	DE Type	DE Length (Min/Max)
HL01	628	Hierarchical ID Number	M	AN ¹	1/12
HL02	734	Hierarchical Parent ID Number	O	AN	1/12
HL03	735	Hierarchical Level Code	M	ID ²	1/2
HL04	736	Hierarchical Child Code	O	ID	1/1

Note¹ – A string data element is a sequence of any characters from the character set and contains at least one non-space character. The significant characters shall be left justified. Leading spaces, when they occur, are presumed to be significant characters. In the actual data stream, trailing spaces should be suppressed. The representation for this data element is AN.

Note² – An identifier data element always contains a unique value from a single, predefined list of values that is maintained in ASC X12 or some other body recognized by ASC X12 and identified by a reference in Appendix A of X12.3 Data Element Dictionary. Trailing spaces should be suppressed. The representation for this data element type is ID.

The X12 856 HL segment is used to identify levels of detail information using a hierarchical structure, such as relating line-item data to shipment data and packaging data to line-item data. The 856 HL segment defines a top-down/left-right ordered structure.

HL01 shall contain a unique alphanumeric number for each occurrence of the HL segment in the transaction set. For example, HL01 could be used to indicate the number of occurrences of the HL segment in which case the value of HL01 would be “1” for the initial HL segment and would be incremented by one in each subsequent HL segment within the transaction.

HL02 identifies the hierarchical ID number of the HL segment to which the current HL segment is subordinate.

HL03 indicates the context of the series of segments following the current HL segment up to the next occurrence of an HL segment in the transaction. For example, HL03 is used to indicate that subsequent segments in the HL loop form a logical grouping of data referring to shipment, order, or item-level information.

HL04 indicates whether or not there are subordinate (or child) segments related to the current HL segment. ("0" indicates that there are no subordinate segments; "1" indicates that there are subordinate segments)

It would be possible to encode an entire EDI transaction into a machine-readable medium, however there is substantial overhead within EDI to facilitate the routing of the message. Since, in the case of machine-readable media, the medium accompanies the routed item the overhead is unnecessary information. And while the data carrying capacity of machine-readable media has increased substantially, wherever a systems designer can reduce the number of encoded characters, the better is the design.

ASC MH 10/SC 8 took the basic Hierarchical Level (HL) structure and made two modifications. Both involved the variable length nature of the EDI HL with each of the data elements separated by a data element separator versus a machine-readable media requirement for defined lengths and short fields. Since the committee did not wish to use separator characters, because of increasing the length of the field, fixed length data elements were used where ever possible. Further, the length of the Hierarchical ID Number was fixed at two (2). With the character set of 0-9 and A-Z, a length of 2 characters yields 1,296 permutations. Ninety-nine (99) and even thirty-six (36) permutations were considered ample in most cases, however, several real-life examples of different parts with associated serial numbers caused the ASC MH 10/SC 8 to go to a second character position. The Hierarchical Child Code identifier and the Hierarchical Level Code identifier were swapped positionally since the Hierarchical Level Code was variable length. Placing the variable length field at the end of the composite field provided unambiguous meaning to each of the sub-fields.

This yielded the format for the Hierarchical Level Data Identifier "F". The purpose of Data Identifier "F" is to identify dependencies among the content of hierarchically related groups of data segments. The structure of this DI is as follows with all parts required:

Table VI – 2 – ANS MH10.8.2 Data Identifier "F" Structure

Part	String (AN) or Identifier (ID)	Length
Hierarchical ID Number	AN	2 ¹
Hierarchical Parent ID Number	AN	2 ¹
Hierarchical Child Code	ID	1
Hierarchical Level Code	ID	1/2
<i>Note¹ – With the character set of 0-9 and A-Z, a length of 2 characters yields 1,296 permutations</i>		

While the complete set of Hierarchical Level Code identifiers can be found in ANS X12, Data Element 735, the following represent what ASC MH 10/SC 8 considers to be the most commonly used identifiers:

Table VI – 3 – Commonly Used Hierarchical Level Codes

Level	Identifier	Description
Shipment	S	Data that applies to the whole shipment, such as bill of lading number, lading quantity, supplier code, etc.
Order	O	Data related to the sender's order and the associated receiver's original purchase order.
Tare	T	The tare level is used to identify pallets. If there are no identifiable pallets, this level may be omitted.
Pack	P	The pack level is used to identify the cartons within which the item is shipped, e.g., label serial numbers. In most cases there will be some sort of packs.
Sub-pack	Q	Data related to a grouping of identifiable packages within the pack level. Note that this level is only used when the inner pack has identifiable numbers for each inner pack.
Item	I	Stock keeping unit (SKU) identification data.
Component	F	Data related to the manufacturer's component
Serial #	X	Data related to the manufacturer's serial number

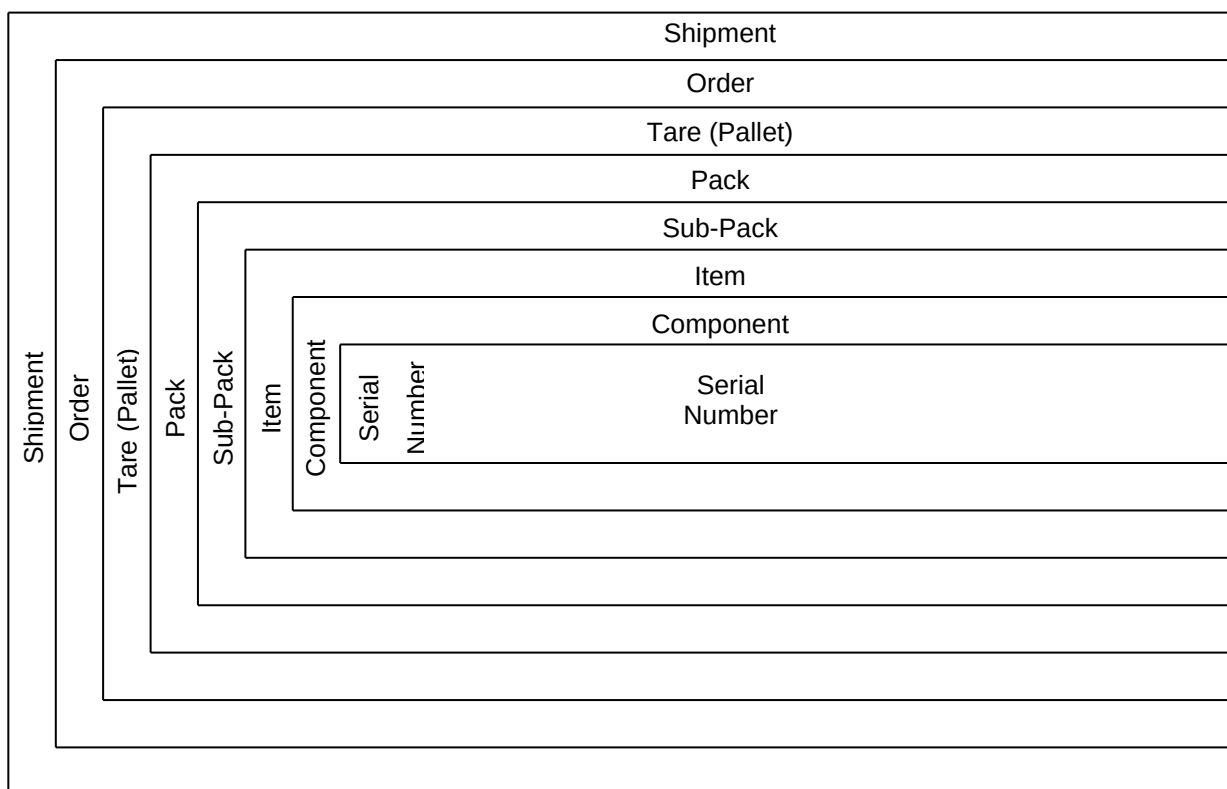


Figure VI – 2 – Typical MH 10/SC 8 Nested Packaging Levels

Consequently, for machine-readable media the structure “F08041P” would mean:

- F Data Identifier
- 08 Level of this hierarchy, e.g., a case on a pallet
- 04 Level of the parent hierarchy, e.g., the pallet
- 1 Yes, there are children to the case
- P Pack

Consider the following structure:

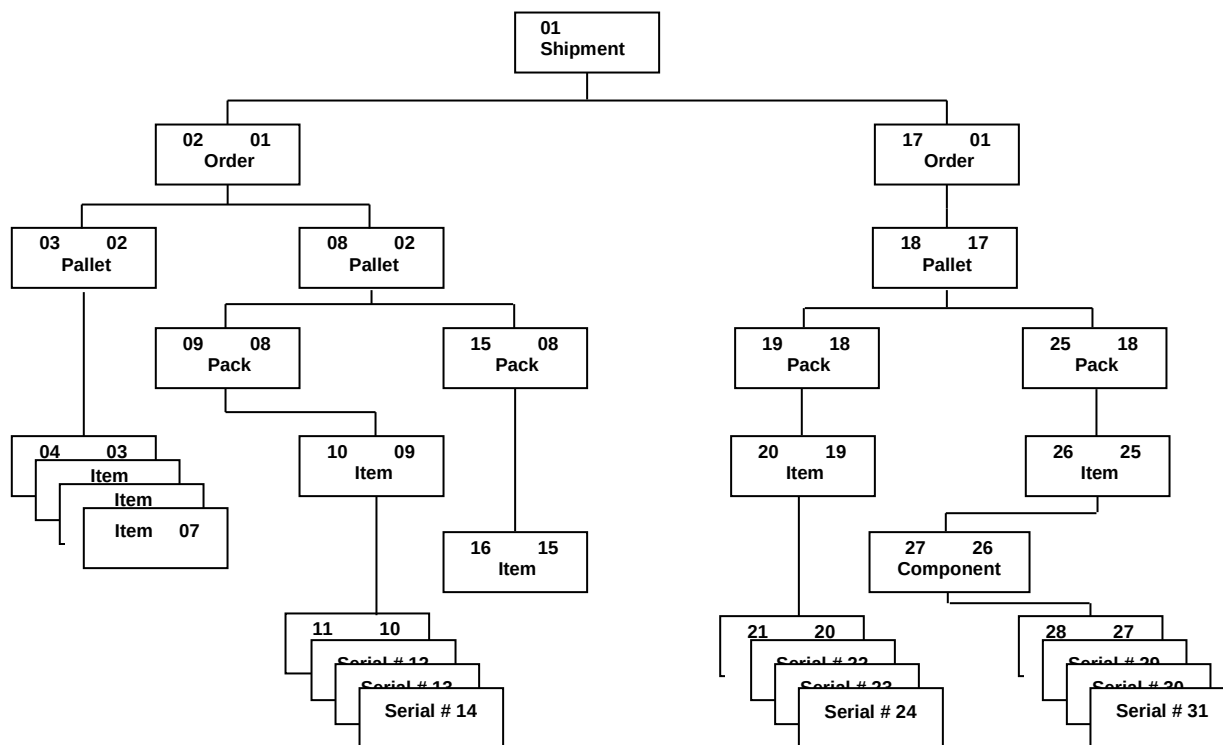


Figure VI – 3 – Hierarchical Levels

Using DI “F” the data would be as shown in Table VI – 4, below:

NOTE: Table VI – 4 has been oriented in a “visual” mode, not a data-transmission-form; the data order is upper left to lower right, left column first. The data transmission form of one contiguous data stream follows the table (Figure VI – 4).

Table VI – 4 – Reducing Figure VI – 3 to a Data Stream Using DI “F”

] > ^R _S 06 ^G _S	F11100X ^G _S	25SUN043325711Serial21 ^G _S
F01001S ^G _S	25SUN043325711Serial11 ^G _S	25SUN043325711Serial12 ^G _S
2QShipment01 ^G _S	25SUN043325711Serial12 ^G _S	25SUN043325711Serial23 ^G _S
F02011O ^G _S	25SUN043325711Serial13 ^G _S	25SUN043325711Serial44 ^G _S
KOrder02 ^G _S	25SUN043325711Serial14 ^G _S	F25181P ^G _S
F03021T ^G _S	F15081P ^G _S	JUN043325711Pack25 ^G _S
JUN043325711Pallet03 ^G _S	JUN043325711Pack15 ^G _S	F26251I ^G _S
F04030I ^G _S	F16150I ^G _S	25PUN043325711Item26 ^G _S
25PUN043325711Item04 ^G _S	25PUN043325711Item16 ^G _S	F27261F ^G _S
25PUN043325711Item05 ^G _S	F17011O ^G _S	25PUN043325711Comp27 ^G _S
25PUN043325711Item06 ^G _S	KOrder17 ^G _S	F28270X ^G _S
25PUN043325711Item07 ^G _S	F18171T ^G _S	25SUN043325711Serial28 ^G _S
F08021T ^G _S	JUN043325711Pallet17 ^G _S	25SUN043325711Serial29 ^G _S
JUN043325711Pallet08 ^G _S	F19181P ^G _S	25SUN043325711Serial30 ^G _S
F09081P ^G _S	JUN043325711Pack19 ^G _S	25SUN043325711Serial31 ^R _S ^E _{OT}
JUN043325711Pack09 ^G _S	F20191I ^G _S	
F10091I ^G _S	25PUN043325711Item20 ^G _S	
25PUN043325711Item10 ^G _S	F21200X ^G _S	

The following example shows how the data from Table VI - 4 would be encoded – as one continuous data stream;

] >^R_S06^G_SF01001S^G_S2QShipment01^G_SF02011O^G_SKOrder02^G_SF03021T^G_SJUN043325711Pallet03^G_SF04030I^G_S25PUN043325711Item04^G_S25PUN043325711Item05^G_S25PUN043325711Item06^G_S25PUN043325711Item07^G_SF08021T^G_SJUN043325711Pallet08^G_SF09081P^G_SJUN043325711Pack09^G_SF10091I^G_S25PUN043325711Item10^G_SF11100X^G_S25SUN043325711Serial11^G_S25SUN043325711Serial12^G_S25SUN043325711Serial13^G_S25SUN043325711Serial14^G_SF15081P^G_SJUN043325711Pack15^G_SF16150I^G_S25PUN043325711Item16^G_SF17011O^G_SKOrder17^G_SF18171T^G_SJUN043325711Pallet17^G_SF19181P^G_SJUN043325711Pack19^G_SF20191I^G_S25PUN043325711Item20^G_SF21200X^G_S25SUN043325711Serial21^G_S25SUN043325711Serial12^G_S25SUN043325711Serial23^G_S25SUN043325711Serial44^G_SF25181P^G_SJUN043325711Pack25^G_SF26251I^G_S25PUN043325711Item26^G_SF27261F^G_S25PUN043325711Comp27^G_SF28270X^G_S25SUN043325711Serial28^G_S25SUN043325711Serial29^G_S25SUN043325711Serial30^G_S25SUN043325711Serial31^R_S^E_{OT}

Figure VI – 4 – Encodeable Data

The following is another example from the telecommunication industry.

Presume the following EDI data:

EDI DATA	EXPLANATION.
ST~856~000000001	ASN Transaction Set - Transaction Set Control #000000001
BSN~00~000002~20010207~1001	Original Ship Notice #000002Created 02/07/01 at 10:01 am
DTM~011~20010207	Shipped on 2/07/01
HL~001~~S	Hierarchical Segment #1 - Shipment Level
TD5~~2~HMES	Shipped via USF Holland
REF~BM~104462	Bill of Lading = 104462
REF~CN~6783222	Carrier Pro # = 6783222
NI~ST~SBC	Ship to Name

N3~1700 HAZEL DELL RD	Ship to Address
N4~SPRINGFIELD~IL~627035258	Ship to City, State, Zip Code (9 digits)
HL~002~001~O	Hierarchical Segment #2- Order Level Subordinate to HL001
PRF~AB~347554	P.O. Number = AB347554
REF~VN~V11234345	Supplier's Order No. = V11234345
REF~IV,A00001	Invoice No. A00001
REF~PK~B12456	Packing List No. B12456
HL~003~002~I	Hierarchical Segment #3, Item Level Subordinate to HL002
LIN~0001~IN~102421930	SBC's Product Identifier = 102421930
SN1~001~600~FT	Total qty. shipped = 600 ft
PRF~AB347554~~~~0001	P.O. Number = AB34755 - Item = 0001
CLD~02~600	Number of reels = 2
	Number of units shipped on reels = 600 (feet as in SN103),
REF~LS~ABCD+40000	3S Bar Code Label = ABCD+40000,
REF~SE~AS23D145	Cable Reel Serial # = AS23D145,
REF~MR~EEE	Cable Reel Type = EEE,
REF~LS~ABCD+40001	3S Bar Code Label = ABCD+40001,
REF~SE~AS23D146	Cable Reel Serial # = AS23D146,
REF~MR~EEE	Cable Reel Type = EEE,
HL~004~002~I	Hierarchical Segment #4 -
	Item Level Subordinate to HL002,
LIN~0002~VN~TLT395	Vendor Part #TLT395,
SN1~002~2~EA	Total qty. shipped = 2EA,
PRF~AB347554~~~~0003	P.O. Number = AB34755, Item = 0003,
CLD~02~2	Number of containers = 2
	Number of units shipped in containers = 2,
REF~LS~ABCD+40002	3S Bar Code Label = ABCD+40002,
REF~LS~ABCD+40003	3S Bar Code Label = ABCD+40003,
CTT~4~602	HL Segments = 4
	Total Shipped Quantities = 602,
SE~35~0000000001	Segments = 35
	Transaction Set Control # = 000000001,

Figure VI – 5 – EDI Message

Table VI-6, below, uses the data from Figure VI - 5 and shows DI equivalence.

Table VI – 6 – Associating Data Identifier Data with X12 EDI Data

X12 856 EDI DATA	EXPLANATION	DATA IDENTIFIER DATA
DTM~011~20010207	Shipped on 2/07/01	5D010207011
HL~001~~S	Hierarchical Segment #1-Shipment Level	F01001S
TD5~~2~HMES	Shipped via USF Holland	(See Pro #)
REF~BM~104462	Bill of Lading = 104462	3K104462
REF~CN~6783222	Carrier Pro # = 6783222	12KHMES6783222
NI~ST~SBC	Ship to Name	NI~ST~SBC
N3~1700 HAZEL DELL RD	Ship to Address	N3~1700 HAZEL DELL RD

N4~SPRINGFIELD~IL~627035258	Ship to City, State, Zip Code (9 digits)	N4~SPRINGFIELD~IL~627035258
HL~002~001~O	Hierarchical Segment #2- Order Level Subordinate to HL001	F02011O
PRF~AB~347554	P.O. Number = AB347554	KAB347554
REF~VN~V11234345	Supplier's Order No. = V11234345	1KV11234345
REF~IV~A00001	Invoice No. A00001	10KA00001
REF~PK~B12456	Packing List No. B12456	11KB12456
HL~003~002~I	Hierarchical Segment #3 - Item Level Subordinate to HL002	F03020I
LIN~0001~IN~102421930	SBC's Product Identifier = 102421930	P102421930
SN1~001~600~FT	Total qty. shipped = 600 ft	7Q600FT
PRF~AB347554~~~~0001	P.O. Number = AB347554Item = 0001	14KAB347554+0001
CLD~02~600	Number of reels = 2 - Number of units shipped on reels = 600 (feet as in SN103)	7Q2RE 7Q600FT
REF~LS~ABCD+40000	3S Bar Code Label = ABCD+40000	3SABCD+40000
REF~SE~AS23D145	Cable Reel Serial # = AS23D145	SAS23D145
REF~MR~EEE	Cable Reel Type = EEE	BEEE
REF~LS~ABCD+40001	3S Bar Code Label = ABCD+40001	3SABCD+40001
REF~SE~AS23D146	Cable Reel Serial # = AS23D146	SAS23D146
REF~MR~EEE	Cable Reel Type = EEE	BEEE
HL~004~002~I	Hierarchical Segment #4 - Item Level Subordinate to HL002	F04020I
LIN~0002~VN~TLT395	Vendor Part #TLT395	1PTLT395
SN1~002~2~EA	Total qty. shipped = 2EA	Q2
PRF~AB347554~~~~0003	P.O. Number = AB347554 - Item = 0003	14KAB347554+0003
CLD~02~2	Number of containers = 2\ - Number of units shipped in containers = 2	7Q2CH
REF~LS~ABCD+40002	3S Bar Code Label = ABCD+40002	3SABCD+40002
REF~LS~ABCD+40003	3S Bar Code Label = ABCD+40003	3SABCD+40003
484 characters (not including address information [N1, N3, N4])		285 characters (not including address information [N1, N3, N4])

The telecommunication industry concluded that they do not require the Ship To information encoded in the machine-readable media that would accompany the shipment. If the complete EDI transaction were encoded, including the 71 characters associated with the ST, BSN, CTT, and SE segments and the 161 characters associated with the ISA, GS, GE, and IEA segments the complete EDI message would have been 716 (484+71+161) characters in length as opposed to the 285 when encoded with Data Identifiers.

Encoding this data using the Hierarchical Looping Data Identifier “F”, it would appear as follows;

(NOTE: The example below shows the data stream in a “visual” mode – this is NOT how it will appear when encoded)

TABLE VI – 7 – EDI Data using DI “F” – showing data in “visual” mode

$\text{J}>^{\text{R}}_{\text{S}}06^{\text{G}}_{\text{S}}$ $\text{F01001S}^{\text{G}}_{\text{S}}$ $5\text{D010207011}^{\text{G}}_{\text{S}}$ $3\text{K104462}^{\text{G}}_{\text{S}}$ $12\text{KHMES6783222}^{\text{G}}_{\text{S}}$ $\text{F02011O}^{\text{G}}_{\text{S}}$ $\text{KAB347554}^{\text{G}}_{\text{S}}$ $1\text{KV11234345}^{\text{G}}_{\text{S}}$ $10\text{KA00001}^{\text{G}}_{\text{S}}$ $11\text{KB12456}^{\text{G}}_{\text{S}}$	$\text{F03020I}^{\text{G}}_{\text{S}}$ $\text{P102421930}^{\text{G}}_{\text{S}}$ $7\text{Q600FT}^{\text{G}}_{\text{S}}$ $14\text{KAB347554+0001}^{\text{G}}_{\text{S}}$ $7\text{Q2RE}^{\text{G}}_{\text{S}}$ $7\text{Q600FT}^{\text{G}}_{\text{S}}$ $3\text{SABCD+40000}^{\text{G}}_{\text{S}}$ $\text{SAS23D145}^{\text{G}}_{\text{S}}$ $\text{BEEE}^{\text{G}}_{\text{S}}$ $3\text{SABCD+40001}^{\text{G}}_{\text{S}}$	$\text{SAS23D146}^{\text{G}}_{\text{S}}$ $\text{BEEE}^{\text{G}}_{\text{S}}$ $\text{F04020I}^{\text{G}}_{\text{S}}$ $1\text{PTLT395}^{\text{G}}_{\text{S}}$ $\text{Q2}^{\text{G}}_{\text{S}}$ $14\text{KAB347554+0003}^{\text{G}}_{\text{S}}$ $7\text{Q2CH}^{\text{G}}_{\text{S}}$ $3\text{SABCD+40002}^{\text{G}}_{\text{S}}$ $3\text{SABCD+40003}^{\text{R}}_{\text{S}}^{\text{E}}_{\text{OT}}$
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Figure VI - 6 shows how the data in Table VI – 7 would be encoded – as one continuous data stream;

$\text{J}>^{\text{R}}_{\text{S}}06^{\text{G}}_{\text{S}}\text{F01001S}^{\text{G}}_{\text{S}}5\text{D010207011}^{\text{G}}_{\text{S}}3\text{K104462}^{\text{G}}_{\text{S}}12\text{KHMES6783222}^{\text{G}}_{\text{S}}\text{F02011O}^{\text{G}}_{\text{S}}\text{KAB347554}^{\text{G}}_{\text{S}}1\text{KV11234345}^{\text{G}}_{\text{S}}10\text{KA00001}^{\text{G}}_{\text{S}}11\text{KB12456}^{\text{G}}_{\text{S}}\text{F03020I}^{\text{G}}_{\text{S}}\text{P102421930}^{\text{G}}_{\text{S}}7\text{Q600FT}^{\text{G}}_{\text{S}}14\text{KAB347554+0001}^{\text{G}}_{\text{S}}7\text{Q2RE}^{\text{G}}_{\text{S}}7\text{Q600FT}^{\text{G}}_{\text{S}}3\text{SABCD+40000}^{\text{G}}_{\text{S}}\text{SAS23D145}^{\text{G}}_{\text{S}}\text{BEEE}^{\text{G}}_{\text{S}}3\text{SABCD+40001}^{\text{G}}_{\text{S}}\text{SAS23D146}^{\text{G}}_{\text{S}}\text{BEEE}^{\text{G}}_{\text{S}}\text{F04020I}^{\text{G}}_{\text{S}}1\text{PTLT395}^{\text{G}}_{\text{S}}\text{Q2}^{\text{G}}_{\text{S}}14\text{KAB347554+0003}^{\text{G}}_{\text{S}}7\text{Q2CH}^{\text{G}}_{\text{S}}3\text{SABCD+40002}^{\text{G}}_{\text{S}}3\text{SABCD+40003}^{\text{R}}_{\text{S}}^{\text{E}}_{\text{OT}}$

Figure VI – 6 – EDI Message in Encodeable Form

ANNEX A

Informative

QUICK REFERENCE TO DATA IDENTIFIER (DI) CATEGORIES

Outline of Defined Categories

CATEGORY 0	Special Characters Employed as Data Identifiers
CATEGORY 1	Reserved
CATEGORY 2	Container Information
CATEGORY 3	Field Continuation
CATEGORY 4	Date
CATEGORY 5	Environmental Factors
CATEGORY 6	Looping
CATEGORY 7	Reserved
CATEGORY 8	Human Resources
CATEGORY 9	Reserved
CATEGORY 10	License Plate
CATEGORY 11	Transaction Reference
CATEGORY 12	Location Reference
CATEGORY 13	Maintenance Codes
CATEGORY 14	Industry Assigned Codes
CATEGORY 15	Reserved
CATEGORY 16	Item Information
CATEGORY 17	Measurement
CATEGORY 18	Miscellaneous
CATEGORY 19	Traceability Number for an Entity
CATEGORY 20	Traceability Number for Groups of Entities
CATEGORY 21	UPU / MH 10/SC8/WG2 Agreed Upon Codes
CATEGORY 22	Party to the Transaction
CATEGORY 23	Activity Reference
CATEGORY 24	Reserved
CATEGORY 25	Internal Applications
CATEGORY 26	Mutually Defined

Alphabetical Listings of Assigned DI Categories

ACTIVITY REFERENCE	CATEGORY 23
CONTAINER INFORMATION	CATEGORY 2
DATE	CATEGORY 4
ENVIRONMENTAL FACTORS	CATEGORY 5
FIELD CONTINUATION	CATEGORY 3
HUMAN RESOURCES.....	CATEGORY 8
INDUSTRY ASSIGNED CODES.....	CATEGORY 14
INTERNAL APPLICATIONS.....	CATEGORY 25
ITEM INFORMATION	CATEGORY 16
LICENSE PLATE	CATEGORY 10
LOCATION	CATEGORY 12
LOOPING	CATEGORY 6
MAINTENANCE CODES.....	CATEGORY 13
MEASUREMENT	CATEGORY 17
MISCELLANEOUS	CATEGORY 18
MUTUALLY DEFINED.....	CATEGORY 26
PARTY TO THE TRANSACTION.....	CATEGORY 22
SPECIAL CHARACTERS.....	CATEGORY 0
TRACEABILITY NUMBER FOR AN ENTITY	CATEGORY 19
TRACEABILITY NUMBER FOR GROUPS OF ENTITIES.....	CATEGORY 20
TRANSACTION REFERENCE.....	CATEGORY 11
UPU / MH 10/SC8/WG2 AGREED UPON CODES	CATEGORY 21

ANNEX B

Informative

ANNOTATED ALPHABETICAL LISTING OF ASSIGNED DATA IDENTIFIER (DI) CATEGORIES

Annotated Alphabetical Listing of Assigned DI Categories

ACTIVITY REFERENCE**CATEGORY 23**

This category relates to work activities, such as Work Order, Operation and Sequence numbers. It should not be confused with Category 11 (Transaction Reference), which relates to purchasing transactions.

CONTAINER INFORMATION**CATEGORY 2**

This category relates to identification of returnable containers such as compressed gas cylinders, wire reels, transportation equipment and other returnable-type containers. While many of these numbers are serial numbers, this category exists to provide an easy distinction between container serial number and product or label serial number referenced in Category 19 (Traceability Number for an Entity).

DATE**CATEGORY 4**

This category relates to a variety of date structures, as well as to the significance of the date (e.g., Date of Manufacture or Expiration Date).

ENVIRONMENTAL FACTORS**CATEGORY 5**

This category relates to identification of environmental issues such as temperature, air pressure, wind speed, and similar measurements. Where Category 5 defines environmental measurements, Category 17 defines physical measurements.

FIELD CONTINUATION**CATEGORY 3**

This category relates to the continuation of data from a defined field which must be broken into two symbols because of space or other constraints. Only some of the assigned DI fields have valid continuation assignments.

HUMAN RESOURCES**CATEGORY 8**

This category relates to personnel identification.

INDUSTRY ASSIGNED CODES**CATEGORY 14**

This category relates to code or numbering systems that are controlled by and specific to a specific industry or governmental agency (e.g., NATO Stock Number). This allows for unambiguous identification of those code systems within the ANSI MH10.8.2 DI framework without necessitating the assignment of unique ANSI MH10.8.2 DIs for those items that have little or no relevance to those outside that industry or governmental agency.

INTERNAL APPLICATIONS**CATEGORY 25**

This category relates to the use of DIs for purposes that will remain within a closed system and for which a valid ANSI MH10.8.2 DI cannot provide unambiguous reference. For use within a single manufacturing facility, for example, the use of the Internal Applications DI (Y) could precede any other ANSI MH10.8.2 DI which could be defined, in that instance, for a purpose which is not in conformance with these guidelines.

ITEM INFORMATION**CATEGORY 16**

This category relates to the identification or characteristics of an item (see definitions), such as its Part Number, Manufacturing Revision Level or its Classification as a Hazardous Material. An item is something that is not identified as a unique entity but rather as representative of all other like items (see Definitions for further information). Additional DIs (20P-24P and 30P-34P) are set aside for descriptive information not otherwise provided for and which is defined between trading partners or intended for internal use (but with messages that will leave the system, precluding the use of a Category 25 DI). This category should not be confused with Category 19, Traceability Number for an Entity, nor Category 20, Traceability Number for a Group of Entities, both of which provide for identification of unique entities (see definitions).

LICENSE PLATE**CATEGORY 10**

This category relates to a worldwide unique identification of a transport unit or a unitized load (e.g., shipping container or pallet). Each DI is comprised of a unique Issuing Agency Code (IAC) assigned pursuant to ISO/IEC 15459, a world-wide unique organizational/entity/company identification number assigned by the IAC, and a unique transport unit/unitized load number assigned by the organization, entity, or company¹³.

LOCATION**CATEGORY 12**

This category relates to either a physical location that is used as a reference point (such as a Shelf Location) or to a physical location that is used as a destination reference (such as a Ship To address). Of particular note are the assignments 51L-52L and 54L-55L that relate to postal codes used as shipping addresses. These two sets of DIs provide for both domestic and international use (with an ISO country code suffix).

LOOPING**CATEGORY 6**

This category relates to the parent/child relationship between various fields of data, using pre-existing techniques from electronic data interchange. An example is where a shipment contains multiple orders over multiple pallets, multiple packages, multiple items, with multiple serial numbers. Using techniques described within Annex F of this document it is possible to relate a given serial number with a specific order.

MAINTENANCE CODES**CATEGORY 13**

This category identifies specific codes used in maintenance functions, including those functions expressed over time such as machine-on time, mean-time-between-failure, and the like.

MEASUREMENT**CATEGORY 17**

This category relates to physical dimensions, measures, quantity or monetary value of an item or group of items (may refer to entities as well). Of particular note is the assignment 7Q that is quantity followed by an ANSI Data Element 355 description of unit of measure. To indicate that there are n cartons in the shipment with x items per carton, either two 7Q fields can appear in the same message with appropriate ANSI modifiers or a 7Q can be used with a Q (generic quantity) with the significance mutually defined.

¹³ Note: An exception within the License Plate category is the inclusion of "7J" Vehicle Registration License Plate Number (not unique without identification of country and issuing governmental region/authority)

MISCELLANEOUS**CATEGORY 18**

This category relates to DIs that cannot otherwise be categorized (currently contains Return Authorization Codes).

MUTUALLY DEFINED**CATEGORY 26**

This category relates to data or information which has not been assigned a DI within this document and which trading partners need to include in their automatic identification application. The structure and significance of this information is to be agreed upon by all appropriate parties to the transaction.

PARTY TO THE TRANSACTION**CATEGORY 22**

This category relates to codes that identify all business entities that may be a party to a transaction (e.g., Vendor Number, Customer Number or Carrier Number).

SPECIAL CHARACTERS**CATEGORY 0**

This category relates to the use of a non-alpha, non-numeric character in the first data position of an automatic identification message (e.g., bar code) to identify the message as being controlled by a specific organization (e.g., Health Industry Business Communications Council, Uniform Code Council).

TRACEABILITY NUMBER FOR AN ENTITY**CATEGORY 19**

This category relates to the identification of a specific item (entity) in a unique manner for purposes of tracing that entity. Codes with this category DIs may identify a finished product or they may identify packaging that contains multiple entities if the packaging is what is being tracked. If a DI from this category is used, an identical message on another entity should never be found within the originating system. For example, a television's serial number is a traceability number for an entity, as is a unique number assigned to a carton to identify it in conjunction with an EDI transaction. This category should not be confused with Category 16 (Item Identification), which provides for identification of all like items (where an identical message would certainly be found within the same system), or with Category 20, which provides unique identification for groups of entities (see below).

TRACEABILITY NUMBER FOR GROUPS OF ENTITIES**CATEGORY 20**

This category relates to the identification of a lot, batch or other grouping of entities for purposes of tracing that group. Additional DIs (20T-24T and 30T-34T) have been set aside for additional information which is not otherwise provided for and which is mutually defined between trading partners or intended for internal use (but with messages which will leave the system, precluding the use of a Category 25 DI). This should not be confused with Category 19 (Traceability Number for an Entity) or Category 16 (Item Identification).

TRANSACTION REFERENCE**CATEGORY 11**

This category relates to the identification of agreements or correspondence that is involved in the sale, purchase or transportation of goods or services. This category is distinct from Category 23 (Activity Reference) that relates to the production of such goods and/or services.

UPU / MH 10/SC8 AGREED UPON CODES**CATEGORY 21**

This category relates to a set of identifiers ("5U" to "55U") that may be unique to the nature of the business of the United Postal Union (UPU) postal authorities that might not otherwise be used within the supply chain. The agreement between the UPU and ANSI MH10/SC 8 is such that the UPU will endeavor to use DIs common to the rest of the marketplace. Only where there is a unique postal requirement for a unique DI, UPU may utilize one or more of the Category 21 DIs with the collaboration of ANSI MH10/SC 8.

ANNEX C

Normative

DATA IDENTIFIER (DI) APPLICATION NOTES

APPLICATION NOTES

The ANSI MH10.8.2 DI Standard addresses general requirements for Data Identifiers. Industry- or company-wide application standards will further define and regulate the use of any or all Data Identifiers in this document.

The basic structure of a ANSI MH10.8.2 DI is an alphabetic character preceded by 1, 2, 3 or no numeric digits. To decode an ANSI MH10.8.2 DI, software must parse the data up to the first alphabetic character and then evaluate that alphabetic character and the preceding numeric characters, if any.

However, it is recognized that some organizations will face implementation problems that cannot be fully covered in the general guidelines or that additional information on the intended use of certain of the DIs assigned in this document will be needed. The following application notes may be of some assistance.

The following topics are discussed in this Annex.

C.1 CAUTION ABOUT THE USE OF GS1 NUMBERS

C.2 USE OF ANSI AND ISO DATA ELEMENT IDENTIFIERS

C.3 DATE SIGNIFICANCE

C.4 LOCATION CODING

C.4.1 - Ship From, Ship To, Ship For

C.4.2 - Multiple Levels of Location Marking

C.5 ITEM IDENTIFICATION

C.6 QUANTITY

C.6.1 - Quantity Followed by Unit of Measure

C.6.2 - Monetary Value

C.6.3 - Number of a Carton Within Shipment

C.7 PACKAGING IDENTIFICATION

C.7.1 - Master Pack Identification

C.7.2 - Lowest Level of Packaging

C.8 LOT/BATCH IDENTIFICATION

C.9 ASSIGNMENT "4K"

C.10 ASSIGNMENT "18K" Structured Reference

C.11 Unique Identification of Items

C.1 THE USE OF GS1 NUMBERS

A number of industries have mandated that their members secure GS1 numbers in order to provide a common vendor numbering system. However, many of these organizations also employ alphanumeric coding schemes and/or include additional information that is not provided for by any GS1 standard and, therefore, are not in compliance with GS1 specifications. These applications are valid *only* for the respective industries which have issued standards which accept GS1 numbering in addition to other numbering systems which require alphanumeric symbologies or which have otherwise mandated the use of these numbering systems.

In no case should ANSI MH10.8.2 DIs be used in conjunction with, or in place of, approved GS1 symbols for retail or POS applications for which the GS1 Global Trade Item Number (GTIN) has been mandated. In all such cases, only the appropriate GS1 standards and specifications shall be applicable.

Questions on the use of GS1 numbering systems and standards, as well as the use of the GS1 bar code symbologies, can be obtained from the respective agency (address listed in this document).

C.2 THE USE OF ANSI AND ISO DATA ELEMENT IDENTIFIERS

For some DI assignments, qualifiers (suffixes) are used to provide additional significance to the data in the message. These qualifiers are drawn from ANSI and ISO Electronic Data Interchange (EDI) standards. In all cases the current "Draft Standard Approved for Trial Use" shall be the authoritative document.

In some instances, American National Standards Institute (ANSI) X12.3 Data Element Qualifiers are used. In other instances, International Standards Organization (ISO) codes are used. ANSI MH10.8.2 would prefer to use internationally accepted (ISO) codes for all applications but ISO standards do not exist for all needs.

EDIFACT, the United Nations EDI Standard, does provide codes for applications for which ANSI standards are referenced. However, EDIFACT does not yet have a practical coordination and review body that could assign additional codes as needed.

Insofar as this will be a dynamic document that will evolve new assignments in order to meet the needs of automatic identification users around the world, it was felt that an organization, which did have a professional staff to be responsive to requests, was essential. For this reason, ANSI standards have been used where necessary.

C.3 DATE SIGNIFICANCE

Provisions are made for various data codings in Category 4. Most DIs pertain to a specific date structure (e.g., DDMMYY) but do not specify the significance of the date.

In many applications, the structure of the date is mandated but the significance of the date is mutually agreed between trading partners and assignments ("D"-"5D") may be used in these cases.

Further, when both the structure and significance of the date is mutually defined, the assignment "9D" may be used.

However, when the significance of the date must be included, the assignments "6D" and "7D" use an ANSI Qualifier following the date to indicate its significance. The following examples show how this might be applied.

Example: DI/Data/ANSI Qualifier

Date of Manufacture (2-digit year, month, day)	6D890420049
Expiration Date (4-digit year, month, day)	7D20051231036

C.4 LOCATION CODING

The following topics are covered in this note. Location coding is covered in Category 12.

- "Ship From", "Ship To", "Ship For" location codes
- Multiple levels of location, marking.

C.4.1 Ship From, Ship To, Ship For

In order to facilitate automated sortation and routing of shipments, location codes for shipping locations have been provided. The assignments provides for three possible locations.

- Ship From
- Ship To
- Ship For (mutually defined)

There are two different sets of assignments for "Ship From" and "Ship To" location:

- Mutually defined or industry standard ("3L" & "2L")
- Postal code location ("51L" & "52L" and "54L" & "55L")

The use of mutually defined or industry standards will not be discussed here other than to note that the "Ship For" destination code (5L) will generally be printed by the supplier at the customer's request and used by the customer to facilitate automated internal routing of shipments. The "Ship For" code will, therefore, have significance only to the customer.

The use of postal authority codes (postal codes) does, however, merit some discussion. In this section, only the "open system" ANSI MH10.8.2 DIs will be discussed.

General Considerations

When postal codes are used which fall within the jurisdiction of a single postal authority (generally, the same country), there is no ambiguity of the location referred to. These are assignments "51L" ("Ship From") and "52L" ("Ship To").

However, postal coding systems around the world may present ambiguities to computer systems. For example, many European postal codes, as well as others around the world, are 4-digit numeric codes. Thus, the postal code "6300" could exist in more than one country. Without a means for identifying the country that administers that particular postal code, the data is meaningless.

Within Europe there is a postal convention that allows the inclusion of a country code preceding the numeric code. Following this convention, "CH-6300" refers to a Swiss (Confederation Helvetia) postal code.

Most postal authorities do not have such conventions, however, so another means of identifying the postal authority is needed - one that does not conflict with the European convention.

The ANSI MH10.8.2 DI Standard uses 2-character ISO country codes following the postal code for this purpose.

The following protocol is used in the ANSI MH10.8.2 DI Standard.

"Domestic" Postal Codes

If the shipment is within a single country or postal authority, use of the appropriate "domestic" DI ("51L" or "52L") preceding the postal code is all that is required. *For the European postal convention countries, the "domestic" postal code DIs may be used with the proper country prefix included in the data portion of the labeling device (e.g., bar code label, RFID tag).*

"International" Postal Codes

If the shipment is to move between countries or postal authorities, the appropriate "international" DI ("54L" or "55L") preceding the postal code will indicate that an ISO country code follows the postal code.

Example 1: "Domestic" shipments (within the same postal authority).

<u>Location</u>	<u>City, Country</u>	<u>DI/Postal Code</u>
"Ship From"	Zug, Switzerland	51L6300
"Ship To"	Geneva, Switzerland	52L1216

Example 2: "Domestic" shipments (within the European postal convention)

<u>Location</u>	<u>City, Country</u>	<u>DI/Country/Postal Code</u>
"Ship From"	Zug, Switzerland	51LCH6300
"Ship To"	Brussels, Belgium	52LB1150

Example 3: Shipment between postal authorities

<u>Location</u>	<u>City, Country</u>	<u>DI/Postal Code/ISO Qualifier</u>
"Ship From"	Zug, Switzerland	54L6300CH
"Ship To"	Morley, Western Australia	55L6062AU

NOTE: Italics and bold are used for emphasis and are not used in actual coding.

It should also be noted that the longest possible bar code (excluding start, stop and symbology check characters) will be 14 characters (3-character DI, 9-digit U.S.A. Zip Code, 2-character ISO country code).

C.4.2 Multiple Levels of Location Marking

Provision is made in this document for multiple levels of location marking ("1L", "20L"- "24L"). These are considered to be for internal or mutually defined use.

The "Location" assignment is considered to be "generic" and is kept to two characters to reduce symbol length.

For applications that require further differentiation or a hierarchical method of indicating location *and require that information in the DI*, the "First Level" through "Fifth Level" assignments is provided. An example of their use is indicated below.

Example 1: Hierarchical Location

<u>Location Description</u>	<u>DI</u>
Building Number	1L
File Storage Room Number	20L
File Cabinet Row Number	21L
File Cabinet Number	22L
File Cabinet Drawer Number	23L
File Number	24L

Example 2: Location Differentiation

<u>Location Description</u>	<u>DI</u>
Building Number	1L
Machine Tool Location Number	20L
Physical Plant Equipment Location Number	21L
Routing Location Number	22L
Asset Control Room/Location Number	23L
Mail Stop	24L

C.5 ITEM IDENTIFICATION

Product identification DIs are assigned in Category 16, "Item Information." Multiple levels of identification are provided for both supplier ("1P" and "30P"- "34P") and customer ("P" and "20P" - "24P").

The "1P" and "P" assignments are intended to be the most commonly used DIs. However, many business entities have additional requirements that suggest the use of additional DIs for product identification. The following examples show how they might be used.

Example 1: Multiple Product Characteristics (as assigned by supplier)

<u>Description</u>	<u>DI</u>
Shoe Style Number	1P
Length	30P
Width	31P
Color	32P
Material	33P
Trim	34P

Example 2: Multiple Product Identifications (as assigned by customer)

<u>Description</u>	<u>DI</u>
Part Number	P
Old Catalog Number	20P
New Catalog Number	21P
<i>Additional information</i>	22P
<i>Additional information</i>	23P
<i>Additional information</i>	24P

C.6 QUANTITY

Quantity, number of pieces, or “amount” DIs are assigned in Category 17, “Measurement.” The following topics are covered in this section.

- Quantity followed by a unit of measure
- Monetary value followed by a unit of measure
- Number of cartons within a shipment

C.6.1 Quantity Followed by Unit of Measure

For applications in which the quantity that will be referred is unambiguous and clearly understood between trading partners, the assignment “Q” should be used. Other assignments exist for other, defined, measures.

However, where there is more than one quantity or where the unit of measure needs to be specified, the assignment “7Q” allows for qualification of the value through the use of an 2-digit ANSI Unit of Measure Code.

The unit of measure code immediately follows the data. Because of the ability to qualify the amount, more than one “7Q” message may be found on a single labeling device.

The following examples show how this could be applied.

Example: Quantity, Measure

<u>Measure</u>	<u>DI/Data/ANSI Qualifier</u>
Number of pieces in box (mutually defined)	Q144 (no qualifier)
Weight of each piece (in kilograms)	7Q21.25KG
Rated capacity (in kilowatt hours)	7Q12KH
Overall length (in inches, decimal, nominal)	7Q35.6ED

C.6.2 Value

Provision is made for the definition of unit of value (“12Q”) by using an ISO country/currency code following the data. The use of this DI must be mutually defined between trading partners. The following examples show how this could be applied.

Example: Value of Item

<u>Description</u>	<u>DI/Data/ISO Qualifier</u>
Value of each piece in U.S. Dollars (\$12.75)	12Q12.75USD
or	
Value of shipment in U.S. Dollars (\$14,500)	12Q14500USD

C.6.3 Number of Carton Within Shipment

A DI has been assigned to allow information concerning the number of a carton within a shipment ("13Q"). The structure of the data follows the format:

$$n/x$$

where: n is the number of the carton within the shipment

$/$ is the separator between numeric fields (must be encoded)

x is the total number of cartons in the shipment.

Examples:

<u>Description</u>	<u>DI/Data</u>
5th carton in shipment of 6 cartons	13Q5/6
127th carton in shipment of 127 cartons	13Q127/127

C.7 PACKAGING IDENTIFICATION

Package Identification DIs are assigned in Category 19, "Traceability Number for an Entity." These identifiers are used on labeling devices (e.g., trading partner bar code transaction labels) attached to packaging.

The following topics are discussed in this section.

- Master packaging identification - customer order reference
- Identification of lowest level of packaging

Packaging identification generally is a unique number that identifies that package from all other packages. This number is usually used in conjunction with a supplier identification to provide a completely unique number.

Master packs (sometimes referred to as "unit loads") are transport units either made up of a number of filled transport packages or items held together by pallet, slip sheet, strapping, etc. or comprised of a single large container expressly designed to make items suitable for transportation, stacking, and storage as a unit.

Many industry standards require lower levels of packaging identification (using a lower level DI) within Master Packs to complete a transaction process. Trading partners are encouraged to utilize the lowest level DI and configure shipments accordingly.

C.7.1 Master Pack Identification

Provision is made for identification of the following information on the master packaging label ("4S"-"7S").

- Whether items within the package are the same or different.
- Whether items within the package are covered by one customer order or more than one customer order.

Assignments "4S" and "5S"

Assignments "4S" and "5S" are used when the items in the shipment are covered under the same customer order. The "4S" DI is used when the items are the same. The "5S" DI is used when the items are not the same.

If there is no interest in identifying whether single or multiple customer orders are contained within the packaging *and there is a strong argument against using all four DIs*, then all shipments should be identified as being "on the same customer order" (i.e., not referenced) and "4S" and "5S" can be used.

Assignments "6S" and "7S"

The "6S" and "7S" DIs are used to indicate that the items in the package are covered by multiple customer orders. "6S" is used when the items are the same, "7S" when the items are not the same.

Implicit in the use of "6S" and "7S" is the assumption that "4S" and "5S" DIs will also be encountered by the reading system.

See the next section for information about labels at lower levels within master packaging.

C.7.2 Lowest Level of Packaging

In some instances, packaging identification labels (other than part number) will be affixed to packaging within a master pack. A DI is provided to indicate that no further levels of packaging identification will be found within the package ("3S").

It is assumed that "3S" will be affixed to packaging which contains like items and that no further scanning will be required for package tracking purposes.

It is assumed that "3S" will usually be placed on packaging that is intended for transport or storage and will contain sub-packs on which only item identification is found. "3S" labels will generally be found on intermediate packaging occurring between the Product Identification and Master Packaging. It is further assumed that the "3S" symbol will generally be found within a master pack which contains a DI from the range "4S" - "7S".

C.8 LOT/BATCH IDENTIFICATION

Lot and batch identification DIs are assigned in Category 20, "Traceability Number for Groups of Entities." Multiple levels of identification are provided for both supplier ("T" and "30T"-"34T") and customer ("IT" and "20T"-"24T").

The "T" and "1T" assignments are intended to be the most commonly-used DIs. However, many business entities have additional requirements that suggest the use of additional DIs for product identification. The following examples show how they might be used.

Example: Multiple Lot/Batch Information (as assigned by supplier)

<u>Description</u>	<u>DI</u>
Lot Number	1T
Production Batch Number	30T
Testing Batch Number	31T
Shipment Lot Number	32T
Additional information	33T
Additional information	34T

C.9 ASSIGNMENT "4K"

"Line number of the order assigned by the Customer to identify a Purchasing Transaction." This DI refers to the physical line number of an order on which a large number of items are requested. In some trading relationships, master orders are issued which cover a specified period of time and products are released against the order over time. The process simplifies paperwork for routinely ordered items that are not to be shipped in a single lot.

In such an instance, a simple reference to an order number (e.g., Purchase Order, Work Order, etc.) is not sufficient. For these instances, the "4K" data refers to the specific line of the order in which the product or service is referenced.

The line number and order may refer to electronic or paper transactions.

C.10 **ASSIGNMENT "18K" Structured Reference**

Many data identifier allocations correspond to identifiers, (e.g., bar code) representations that are intended to be engraved or printed on, or affixed to, the physical objects they identify. Container identifiers (category B), License Plates (J), Item Identifiers (P), and Traceability Numbers (S) fall into this category.

These identifiers are also commonly used in communications about the objects they identify. Where such communications are purely electronic, it is self evident that what is communicated is a reference to the object identified. However, particularly in the postal world and in logistics applications, there are situations in which it is desirable to communicate such reference information in the form of a bar code (or 2D symbol or RF tag) that is printed on or attached to a physical object other than the object which is identified.

In such cases, the data identifier corresponding to the type of identifier cannot be used to identify the data, since otherwise, an automated system would be unable to distinguish between the physical object identified and the object carrying a reference to it.

For example, in the domain of license plates, a number of items carrying, say, license plates JJ1, JJ5, JJ7 and JJ10 might be grouped, for transport purposes, into an aggregate carrying license plate 2JJ4 (or put into a container with Container Identifier 5BJ7. Bar codes (or 2D symbols) on the aggregate (or container) may need to list the content of the aggregate. They cannot use the license plate DI for this since, otherwise, an automated system might read one of the reference barcodes, interpret it as the license plate attached to the referenced object, and process the aggregate as if it were the particular individual item concerned. A similar scenario may occur in postal processing, where batch cards (which may be physically indistinguishable from postal items) are used to list the identifiers of the items that comprise the batch.

The solution to this problem requires that there be a clear distinction between an identifier that is part of, or attached to, the object identified and an identifier reference. This can only be achieved by use of a different data identifier. For this, three possibilities have been identified:

1. Create a separate DI, in the category concerned, for each case.
2. Create a separate DI, in category K (transaction reference), for each case.
3. Allocate a single category K DI, embedding both the referenced identifier and its original DI value into the data.

Of these, the first two calls for the allocation of many DI's and risk confusion, since it would be impossible to maintain any consistency of correspondence between the numeric prefixes used for references and the prefixes for the original objects. Approach 3 is therefore proposed as being both simple and elegant.

Structure: identification code, license plate or traceability number for an object or entity, prefixed by the data identifier used for encoding that identification code on the object itself.

Example:

Suppose that a parcel has license plate, issued under the UPU Issuing Agency Code, JGBA123456789.

This will be encoded on the parcel, using Data Identifier J. The parcel label will thus carry a bar code, including the DI, specifically: JJGBA123456789.

The corresponding Structured Reference is thus JJGBA123456789. When encoded in a bar code or other media, it will be prefixed by the DI for a Structured Reference, i.e. as 18KJJGBA123456789.

Similarly, a bar code reference to an aggregate transport unit (DI 2J) with license plate JGBA456789123 would be encoded as 18K2JJGBA456789123.

C.11 **Unique Identification of Items**

The intended use of Data Identifier (DI) 25S is to indicate that the data following the DI represents a concatenated data string that uniquely identifies an item.

The 25S data string is formed from three segments, “IAC” followed by “CIN” followed by “Serial Number” (defined in bullets below). The data string making up DI 25S shall be globally unique.

There are three methods for creating unique serialization:

- Serial Number (unique SN within CIN’s domain)
- Part number + serial number (unique for that part number for the CIN)
- Lot/batch number + serial number (unique within the lot/batch for the CIN)
- Part number + lot/batch number + serial number (unique within the part number and its lot/batch for the CIN)

Data strings following 25S should not be parsed to obtain the component data elements.

ANNEX D

Informative

ANSI X12.3

DATA ELEMENT NUMBER 355

UNIT OF MEASURE CODE

ANSI X12.3 Data Identifier Dictionary

Code List 355 Unit of Measure

CODE	DEFINITION	CODE	DEFINITION
01	Actual Pounds	67	Siemens (A Unit of Admittance)
02	Statute Mile	68	Ampere
03	Seconds	69	Test Specific Scale
04	Small Spray	70	Volt
05	Lifts	71	Volt-Ampere Per Pound
08	Heat Lots	72	Watts Per Pound
10	Group	73	Ampere Turn Per Centimeter
11	Outfit	74	Milli Pascals
12	Packet	75	Cycles
13	Ration	76	Gauss
14	Shot	77	Mil
15	Stick	78	Kilogauss
24	Theoretical Pounds	79	Electron Volt
26	Actual Tons	80	Pounds Per Square Inch (Absolute)
27	Theoretical Tons	81	Henry (A Unit of Inductance)
31	Catchweight	82	Ohm (A Unit of Resistance)
50	Actual Kilograms	83	Farad (A Unit of Capacitance)
51	Actual Tonnes	84	Kilo Pounds Per Square Inch (KSI)
53	Theoretical Kilograms	85	Foot Pounds
54	Theoretical Tonnes	86	Joules
56	Sitas	AA	Ball
58	Net Kilograms	AC	Acre
59	Parts Per Million	AM	Ampoule
60	Percent Weight	AP	Aluminum Pounds Only
61	Parts Per Billion	AS	Assortment
62	Percent Per 100 Hours	AY	Assembly
63	Failure Rate In Time	B2	Bunks
64	Pounds Per Square Inch Gauge	B3	Batting Pound
65	Coulomb (A Unit of Charge)	B4	Barrel, Imperial
66	Oersteds	B5	Billet

B6	Bun	BL	Block
B7	Cycles	BM	Bolt
B8	Board	BN	Bulk
B9	Batt	BO	Bottle
BA	Bale	BP	100 Board Feet
BB	Base Box	BQ	Brake horse power
BC	Bucket	BR	Barrel
BD	Bundle	BS	Basket
BE	Beam	BT	Belt
BF	Board Feet	BU	Bushel
BG	Bag	BV	Bushel, Dry Imperial
BH	Brush	BW	Base Weight
BI	Bar	BX	Box
BJ	Band	BY	British Thermal Unit (BTU)
BK	Book	BZ	Million BTUs
C1	Composite Product Pounds (total weight)	CN	Can
C3	Centiliter	CO	Count
C4	Carload	CP	Crate
C5	Cost	CQ	Cartridge
C6	Cell	CR	Cubic Meter
C7	Centipoise (CPS)	CS	Cassette
C8	Cubic Decimeter	CT	Carton
C9	Coil Group	CU	Cup
CA	Case	CV	Cover
CB	Carboy	CW	Hundred Pound (CWT)
CC	Cubic Centimeter	CX	Coil
CD	Carat	CY	Cubic Yard
CE	Centigrade, Celsius	CZ	Combo
CF	Cubic Feet	C2	Carset
CG	Card	DA	Days
CH	Container	DB	Dry Pound
CI	Cubic Inches	DB	Dry Pounds
CJ	Cone	DC	Disk (Disc)
CK	Connector	DD	Degree
CL	Cylinder	DE	Deal
CM	Centimeter	DF	Dram

DG	Decigram	GI	Imperial Gallons
DH	Miles	GL	Grams Per Liter
DI	Dispenser	GM	Grams Per Square Meter
DK	Kilometers	GN	Gross Gallons
DL	Deciliter	GR	Gram
DM	Decimeter	GS	Gross
DP	Dozen Pair	GY	Gross Yard
DR	Drum	GZ	Gage Systems
DS	Display	HA	Hank (100 feet of rope)
DT	Dry Ton	HB	Hundred Boxes
DZ	Dozen	HC	Hundred Count
E3	Inches, Fraction-Average	HD	Half Dozen
E4	Inches, Fraction-Minimum	HE	Hundredth of a Carat
E5	Inches, Fraction-Actual	HF	Hundred Feet
E7	Inches, Decimal-Average	HG	Hectograms
E8	Inches, Decimal-Actual	HH	Hundred Cubic Feet
E9	English (Feet, Inches)	HI	Hundred Sheets
EA	Each	HJ	Horse power
ED	Inches, Decimal-Nominal	HK	Hundred Kilograms
EF	Inches, Fraction-Nominal	HL	Hundred Feet-Linear
EM	Inches, Fraction-Minimum	HO	Hundred Troy Ounces
EP	Eleven pack	HP	Hundred Pounds
EV	Envelope	HR	Hours
EX	Feet, Inches and Fraction	HS	Hundred Square Feet
EY	Feet, Inches	HT	Half Hour
EZ	Feet and Decimal	HU	Hundred
FA	Fahrenheit	HV	Hundred Weight (Short)
FC	1000 Cubic Feet	HW	Hundred Weight (Long)
FM	Million Cubic Feet	HY	Hundred Yards
FO	Fluid Ounce	IN	Inch
FP	Pounds Per Square Foot	JB	Jumbo
FT	Foot	JO	Joint
GA	Gallon	JR	Jar
GB	Gallons/Day	JU	Jug
GG	Great Gross (Dozen Gross)	KA	Cake
GH	One-half Gallon	KD	Kilograms Decimal

KE	Keg	MI	Metric
KG	Kilograms	MJ	Minutes
KH	Kilowatt Hour	MK	Milligrams Per Square Inch
KI	Kilograms/Millimeter Width	ML	Milliliter
KK	100 Kilograms	MM	Millimeter
KL	Kilograms/Meter	MN	Metric Net Ton
KM	Kilograms/Square Meter,	MO	Months
KN	Kilometer	MP	Metric Ton
KT	Kit	MQ	1000 Meters
KV	Kelvin	MR	Meter
KW	Kilograms Per Millimeter	MS	Square Millimeter
LA	Pounds Per Cubic Inch	MT	Metric Long Ton
LB	Pound	MU	Millicurie
LC	Linear Centimeter	MV	Number of Mults
LE	Lite	MW	Metric Ton Kilograms
LF	Linear Foot	MX	Mixed
LG	Long Ton	MY	Millimeter-Average
LH	Labor Hours	MZ	Millimeter-Minimum
LI	Linear Inch	M1	Milligrams per Liter
LJ	Large Spray	M2	Millimeter-Actual
LK	Link	M3	Mat
LM	Linear Meter	M4	Monetary Value
LN	Length	M6	Milligrams/Square Inch
LO	Lot	NB	Barge
LP	Liquid Pounds	NC	Car
LR	Layer	NL	Load
LS	Lump Sum	NM	Nautical Mile
LT	Liter	NN	Train
LY	Linear Yard	NT	Trailer
MA	Machine/Unit	NV	Vehicle
MB	Millimeter-Nominal	OL	Ounces Liquid
MC	Microgram	OP	Two-pack
ME	Milligram	OT	Overtime Hours
MF	Milligrams/Square Foot Per Side	OZ	Ounces Avoirdupois
MG	Metric Gross Tons	P1	Percent
MH	Microns	P2	Pounds Per Foot

P3	Three-Pack	QU	Quart, Imperial
P4	Four-Pack	RA	Rack
P5	Five-Pack	RD	Rod
P6	Six-Pack	RE	Reel
P7	Seven-Pack	RG	Ring
P8	Eight-Pack	RK	Roll-Metric Measure
P9	Nine-Pack	RL	Roll
PA	Pail	RM	Ream
PB	Pair Inches	RN	Ream-Metric Measure
PC	Piece	RO	Round
PD	Pad	SA	Sandwich
PE	Pounds Equivalent	SB	Square Mile
PF	Pallet (Lift)	SC	Square Centimeter
PG	Pounds Gross	SD	Solid Pounds
PH	Pack (Pak)	SE	Section
PI	Pitch	SF	Square Foot
PJ	Pounds, Decimal-Pounds/Square Foot-Pound Gage	SG	Segment
PK	Package	SH	Sheet
PL	Pallet/Unit Load	SI	Square Inch
PM	Pounds-Percentage	SJ	Sack
PN	Pounds Net	SK	Split Tank Truck
PO	Pounds Per Inch of Length	SL	Sleeve
PP	Plate	SM	Square Meter
PR	Pair	SN	Square Rod
PS	Pounds Per Square Inch	SO	Spool
PT	Pint	SP	Shelf Package
PV	One-half Pint	SQ	Square
PW	Pounds Per Inch of Width	SR	Strip
PX	Pint, Imperial	SS	Sheet-Metric Measure
PY	Peck, Dry US	ST	Set
PZ	Peck, Dry Imperial	SU	Short Ton
Q1	Quarter (Time)	SV	Skid
QD	Quarter Dozen	SW	Skein
QR	Quire	SX	Shipment
QS	Quart, Dry US	SY	Square Yard
QT	Quart	T1	Thousand Pounds gross

TA	Tenth Cubic Foot	TV	Thousand Kilograms
TB	Tube	TW	Thousand Pieces of Sheets
TC	Truck Load	TX	Troy Pound
TD	Therms	TY	Tray
TE	Tote	TZ	Thousand Cubic Feet
TF	Ten Square Yards	UN	Unit
TG	Gross Ton	VI	Vial
TH	Thousand	VT	Voltage
TI	Thousand Square Inches	WB	Wet Pound
TJ	Thousand Square Centimeters	WE	Wet Ton
TK	Tank	WH	Wheel
TL	Thousand Feet-Linear	WI	Weight Per Square Inch
TM	Thousand Feet (Board)	WK	Week
TN	Net Ton	WP	Pennyweight
TO	Troy Ounce	WR	Wrap
TP	Ten Pack	WT	Wattage
TQ	Thousand Feet	YD	Yard
TR	Ten Square Feet	YL	100 Linear Yards
TS	Thousand Square Feet	YR	Years
TT	Thousand Linear Meters	YT	Ten Yards
TU	Thousand Linear Yards	ZZ	Mutually Defined

The list above is not comprehensive, but is representative of codes employed.

A full list of codes representing unit of measurement is available from:

DATA INTERCHANGE STANDARDS ASSOCIATION (X12 DISA)

7600 Leesburg Pike, Suite 430,
Falls Church, VA 22043 USA

ATTN: Manager, Publications and Standards

Voice: 703.970.4480

<http://www.x12.org/>

<http://www.disa.org/Bookstore/Public/ListProducts.cfm?intCategory=3&intCartID=0&intUserID=0/>

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<http://webstore.ansi.org/default.aspx>

ANNEX E

Informative

ANSI X12.3

DATA ELEMENT NUMBER 374

DATE/TIME CODES

ANSI X12.3 Data Identifier Dictionary

Code List 374 Date/Time Codes

CODE	DEFINITION	CODE	DEFINITION
001	Cancel After This Date/Time	032	Promotion Invoice Performance Delivery Date/Time - End
002	Delivery Requested On This Date/Time	033	Promotion Floor Stock Protect Date/Time - Start
003	Invoice Date/Time	034	Promotion Floor Stock Protect Date/Time - End
004	Purchase Order Date/Time	035	Delivered On This Date/Time
005	Sailing Date/Time	036	Expiration Date/Time
006	Solid Date/Time	037	Ship Not Before Date/Time
007	Effective Date/Time	038	Ship Not Later Than Date/Time
008	Purchase Order Received	039	Ship Week Of Date/Time
009	Process Date/Time	040	Status Date (After And Including)
010	Requested Ship Date/Time	041	Status Date (Prior And Including)
011	Shipped On This Date/Time	042	Superseded Date
012	Terms Discount Due Date/Time	043	Publication Date
013	Terms Net Due Date/Time	044	Received On This Date
014	Deferred Payment Date/Time	045	Cumulative Quantity Start Date
015	Promotion Start	046	Cumulative Quantity End Date
016	Promotion End	047	Buyer's Local Time
017	Estimated Delivery Date/Time	048	Seller's Local Time
018	Date/Time Available/Constructive Placement	049	Confirmed Date
019	Date/Time Unloaded	050	Received On This Date
020	Check Date/Time	051	Cumulative Quantity Start Date
021	Charge Back Date/Time	052	Cumulative Quantity End Date
022	Freight Bill Date/Time	053	Buyer's Local Time
023	Promotion Order Date/Time - Start	054	Seller's Local Time
024	Promotion Order Date/Time - End	055	Confirmed Date
025	Promotion Ship Date/Time - Start	056	Estimated Port Of Entry Date
026	Promotion Ship Date/Time - End	057	Actual Port Of Entry Date
027	Promotion Requested Delivery Date/Time - Start	058	Customs Clearance Date
028	Promotion Requested Delivery Date/Time - End	059	Inland Ship Date
029	Promotion Performance Delivery Date/Time - Start	060	Engineering Change Level Date
030	Promotion Performance Delivery Date/Time - End	061	Cancel If Not Delivered By This Date
031	Promotion Invoice Performance Delivery Date/Time - Start	062	Blueprint Date

063	Do Not Deliver After This Date	089	Inquiry Date
064	Do Not Deliver Before This Date	090	Report Start Date
065	1st Schedule Delivery Date	091	Report End Date
066	1st Schedule Ship Date	092	Contract Effective Date
067	Current Schedule Delivery Date	093	Contract Expiration Date
068	Current Schedule Ship Date	094	Manufacturing Date
069	Promised For Delivery (Date/Time)	095	Bill of Lading Date
070	Scheduled For Delivery (After And Including)	096	Date/Time Of Discharge
071	Requested For Delivery (After And Including)	097	Transaction Creation Date
072	Promised For Delivery (After And Including)	098	Bid (Effective) Date
073	Scheduled For Delivery (Prior To And Including)	099	Bid-Open Date (Date Bids Will Be Opened)
074	Requested For Delivery (Prior To And Including)	100	No Shipping Schedule Established As Of Date/Time
075	Promised For Delivery (Prior To And Including)	101	No Production Schedule Established As Of Date/Time
076	Scheduled For Delivery (Week Of)	102	Expect To Ship By Date
077	Requested For Delivery (Week Of)	103	Expect To Ship By Week Of Date
078	Promised For Delivery (Week Of)	104	Revised Expect To Ship By Date
079	Promised For Shipment (Date/Time)	105	Revised Expect To Ship By Week Of Date
080	Scheduled For Shipment (After And Including)	106	Required By Date
081	Requested For Shipment (After And Including)	107	Deposit Date/Time
082	Promised For Shipment (After And Including)	108	Postmark Date
083	Scheduled For Shipment (Prior To And Including)	109	Date/Time Received At Lockbox
084	Requested For Shipment (Prior To And Including)	110	Agreed Upon Scheduled Ship
085	Promised For Shipment (Prior To And Including)	116	Scheduled Interchange Delivery
086	Scheduled For Shipment (Week Of)	214	Date of Repair/Service
087	Requested For Shipment (Week Of)		
088	Promised For Shipment (Week Of)		

The list above is not comprehensive, but is representative of codes employed. A full list of codes representing time is available from:

**DATA INTERCHANGE STANDARDS ASSOCIATION
(X12 DISA)**

**7600 Leesburg Pike, Suite 430,
Falls Church, VA 22043 USA**

ATTN: Manager, Publications and Standards

Voice: 703.970.4480

<http://www.x12.org/>

<http://www.disa.org/Bookstore/Public/ListProducts.cfm?intCategory=3&intCartID=0&intUserID=0/>

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ANNEX F

Informative

ANSI X12.3

DATA ELEMENT NUMBERS 208 & 209

HAZARDOUS MATERIAL CODES

ANSI X12.3 Data Identifier Dictionary Code List 208 and 209 Hazardous Material Codes

208 Hazardous Material Code Qualifier

CODE	DEFINITION
------	------------

4	46 Level DOT Code ¹
6	Airline Tariff 6D ²
9	Title 49, Code of Federal Regulations (CFR) ³
A	International Civil Aviation Organization (ICAO) Code ⁴
B	Uniform Fire Code (UFC) ⁵
C	Storage Compatibility Group ⁶
D	Hazardous Material ID, DOT ⁷
E	Endorsement
F	Air Force Joint Manual 24-204 ⁸
I	Intergovernmental Maritime Organization (IMO) ⁹
R	Bureau of Explosives (BOE) 6000 Tariff ¹⁰
T	International Air Transport Association Dangerous Code List ¹¹
U	United Nations ¹²
X	Hazardous Class or Division ¹³

The list above is not comprehensive, but is representative of codes employed. A full list of Hazardous Material Code Qualifiers is available from:

1 Code of Federal Regulations CFR Title 46

Available from:
Superintendent of Documents
U.S. Government Printing Office
Washington, DC 20402
Abstract: Hazardous materials codes for domestic water shipments

2 Tariff 6D - Official Regulations on Restricted Articles

Available from:
Airline Tariff Publishing Co.
Dulles Airport
Washington, DC
Abstract: Hazardous materials codes for domestic air shipments

3 Hazardous Material Code (49 Level)

Available from:

Standard Transportation Commodity Code (STCC)/Hazardous Materials Shipping Description
Railinc/Association of American Railroads
7001 Weston Parkway – Suite 200
Cary, NC 27513

Abstract: The hazardous materials section (Group 49) of the STCC is organized according to the kind and degree of hazard associated with hazardous materials or hazardous substances, with special provisions to relate the identified commodity to its product class with the established commodity code structure.

4 IATA Restricted Articles Regulation

Available from:

International Air Transport Association (IATA)
Publications Department
800 Place Victoria - PO Box 113
Montreal, Quebec H4Z 1M1 Canada
Voice: +1 514 874 0202

Abstract: Hazardous materials codes for international air shipments

5 Uniform Fire Code (UFC)

Available from:

International Fire Code Institute (IFCI)
5360 Workman Mill Road
Whittier, CA 90601-2298

**6 Storage Compatibility Group Designator
Code of Federal Regulations
Transportation, Title 49, Section 172
October 1, 1992, pages 328-329**

Available from:

Superintendent of Documents
U.S. Government Printing Office
Washington, D.C. 20402

Abstract: Provides storage group designators, as established by the U.S. Department of Transportation, which specify special storage provisions for hazardous materials for the purpose of transportation in commerce.

**7 Hazardous Materials ID, DOT
Code of Federal Regulations
Transportation, Title 49, parts 100 to 177
revised as of November 1, 1983, pages 75-170**

Available from:

Superintendent of Documents
U.S. Government Printing Office
Washington, D.C. 20402

Abstract: Provides codes, names, and hazard classes for materials designated by the U.S. Department of Transportation as hazardous for purposes of transportation in commerce. The identifier of the materials listed is alphanumeric of the form: "AAdddd". The numeric ("d") portion of the identifier has no significance. The alphabetic prefix may be "UN" for materials appropriate for both international and

domestic shipments; or "NA" for materials appropriate only for domestic shipments and shipments to and from Canada.

**8 Air Force Joint Manual 24-204:
Preparing Hazardous Materials for Military Air Shipments
United States Air Force Material Command**

Available from:

Defense Automated Printing Service
Bldg. 4D, 700 Robins Avenue
Philadelphia, PA 19111-5094

URL: <http://www.wpafb.af.mil/shared/media/document/AFD-121204-014.pdf>

Abstract: This manual provides guidance and procedures for preparing hazardous materials for shipment aboard military aircraft to ensure that such materials are packaged, marked, labeled, and prepared properly for transportation

**9 Intergovernmental Maritime Organization (IMO)
Dangerous Goods Code**

Available from:

Intergovernmental Maritime Consultative Organization (IMCO)
101-104 Piccadilly
London W1 VOA England

Abstract: Dangerous materials codes for international ocean shipments.

**10 Bureau of Explosives (BOE) 6000 Tariff
Hazardous Materials Regulations of the Department of Transportation by Air, Rail, Highway, and Water**

Available from:

Association of American Railroads
Publications
P.O. Box 1265
Evans City, PA 16033

Abstract: Regulations and restrictions covering the acceptance and transportation of explosives and other dangerous articles by carriers.

**11 International Air Transport Association (IATA) Dangerous Goods Code
Dangerous Goods Regulations**

Available from:

International Air Transport Association (IATA)
Publications Department
800 Place Victoria - PO Box 113
Montreal, Quebec H4Z 1M1 Canada
Voice: +1 514 874 0202

Abstract: Air courier regulations for the shipping and acceptance handling of dangerous goods. Based on the International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air.

**12 United Nations Number (Dangerous Goods) *Transportation of Dangerous Goods*,
Recommendations of the Committee of Experts of the Transport of Dangerous Goods, Third Revised
Edition United Nations ST/SG/AC10/1REV.3, 1983, SALES NO.E.83 VIII.1**

Available from:

United Nations Publications
Polaris des Nations

CH – 1211 Geneva 10 Switzerland

Abstract: Provides codes, names and hazard classes for materials designated as dangerous for purposes of transport in commerce. The identifier of the dangerous goods listed is numeric of the form "dddd".

13 Hazardous Class or Division

**Code of Federal Regulations, Transportation, Title 49,
Subchapter C, Subpart B, Table of Hazardous Materials and Special Provisions
October 1, 1992 Version, pages 120-238**

Available from:

Superintendent of Documents
U.S. Government Printing Office
Washington, D.C. 20402

Abstract: Provides classes and divisions for Hazardous Materials as established by the U.S. Department of Transportation for the purpose of transportation in commerce.

ANNEX G

Informative

ISO 4217

CURRENCIES AND FUNDS CODES

ISO 4217

Currencies and Funds Codes

<u>DEFINITION</u>	<u>ALPHABETIC CODE</u>	<u>NUMERIC CODE</u>
Algerian Dinar	DZD	012
Argentine Peso	ARS	032
Australian Dollar	AUD	036
Brazilian Real	BRL	986
Canadian Dollar	CAD	124
Chinese Yuan	CNY	156
Danish Krone	DKK	208
Egyptian Pound	EGP	818
European Euro	EUR	978
Hong Kong Dollar	HKD	344
Iceland Kronur	ISK	352
Indian Rupee	INR	356
International Monetary Fund (SDR)	XDR	960
Israeli Shekel	ILS	376
Japanese Yen	JPY	392
Korean (South) Won	KRW	410
Mexican Peso	MXN	484
New Zealand Dollar	NZD	554
Norwegian Krone	NOK	578
Paraguayan Guarani	PYG	600
Polish Zloty	PLN	985
Romanian Leu	RON	946
Russian Rouble	RUB	643
Saudi Riyal	SAR	682
Singapore Dollar	SGD	702
South African Rand	ZAR	710
Swedish Kronor	SEK	752
Swiss Franc	CHF	756
Syrian Pound	SYP	760
Thailand Baht	THB	764
Turkish Lira	TRY	949

United Arab Emirates Dirham	AED	784
United Kingdom Pound Sterling	GBP	826
United States Dollar	USD	840
Uruguayan Peso	UYU	858
Venezuelan Bolivar	VEF	937
Vietnamese Dong	VND	704
Gold	XAU	959

The list above is not comprehensive, but is representative of codes employed. A full list of codes is given in Currency and funds name and code elements (ISO 4217) that is available from:

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New York, NY 10036
(212) 642-4900
<http://webstore.ansi.org/default.aspx>

ANNEX H

Informative

ISO 3166-1 COUNTRY CODES

ISO 3166-1
Country Codes

<u>DEFINITION</u>	<u>ALPHA-2 CODE</u>	<u>ALPHA-3 CODE</u>	<u>NUMERIC CODE</u>
Non-specific Country	AA	AAA	000
Argentina	AR	ARG	032
Australia	AU	AUS	036
Austria	AT	AUT	040
Belgium	BE	BEL	056
Brazil	BR	BRA	076
Canada	CA	CAN	124
Denmark	DK	DNK	208
Finland	FI	FIN	246
France	FR	FRA	250
Germany	DE	DEU	276
Greece	GR	GRC	300
Holy See (Vatican City State)	VA	VAT	336
Iceland	IS	ISL	352
India	IN	IND	356
Ireland	IE	IRL	372
Israel	IL	ISR	376
Italy	IT	ITA	380
Japan	JP	JPN	392
Luxembourg	LU	LUX	442
Mexico	MX	MEX	484
Netherlands	NL	NLD	528
New Zealand	NZ	NZL	554
Norway	NO	NOR	578
Paraguay	PY	PRY	600
Poland	PL	POL	616
Portugal	PT	PRT	620
Romania	RO	ROM	642
Saudi Arabia	SA	SAU	682
Singapore	SG	SGP	702
South Africa	ZA	ZAF	710

Spain	ES	ESP	724
Sweden	SE	SWE	752
Switzerland	CH	CHE	756
Turkey	TR	TUR	792
United Kingdom	GB	GBR	826
United States of America	US	USA	840
Uruguay	UY	URY	858
Venezuela	VE	VEN	862

The list above is not comprehensive, but is representative of codes employed. A full list of Codes for the representation of names of countries and their subdivisions -- Part 1: Country codes (ISO 3166-1), except "Non-specific Country" defined in N664 from ISO 3166/MA, is available from:

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25 West 43rd Street
New York, NY 10036
(212) 642-4900
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ANNEX I

Normative

DATA IDENTIFIER REQUEST FORM

Following is a request form for new Data Identifiers. Where the end user finds that the identifiers described in this document are insufficient, this request form should be used as appropriate.

Rev. ANS MH10.8.2-7262 (DI)

Reference: _____

Date: _____

ANS MH10.8.2 DATA IDENTIFIER REQUEST FORM

Complete all parts. Submit to:

EMAIL: DIRequests@MHI.org

MAIL: DIMC
c/o MHI
8720 Red Oak Blvd – Suite #201
Charlotte, NC 28007-3992
USA

PHONE: +1 704.676.1190

Ask For Data Identifier Request Desk

Incomplete forms or those with inadequate support for the change requested will be returned to the submitter. The ANS MH10.8.2 DI Maintenance Committee will notify submitters of the status of the work request following their review.

URL for Publicly Available Continuous Maintenance version of ASC MH10.8.2 Data Identifiers and Application Identifiers for current version of ANSI MH10.8.2: <http://www.mhi.org/standards/di>

NOTE: Please save the document in “.doc” format (Word 2003-compatible), not “.docx” format (Word 2007 and above).

REQUEST FOR: ☐ New Data Identifier
 ☐ Data Identifier Interpretation

ORGANIZATION: _____

CONTACT PERSON: _____

ADDRESS: _____

TELEPHONE: _____

EMAIL ADDRESS: _____

1. PROPOSED DATA IDENTIFIER

Provide a short description (20 words or less) which would be included as a description for the proposed Data Identifier. For an interpretation, provide a comprehensive description of the aspect of the identifier that needs interpretation.

PAGE 2 (DATA IDENTIFIER REQUEST)

2. BUSINESS CASE

Explain why you need the proposed assignment. Provide a complete scenario that tells what the business function, operation, or problem is that will be satisfied by a new assignment to the ANS MH10.8.2 Data Identifier Standard. If the proposed DI is already in use by your organization, please identify how long this identifier has been in use and other organizations you are aware of who employ the same identifier. The ANS MH10.8.2 DI Maintenance Committee requires enough information to be able to propose an alternate solution if necessary. Be specific because this will also appear in the ANS MH10.8.2 Voting Package and will be the only information that voters have on which to base their vote.

PAGE 3 (DATA IDENTIFIER REQUEST)

3. DEFINITIONS

Definitions for new assignments and for industry-specific terms must be complete. For new ANS MH10.8.2 DI, provide a proposed assignment and a DI definition. RULES: (1) Acronyms/abbreviations cannot be added to the standards - they must be spelled out. (2) Provide an expanded assignment definition for each DI which is not completely self-explanatory, that is, terms that are not in general business use or that are industry specific. (3) Provide code source references for all externally published (non-ANS MH10.8.2) code lists cited (use the Form for New or Revised Code Source Reference). If one exists, provide a precise description of the structure of the data as foreseen by your organization for this application. Indicate data elements involved and their format (numeric, alphanumeric, fixed or variable length, number of decimals). Indicate the business function of each data element in the application.

NOTE: MH10.8.2 restricts the character set – UNLESS OTHERWISE STATED WITHIN THE DI – to upper-case alphanumeric letters and 0 through 9 numbers. If other than these characters are needed for use, THEY MUST BE SO STATED WITHIN THIS REQUEST, otherwise they cannot be used.

PAGE 4 (DATA IDENTIFIER REQUEST)

4. MEDIA AND APPLICATION USE

- *With what media (e.g., bar code, 2D symbol, RF tag, etc.) do you intend to use the proposed Data Identifier?*

- *At what stage will the Data Identifier and data be created and applied?*

- *On to what and when will the media be applied (package, label, tag, document, . . .)?*

- *Why does the information need to be machine-readable?*

- *When and where is the media read?*

- *Describe the use of the Data Identifier by other users than the originator:*

- *What is the number of potential users?*

5. Justification

Describe the benefits (hard and soft savings) expected from the application.

6. Additional Information

Feel free to attach any addition information related to your organization and the application.

Date: _____

Signature: _____

PAGE 5 (DATA IDENTIFIER REQUEST)

Data Identifier Data Dictionary Record

NOTE: When multiple options follow a heading, choose one and remove the others.

Example:

As currently listed: **Class:** Numeric / Alpha / Alphanumeric / Binary

After selection: **Class:** Alpha

DATA DICTIONARY DETAILED ENTRY			
Name:		Version:	
Key:			
XML Tag:		Proposed DI:	
Definition:			
Class: Numeric / Alpha / Alphanumeric / Binary Remarks: Decimals: Yes / No Min_Length: Max_Length: Case Sensitive: Yes / No Characters other than upper-case alpha and 0-9 numbers needed: Yes / No If Yes – list the characters required.			
Business Rules:			
Data Element Source / Authority:			
APPLICATION AREAS			
Area	Application	Category	Remarks
USES			
Application Area	Usage	Type	Specific Use
ALIAS: Production Date			
^a Table footnote.			

ANNEX J

Informative

SYSTEM IDENTIFIERS

Annex J (Informative) System Identifiers

General

Section I, Data Identifiers, lists a Category “0” as Special Characters Not Assigned or Control by ASC MH 10/SC 8. These characters, in a leading position of the data structure, are sometimes referred to as system identifiers, denoting a data structure maintained by the organization claiming this system identifier.

Table J.1: Category “0” - System Identifiers

System Identifier (See Notes)	Data Structure Usage
+	Plus sign. Health Industry Business Communications Council (HIBCC)
&	Ampersand. ICCBBA
=	Equal sign. ICCBBA
FNC1	Function 1. Appears in the first position following the symbology start character of a Code 128, Code 49, or Code 16K Symbol to signify a GS1-controlled symbol
[]> ^{Rs}	Left square bracket, right parenthesis, greater than sign, record separator character. Data structure compliant to ISO/IEC 15434, <i>Information technology — Automatic Identification and Data Capture Techniques — Syntax for High Capacity ADC Media</i>
-	Hyphen – Minus. Pharmaceutical Central Number (PZN), controlled by IFA-ABDATA, Germany (Registration of this system identifier expires on 2016-07-01). Replaced by “9N”.
!	Exclamation mark. Eurocode-IBLS

Notes:

Certain characters, e.g., FNC 1, have no ISO/IEC 646 (ASCII) equivalent and require special processing for human-readable and universal AIDC media encoding.

Certain characters, e.g., the ^{Rs} in []>^{Rs}, are difficult to represent in human-readable and may require mutually agreed upon dingbats for the representation in human-readable text.

Certain characters, e.g., the exclamation mark, are not universally encodable in the basic character set of all symbologies, e.g., Code 39.

Controlling authority

None of these character uses are covered or controlled by this standard, ANS MH10.8.2. Neither does this standard recommend the use of these system identifiers.

Minimum requirements for inclusion within this Annex

For a system identifier to be listed in this annex requires two basic principles:

1. The system identifier must be integral in a specification approved by the governing organization of which the system identifier refers.
2. The specification within which the system identifier is integral must have a maintained URL, permitting open ordering of the specification. Ideally, the specification would be available at no charge.

Specification availability

Specifications for the system identifier contained within this informative annex can be accessed at the following URLs.

Table J.2: URLs for additional information on System Identifier

System Identifier	Controlling Specification	URL
+	ANSI HIBC 2.4 Supplier Labeling Standard	http://www.hibcc.org
&	ISBT 128 Standard Technical Specification	http://www.iccbba.org
=	ISBT 128 Standard Technical Specification	http://www.iccbba.org
FNC1	GS 1 General Specifications	http://www.gs1.org/barcodes-epcrfid-id-keys/gs1-general-specifications
D> ^{RS}	ISO/IEC 15434, Information technology — Automatic Identification and Data Capture Techniques — Syntax for High Capacity ADC Media	http://www.iso.org
- (Minus sign)	Pharmaceutical Central Number (PZN)	Organization: http://www.ifaffm.de Document: http://www.ifaffm.de/en/ifa-codingsystem.html (temporarily, text is available only in German) (Registration of this system identifier expires on 2016-07-01. Replaced by “9N”.)
! (Exclamation mark)	Eurocode-IBLS	Organization: http://www.eurocode.org Document: http://www.eurocode.org/guides/index.html

The on-line listing of the continuous maintenance version of ANS MH10.8.2, Data Application Identifiers can be found at:

- <http://www.mhi.org/standards/request>

ANNEX K

Normative

DATA IDENTIFIERS FOR RETURNABLE PACKAGING ITEMS (RPIs)

Annex K (Normative) Data Identifiers for Returnable Packaging Items (RPIs)

K.1 General

The concepts of returnable, reusable, and recyclable are frequently used interchangeably, though conceptually they are quite different. A key underlying concept of difference is ownership, whereby returnable items maintain the original ownership, while the ownership of reusable and recyclable items is transferred between parties. Figure K.1 shows a consumer lifecycle explaining the differences.

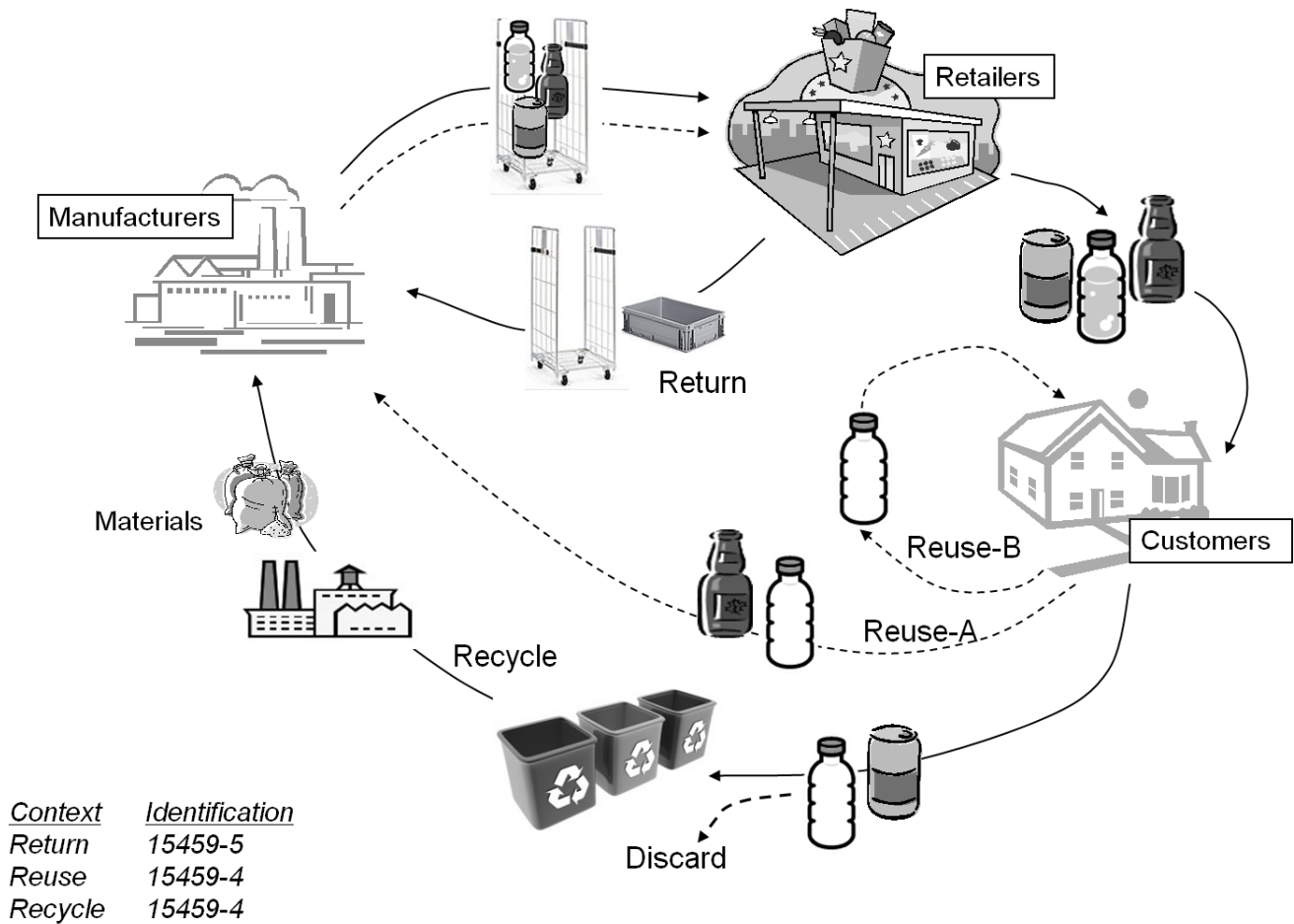


Figure K.1 – Returnable, reusable, and recyclable lifecycle

K.2 Recyclable item

With a recyclable item, the ownership of the item is transferred when the item is sold. A typical use recyclable item is a battery, which can be disposed at a recycling station and parts of the content can be recycled and used for manufacturing of new batteries.

K.3 Identification

ISO/IEC 15459-4 addresses unique identification for individual recyclable items (products).

K.4 Reusable item

With a reusable item the ownership of the item is transferred when the item is sold. In some jurisdictions an incentive to return the reusable packaging is provided at the time of purchase, a deposit, which can be recovered when the item is returned to the location from which the item was purchased.

A typical use recyclable item is a hard plastic bottle, which can either be reused by the user (i.e. filling the bottle with new content after cleaning it) or disposed at a recycling station and depending on whether its constituents parts can be reused (i.e. cleaned and refilled) or recycled and used for “manufacturing” of new bottles.

K.5 Identification

ISO/IEC 15459-4 addresses unique identification for individual reusable items (products).

K.6 Returnable items

With a returnable item the ownership of the item remains with the party providing the item, even though the item is sent to a customer. The supplier retains ownership of the asset with the anticipation that the customer will return the asset once it has served its original purpose

A typical use of a returnable item is for transportation of goods where the item can be reused in terms of that the content and carrier can change but the owner is still the same.

K.7 Identification

ISO/IEC 15459-5 addresses unique identification for returnable items.

K.8 Returnable Transport Items and Returnable Packaging Items

K.9 Overview

Some pallets and returnable boxes are equipped with shock absorbing material to protect them from potential damage occurring during in the transportation and handling process. An effective solution is the use of partitions or sorting boards for separating the contents into appropriate groups, making it possible to place many items on a single pallet or returnable box. This kind of accessory for a pallet or returnable box is defined as a “partition”. The typical example of this is a post-type partition used with the post pallet. Also included in this group is packing material used to place or arrange the contents between the posts, or a packaging material for dividing the inside of the returnable box into several smaller sections.

K.10 Partitions

Some pallets and returnable boxes are equipped with shock absorbing material to protect them from potential damage occurring during in the transportation and handling process. An effective solution is the use of partitions or sorting boards for separating the contents into appropriate groups, making it possible to place many items on a single pallet or returnable box. This kind of accessory for a pallet or returnable box is defined as a “partition”. The typical example of this is a post-type partition used with the post pallet. Also included in this group is packing material used to place or arrange the contents between the posts, or a packaging material for dividing the inside of the returnable box into several smaller sections.

K.11 Posts

Figure K.2 shows a post that is normally used to securely fix packing materials or returnable box on the pallet. Most of these posts are made of high durable substances like plastic or metal.

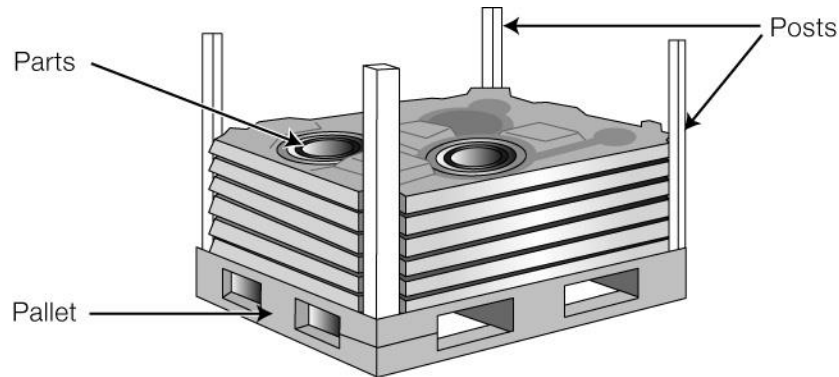


Figure K.2 – Post

K.12 Packing materials

Some kind of packing materials should be provided to protect the items from a shock or vibration that may be encountered during transportation, or protect them from being touched or hit by the pallet or returnable box in which they are placed. Most of the packing materials are made of high resilient flexible substances like plastic, urethane, and polystyrene foam. This guideline is applicable to these kinds of packing materials (see Figure K.3 and Figure K.4).

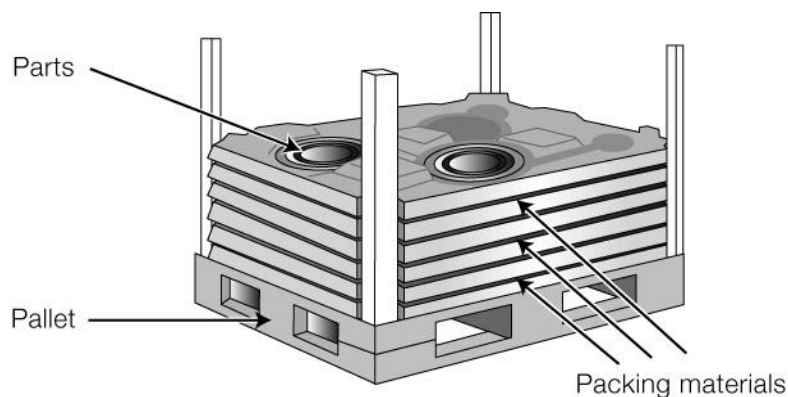


Figure K.3 – Packing material

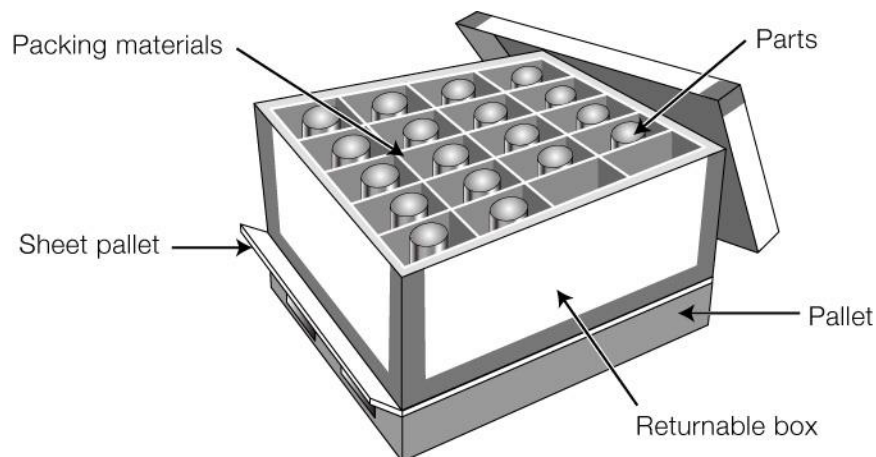


Figure K.4 – Packing material

K.13 Identification

In Figure K.3, the base pallet is the actual RTI and the moulded plastic layers and posts are the RPIs. If each is serialized, it may be important to associate the RPIs with the parent RTI. In this example, the RTI may have a unique identity of “25BUN0433257110000001”. The four posts might have a unique identity of

“55BUN043325711P000001”

“55BUN043325711P000002”

“55BUN043325711P000003”

“55BUN043325711P000004”

... and the six plastic layers might have a unique identity of

“55BUN043325711L000001”

“55BUN043325711L000002”

“55BUN043325711L000003”

“55BUN043325711L000004”

“55BUN043325711L000005”

“55BUN043325711L000006”

K.14 My parent is ...

One possibility to associate the RPIs with the parent RTI is with the use of the Data Identifier “1F” which declares, “My parent is ...”. Using this example the 3rd plastic layer would be encoded 55BUN043325711L000003<GS>1F25BUN0433257110000001.

The other layers and posts would be similarly encoded.

K.15 I have ____ children.

Yet another possibility is to simply identify the number of RPIs associated with the parent RTI using the Data Identifier "3F" which declares, "I have ____ children". Using the same example the base pallet would be encoded

25BUN0433257110000001<GS>3F10

K.16 I have __children and they are...

Yet another possibility is to simply identify the number of RPIs associated with the parent RTI using the Data Identifier "5F" which declares, "I have ____ children and they are...". Using the same example the base pallet would be encoded

25BUN0433257110000001<GS>5F10<GS>55BUN043325711L000001<GS>55BUN043325711L000002<GS>55BUN043325711L000003<GS>55BUN043325711L000004<GS>55BUN043325711L000005<GS>55BUN043325711L000006<GS>55BUN043325711P000001<GS>55BUN043325711P000002<GS>55BUN043325711P000003<GS>55BUN043325711P000004

ANNEX L

Normative

MATERIAL CATEGORIES AND MATERIAL CODES

Annex L (Normative) Material categories and material codes

Material categories and material codes are used to identify the material from which an item is made, to facilitate easier recycling or other reprocessing. Such symbols have been defined for batteries, biomaterial/organic material, glass, metals, paper, and plastics.

L.1 Material categories

The code shall be used in relation to the product identification of the actual product, when categorization of material is to be declared.

Table L.1 — Material categories

Code	Description
10	Plastic
20	Paper and cardboard
40	Metal
50	Wood
60	Textile
70	Glass
80	Composite of packaging materials
00	Other

L.2 Material types and coding

It is recommended that the codes in the tables below be placed below the Recycling Symbol for a recyclable material, i.e., where it is most suitable.

(Reference: http://en.wikipedia.org/wiki/Recycling_codes)

L.2.1 Plastics (10)

Table L.2 — Coding of plastic

Code	Description	Examples
PET	Polyethylene terephthalate	Polyester fibers, soft drink bottles
HD-PE	High-density polyethylene	Plastic bottles, plastic bags, trash HDPE cans, imitation wood
LD-PE	Low-density polyethylene	Plastic bags, buckets, soap dispenser bottles, plastic tubes

L.2.2 Paper (20)

Table L.3 — Coding of paper

Code	Description	Examples
CPAP	Cardboard	Transport packaging, moving boxes
PAP	Paper	Mixed paper magazines, mail, newspapers
PBD	Paperboard	Greeting cards, food boxes, shoe boxes

L.2.3 Metals (40)

Table L.4 — Coding of metals

Code	Description	Examples
FE	Steel	---
ALU	Aluminum	---

L.2.4 Wood (50)

Table L.5 — Coding of wood

Code	Common name	Genus
FOR	Wood	---
NTR	Impregnated Wood	---

L.2.5 Textile (60)

Table L.6 — Coding of textiles

Code	Description	Examples
TEX	Textile	---

L.2.6 Glass (70)

Table L.7 — Coding of glass

Code	Description	Examples
GL1	Clear Sort Glass	---
GL2	Dark Sort Glass	---

L.2.7 Composite (80)

Table L.8 — Coding of composite

Code	Description	Examples
PAPALU	Cardboard + Aluminum	Liquid storage containers, juice boxes, cardboard cans, Cigarette pack liners, gum wrappers, cartage shells for blanks, fireworks coloring material.
PAPPET	Cardboard + Plastic	Consumer packaging, pet food bags, cold store grocery bags, ice cream containers, cardboard cans, disposable plates



Figure L.1 — Examples of encoding recyclable materials